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"CGLOPS-2"

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QUALITY ASSESSMENT REPORT

LAKE WATER QUALITY

300M PRODUCT

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draft – pending review

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List of Acronyms

ATBD	Algorithm Theoretical Basis Document
BC	Brockmann Consult
CCI	Climate Change Initiative
C-GLOPS	Copernicus Global Land Operations
FTP	File Transfer Protocol
GCOS	Global Climate Observing System
GIO	GMES Initial Operations
GPT	graph processing tool
LSR	Lake Surface Reflectance
MERIS	Medium Resolution Imaging Spectrometer on Envisat
MSI	Multispectral Instrument on Sentinel-2
NERC	National Environment Research Council (UK)
NRT	Near Real Time
NTU	Nephelometric Turbidity Unit
obs	Observation
OC	Ocean Colour
OLCI	Ocean and Land Colour Instrument on Sentinel-3
OWT	Optical Water Type
PML	Plymouth Marine Laboratory
PUM	Product User Manual
QAA	Quasi-Analytical Approach
QAR	Quality Assessment Report
rep	Representative
R_w	Water leaving reflectance
TS	Trophic State
TSI	Trophic State Index
TSM	Total Suspended Matter
TUR	Turbidity

WGS84	World Geodetic System 1984

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1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Copernicus Global Land Service – Lake Water provides optical and temperature observations of global inland water bodies that are either the largest or most strategic for monitoring. The set of water bodies that are included in the observations consist for the largest part of natural lakes and further includes reservoirs, enclosed lagoons and some wide river sections. The products contain six (sets of) variables: lake water-leaving reflectance (all wavebands available after atmospheric correction), turbidity (TUR), total suspended matter concentration (TSM), a trophic state index (TSI) derived from phytoplankton biomass by proxy of chlorophyll-a, chlorophyll-a concentration and a floating cyanobacteria risk index. This document describes all product from observing > 4000 inland waterbodies using version 2.0 of the processing chain. Production and delivery of the parameters are over set 10-day intervals (starting the 1st, 11th and 21st day of each month) and mapped to a common global spatial grid at nominally 300m (~0.0022°) resolution. The input products are Envisat MERIS and Sentinel-3 OLCI L1B data. The algorithms used to derive the input for the optical lake water products are implemented in the *Calimnos* processing chain and were tuned and validated against 13 predefined optical water types (OWT) in the NERC (UK) GloboLakes project, plus two additional OWTs for identifying pixels contaminated by land adjacency. This Quality Assessment Report (QAR) describes the performance of the algorithms and the products delivered within this service.

1.2 SCOPE AND OBJECTIVES

One of the main objectives of the Copernicus programme is to provide to the scientific community and other stakeholders, including policy makers, state-of-the-art information required for their respective applications. Prior to product version 2.0.0, the Lake Water Quality (LWQ) products turbidity (TUR) and Trophic State Index (TSI) were already in pre-operational mode, whereas the water-leaving reflectance was in demonstration mode due to suspected limitations which do not ultimately affect the derived turbidity and TSI. New variables added in this version include Total Suspended Matter (TSM) concentration, chlorophyll-a concentration and a floating cyanobacteria index, and are in demonstration mode pending further review. The original products are provided for the full Envisat MERIS and Sentinel-3 OLCI (A/B) archives as versions 1.3.0 and 1.4.0. The new version enters near real-time (NRT) using OLCI. The objective of this document is to present the results of the quality assessment of the Lake Water Quality products of versions 1.3.0-2.0.0 (300m products). The assessment is performed at the level of single-lake time-series as well as overall algorithm performance through matchup analysis of water-leaving reflectance, TUR and TSI.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements and the expected performance
- Chapter 3 describes the methodology for quality assessment, the metrics and the criteria of evaluation
- Chapter 4 presents the results of the analysis
- Chapter 5 summarizes the main conclusions of the study
- Chapter 6 makes recommendations based upon the results

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: Call for tenders JRC/2020/OP/3213 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2020)7715271, dated 17/12/2020

AD4: Call for tenders JRC/IPR/2023/CPN/2325 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2023)6699927, dated 03/10/2023

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service.
CGLOPS2_SVP	Service Validation Plan of the Global Land Service
CGLOPS2_ATBD_LWQ300_v2.0.0_I1.01	Algorithm Theoretical Basis Document of the Lake Water Quality products, demonstration

	product, historic and NRT data

1.4.3 Output

Document ID	Descriptor
CGLOPS2_QAR_LWQ300_v2.0.0_I1.01	Report describing the results of the scientific quality assessment of the Lake Water Quality products, demonstration and pre-operational products, historic and NRT data

1.4.4 External documents (if any)

N/A

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2 REVIEW OF USER REQUIREMENTS

According to the applicable document [AD2], the user requirements relevant for Lake Water Products v2.0 are:

Definition:

The Lake Water Products are lake turbidity (TUR) in NTU, total suspended matter (TSM) in g/m^3, Trophic State Index (TSI), chlorophyll-a concentration (Chl-a) in g/m^3, floating cyanobacteria index and lake surface reflectance spectra (LSR, given in terms of fully normalised water reflectance R_w; a dimensionless quantity) from which TUR, TSM, Chl-a, TSI and the floating cyanobacteria index are derived. In addition, the mean of relative uncertainty of Chl-a and TSM in the observation period, and the relative uncertainty and unbiased uncertainty in LSR at reflectance peak are included. The products shall be provided as 10-day averages of the respective parameters, except for LSR for which the observation that is statistically most representative for the observed time period is given. The reason for the latter is that an average of each LSR waveband may yield a physically inaccurate spectrum of reflectance and would, thus, not be useful to feed into alternative algorithms for the retrieval of optical water quality parameters. Turbidity and TSM are key indicators of water clarity. Turbidity quantifies the haziness of the water and acts as an indicator of underwater light availability. TSM quantifies the concentration of suspended particles in the water body. TSI, as defined by Carlson (1977), refers to the degree at which organic matter accumulates in the water body and is most commonly used in relation to monitoring eutrophication. Chl-a is the precursor product to Trophic State Index and is commonly used as a proxy to phytoplankton biomass. The cyanobacteria risk index indicates the detection of cyanobacteria surface blooms in the water body.

Geometric properties:

The baseline datasets pixel size shall be provided at resolutions of 100m and/or 300m. The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view. Pixel coordinates shall be given for the pixel centre.

Geographical coverage:

Global window

The initial window definition is aligned to the global datasets produced during the GMES Initial Operations (GIO) phase for the most widely used output data. The following output specifications are further optimised with respect to the requirements:

- pixel size at 300m: 0.25/112°
- global window coordinates: UL: 180°W-90°N, BR: 180°E, 90°S
- global grid size at 300m: 161280 columns, 80640 lines

Continental windows:

Continental windows may be asked by the contracting authority to satisfy specific user needs. Wherever applicable, continental windows shall be drawn from global output data.

Accuracy requirements:

Lake Water Reflectance is an Essential Climate Variable (GCOS 244, 2022) with an associated accuracy requirement threshold of 30% (breakthrough: 20% and target: 10%) at the reflectance peak (GCOS 245, 2022). Turbidity has an uncertainty threshold of 50% (goal: 35%) and TS has an uncertainty threshold of 0.3 in log10 or 50% [AD4].

Temporal Definition

As a baseline, the biophysical parameters are computed by and representative of *dekads*, i.e. periods of nominally 10 and varying from eight to eleven days, defined to start on days 1, 11 and 21 of each month of the year.

3 QUALITY ASSESSMENT METHOD

3.1 OVERALL PROCEDURE

Multiple methods were applied in order to assess the quality of the lake water products. First, comparisons were performed between the 300m OLCI products for different product versions (v.1.2 and v.1.3.0) and between 300m and 100m lake water products. The 100m lake water products are based on Sentinel-2 MSI data and are provided as demonstration products within the Copernicus Global Land Service. In a second step, spatial and temporal patterns were investigated by visual inspection of maps and time series. Maps showing dekads of one year are used to check the consistency of products in space and time. Time series of selected positions in different lakes were extracted and plotted for several parameters in order to assess the reliability of seasonal trends, outliers, unexpected patterns.

The retrieved LSR and water-column parameters have also been validated against in situ data. Here, the assessment is performed by a) matchup analysis and b) time series (from dekads) at selected measurement stations. The matchup analysis is also based on the 10-day product. The time difference between an in situ measurement and the averaged day of the dekads can be up to 3 days. Such long intervals are nevertheless usually necessary to collect a sufficient number of matchups, because the in situ observation data archive for inland waters is small. The matchup analysis of L2 products which was already performed within UK NERC GloboLakes allows a time difference of 1, 3, or 7 days depending on measurement parameter, to optimise between introducing errors due to spatiotemporal dynamics and limited availability of in situ data.

For most of the comparison exercises with in situ data, a window of 3 x 3 pixels (macro-pixel) around the location of the measurement stations has been extracted from the satellite products

and filtered for invalid pixels before producing the average and standard deviation. For filtering, the following flags are applied:

l1_flags.INVALID or idepix.F_LAND or idepix.F_CLOUD or idepix.F_CLOUD_BUFFER or idepix.F_CLOUD_SHADOW or idepix.F_MIXED_PIXEL

The same valid pixel expression is applied for the generation of the Level 3 products.

The newly introduced flag for floating cyanobacteria and macroalgae is verified by comparing the shape of reflectance spectra within a surface bloom and its surrounding with reports of blooming events given in social and algal bloom warning platforms.

3.2 SATELLITE REFERENCE PRODUCTS

The analyses are all based on the CGLOPS 300m Lake water Level 3 products, which are the 10-day averaged products and the representing spectrum for the lake surface reflectance (in the following called *dekads*). The products provided since 2016 and which are based on OLCI 300m products are assessed. One exercise compares the 300m products with the lake water products at 100m resolution, which are based on Sentinel-2 MSI observations. The aim is to provide consistent products at both resolutions.

3.3 IN SITU REFERENCE PRODUCTS

Multiple data sources have been used for the quality assessment:

- US data bases STORET (http://www3.epa.gov/storet/)
- Water Quality Portal (WQP) (http://waterqualitydata.us/portal/)
- Swedish Environmental Monitoring Programme
- Lake Garda in situ measurements conducted by the institute for Electromagnetic Sensing of the Environment IREA at the Italian National Research Council (CNR)
- Limnades (https://www.globolakes.stir.ac.uk/limnades.html)
- GLORIA (Lehmann et al. 2023): globally representative hyperspectral in situ dataset for optical sensing of water quality
- WISPCloud (Water Insight, H2020 MONOCLE)
- So-Rad (PML, H2020 MONOCLE)

Limnades has been established within the UK NERC GloboLakes project and is held at the University of Stirling, UK. It is a centralised database of research-quality bio-optical measurements of global lakes through voluntary cooperation across the international scientific community. The full dataset was available in the GloboLakes project, from which some validation results are adopted here. Part of the Limnades dataset is also available to the CGLOPS Lake water Team to evaluate the algorithms. Limnades remains the primary repository for inherent and apparent optical property

datasets and associated water constituent measurements, and in situ water constituent measurements for satellite validation.

For the verification of the floating cyanobacteria and macroalgae flag ground-based information is used from reports and social media.

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4 RESULTS

The results are presented starting with the comparison of previous versions of the lake water products (v1.2-v1.4) to current v2.0 (chapter 4.1). Chapter 4.3 compares the 300m lake water products based on OLCI with the 100m lake water products based on Sentinel-2 MSI. This analysis is accompanied by the consistency check of the time series for several lakes for the time of the NRT service provided based on S3-OLCI data (chapter 4.44.2). In chapter 4.4, a comparison with in-situ data is performed for selected lakes.

4.1 EVOLUTION OF 300M PRODUCTS

4.1.1 Version overview

Major product versions are associated with algorithm changes which are likely to introduce discontinuities in time-series. These upgrades are typically associated with archive reprocessing. Minor product versions are introduced for the introduction of new product variables, format changes, and quality assurance improvements such as the introduction of masks or quality labels.

Since the start of the service only minor upgrades have been introduced. At present, two major product versions are planned (Table 1). The upgrades are detailed in the following sections. In summary, upgrades from version 1.3-1.4 to v2.0 introduced multiple new variables and masking criteria, and additionally change the definition of the existing Turbidity product. From v3.0.0, re-calibration of algorithms is expected following upgrades to the atmospheric correction algorithm which causes changes in the LSR variables.

Table 1: Overview of product versions for the 300m product. Polymer versions marked with * are equivalent for OLCI so produce cross compatible R_w products. Product versions in italics are proposed future releases

Product version	IdePix version	Polymer version	Other changes
1.2	6.x		
1.3	7.0.4	4.12*	OWT matching uses spectral angle not mahalanobis distance
1.4	7.0.5	4.12*	Improved ice masking in *IdePix*
2.0	7.0.5	4.14*	Floating cyanobacteria product Separate Turbidity and TSM algorithms Chl_a supplied as well as trophic state

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			Per-pixel uncertainty provided (Chl_a, TSM) Additional mask bands Polymer out-of-bounds masking applied
3.0	*9.x*	*4.16*	*Major Polymer version change* *Algorithms retuned for Polymer v4.16* *Major IdePix update (not part of this evolution)* *Reprocessing needed for consistent time series*

4.1.2 Evolution from v1.2 to v1.3 and v1.4

This section demonstrates the improvements/changes of the results of the current version in comparison with previous versions (v.1.2-1.3). Version 1.4 includes corrections for ice masking with the OLCI sensor which affected v1.3 and was addressed by applying a monthly-varying but otherwise static multi-year ice mask. Ice masking has been consistent for the MERIS sensor. In regions where ice, and particularly thin ice during freeze up and melting periods, is expected, additional filtering of the products using the n_obs_quality_risk_sum variable is advised.

The product evolution from v1.2.0 to v1.3.0 and above is demonstrated for a selection of nine lakes (i.e. Turkana, Balaton, Müritz, Apopka, Paejaenne, Vänern, Champlain and Garda), for which the values for several positions (3-5) in each lake over all processed time periods have been extracted. Figure 1 shows reflectance comparisons between processing versions for selected wavebands, for a set of nine optically and morphologically distinct lakes. Version 1.3.0 uses updated pixel identification and the latest version of the Polymer atmospheric correction processors (v4.12, details in the ATBD). Water-leaving reflectance in version 1.3.0 (vertical) is plotted against version 1.2 (horizontal). Discrepancies are most visible in the blue bands where a group of samples from primarily Lake Müritz (Germany, green crosses) and Lake Apopka (USA, violet crosses) shows higher reflectance values in v1.3.0 (Figure 1).

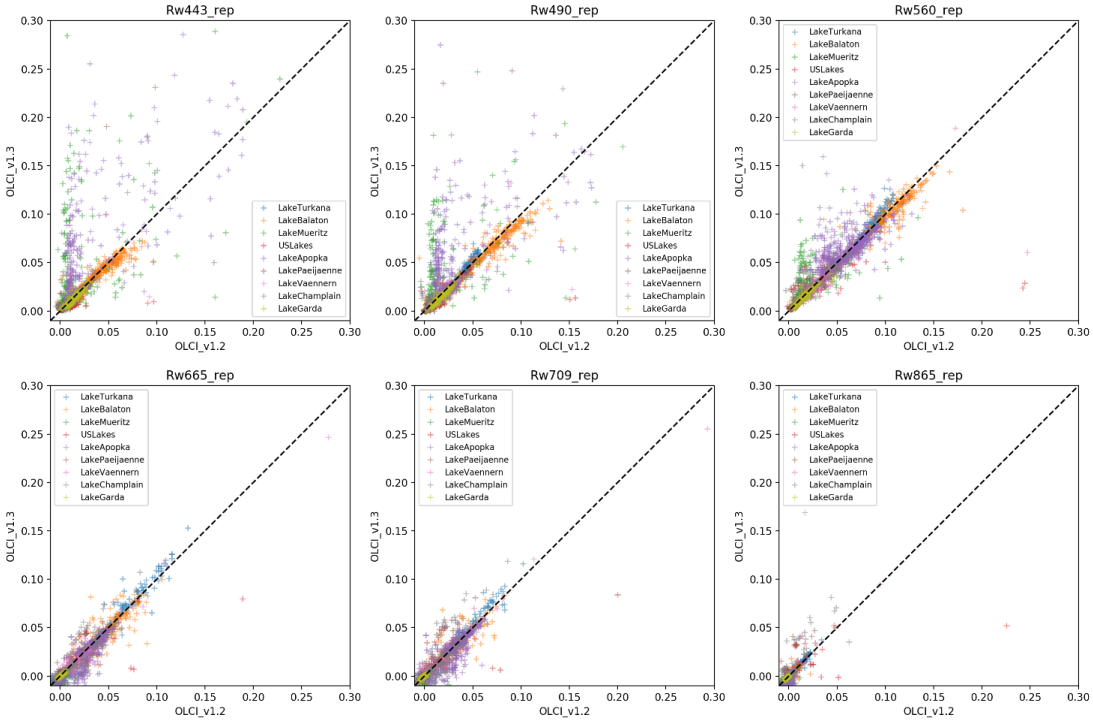


Figure 1: Water-leaving reflectance in Calimnos-OLCI v1.3.0 vs v1.2 from selected locations in nine lakes.

The difference in reflectance at 443nm, 560nm and 665nm is shown in more detail in Figure 2 for a selection of four lakes. Lake Turkana, a turbid lake shows similar results in both processing versions, while the values rise slightly for very high reflectance. Lake Vänern and Lake Garda – oligotrophic to mesotrophic lakes show more scatter between the versions, for Lake Garda especially for the blue band and for all bands in Lake Vänern. The fourth lake, Lake Apopka shows largest differences in the blue band at 443nm, with v1.3 showing higher values than v1.2 and larger scatter in green and red band than the other lakes.

Systematic differences between v1.2 and v1.3 were not observed, and user can therefore continue to use data in the series. For consistency, a v1.3 reprocessing will be completed as the operational processing chain is upgraded. Without more in situ reference data, it is not possible to say whether the departure observed in the blue reflectance bands is an improvement or not. However, it is known that uncertainty is highest in the blue band (443nm) and this may impact the ability to capture water-column properties related to phytoplankton abundance in relatively clear lakes.

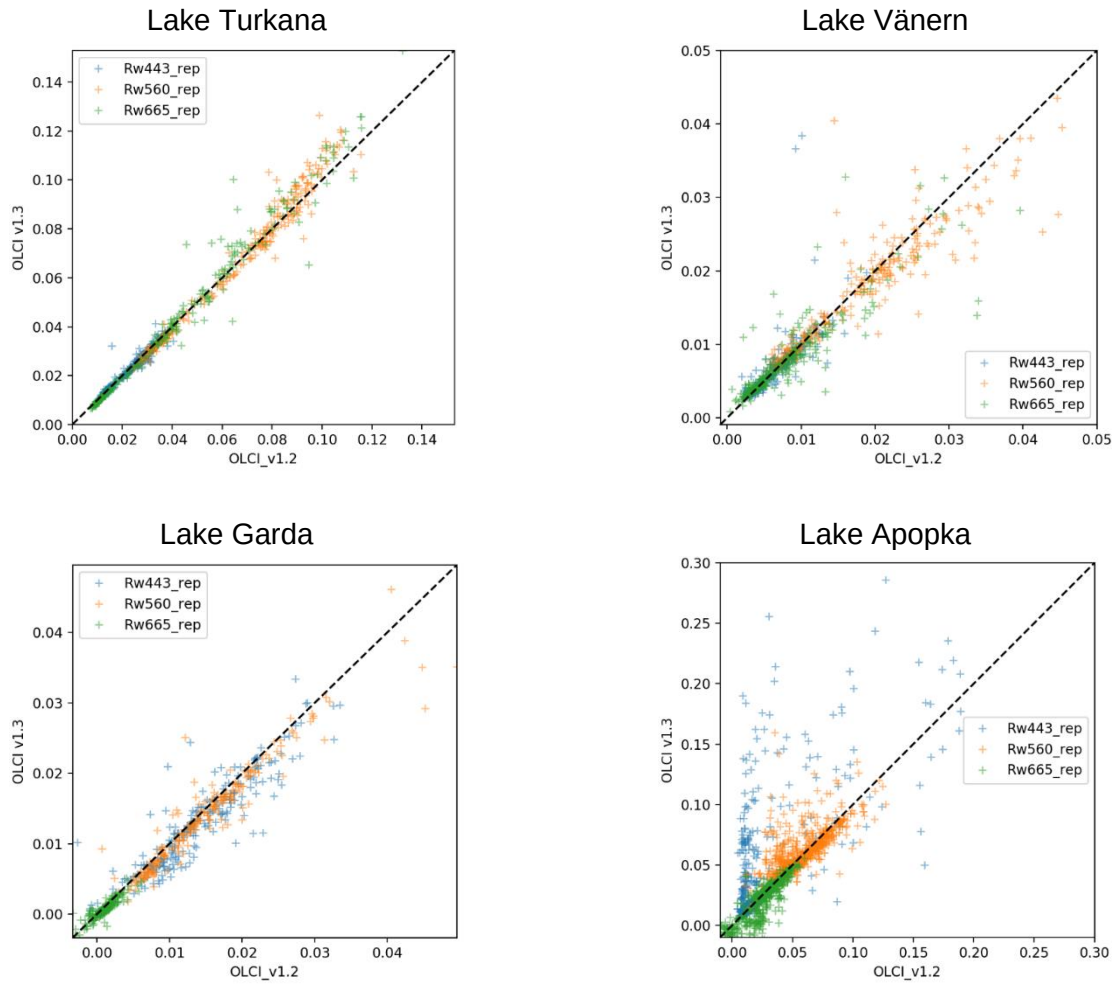


Figure 2: Comparison of water-leaving reflectance in four different lakes between v1.2 and v1.3; for 443, 560, 665nm

The comparison of product versions is further performed for time series of the different parameters: Turbidity, water-leaving-reflectance at 443nm, 560nm and 665nm and finally TSI. The TSI classes range between 0 and 100 with oligotrophic between 0 and 30, mesotrophic between 40 and 50, eutrophic 60-80, and hypereutrophic 80-100.

The first example (Figure 3) shows the results for Lake Turkana, Kenya. In this highly turbid lake, turbidity and reflectance result in an extended dynamic range in v1.3.0 and higher consistency between Reflectance and Turbidity than was observed in v1.2. TSI benefits from a large improvement in the number of valid observations, while the overall retrieved TSI range remains the same.

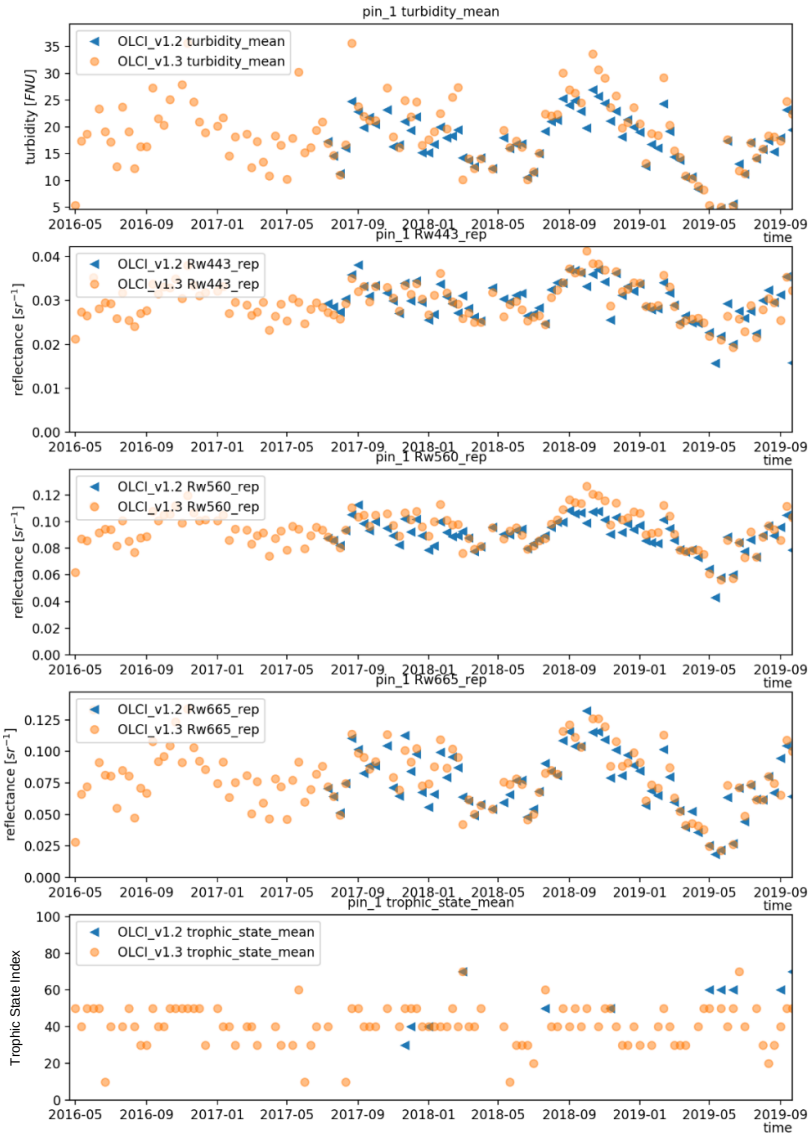


Figure 3: Time series in Lake Turkana (Kenya) showing the consistency in the northern part of the lake of parameters (turbidity, R_w443, R_w560, R_w665 and Trophic State Index) and the evolution of the processing from v1.2 to v1.3.0

The example of a clearer lake is demonstrated with Figure 4 for Lake Vänern, Sweden. Lake Vänern is mostly oligotrophic except for some mesotrophic bays and coastal areas. Outliers in the data are slightly reduced in v1.3.0 compared to v1.2. This is particularly the case for blue and green reflectance and in late 2017. Whereas such outliers in reflectance did not produce valid turbidity values in v1.2, turbidity is now provided in v1.3.0 for the same observations. TSI also shows more valid observations, whereas the overall range of TSI is reduced in v1.3.0, likely owing the extended number of observations in relatively clear waters.

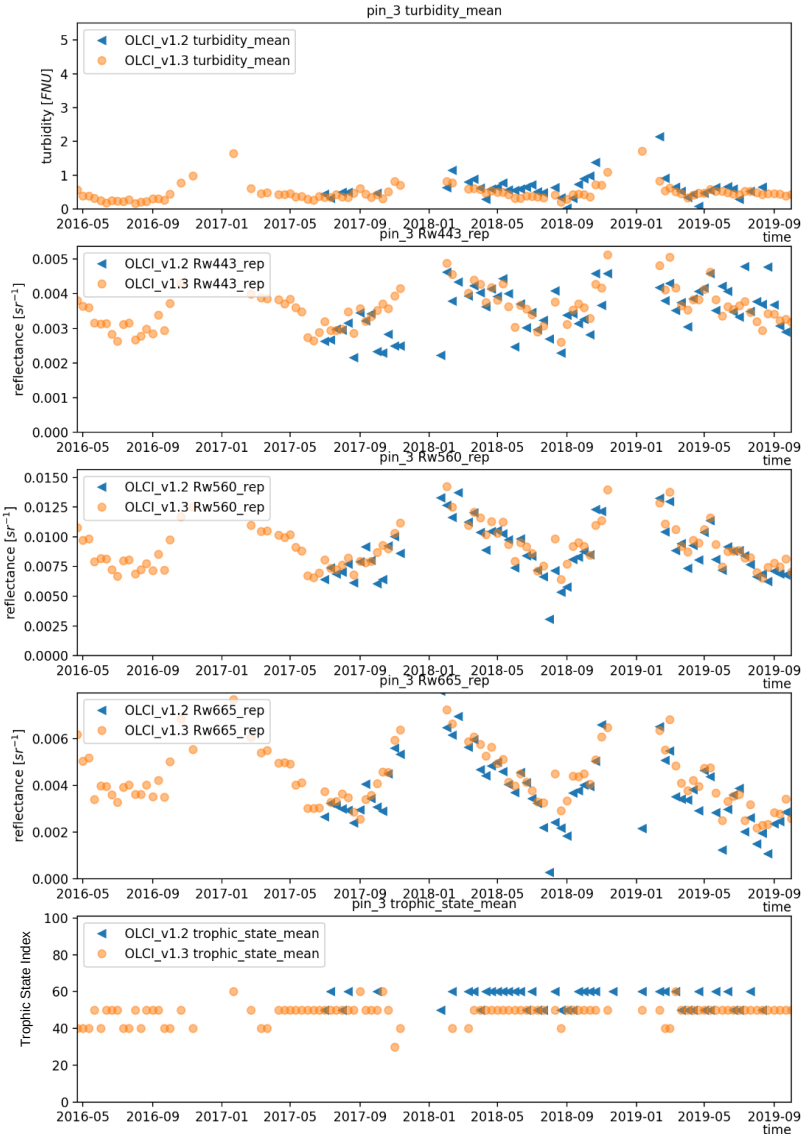


Figure 4: Time series in Lake Vänern showing the consistency of parameters (turbidity, R_w443, R_w560, R_w665 and trophic state) and the evolution of the processing from v1.2 to v1.3.0

In the third example a time series is shown for Lake Garda, a mesotrophic lake. TSI has slightly increased from product v1.2 to v1.3.0 while the turbidity stays at a similar level.

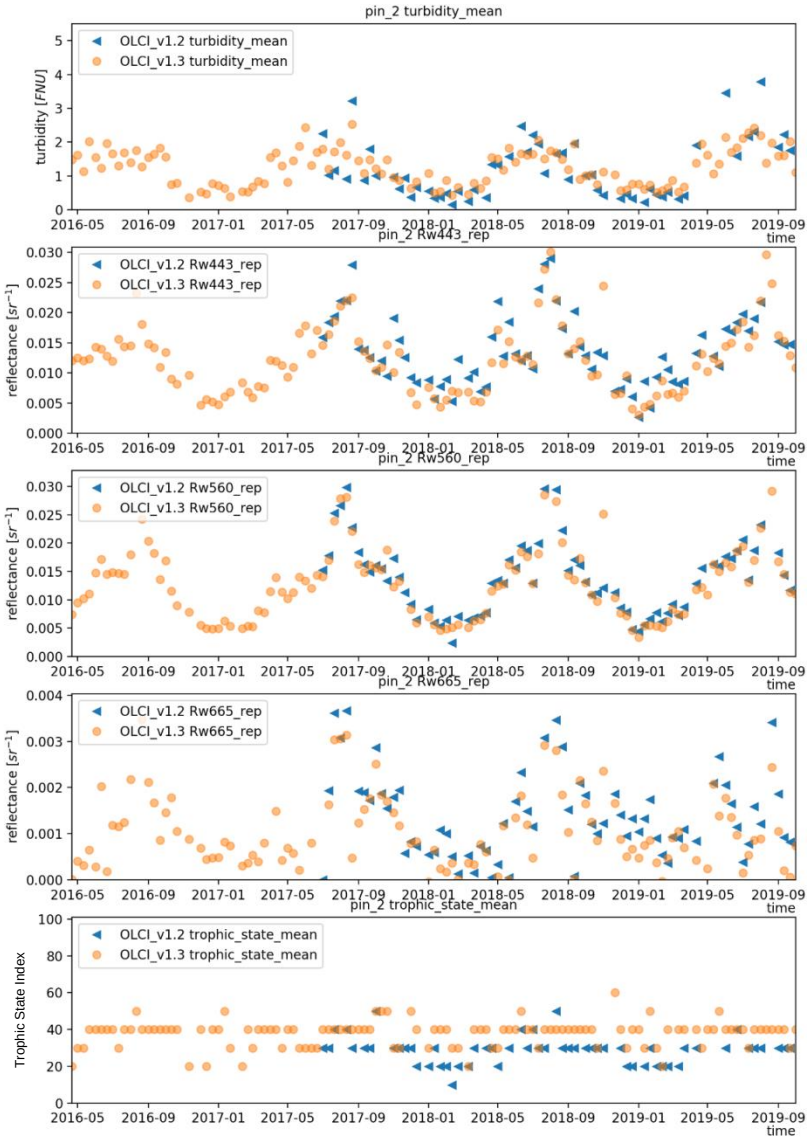


Figure 5: Time series in Lake Garda showing the consistency the Trophic State Index in comparison of processing v1.2 and v1.3.0

Compared to v1.2, the number of valid pixels increased in v1.3, particularly for the trophic state index and turbidity products. This is already demonstrated in the time series plots above and is also shown in the maps of Figure 6. Data gaps in turbidity in version 1.2 are filled in version 1.3, which is the result of adopting a new distance metric to determine the similarity of each pixel with known Optical Water Types. The previous version used Mahalonobis distance which was poor at distinguishing very clear waters. Using the spectral angle this issue is resolved.

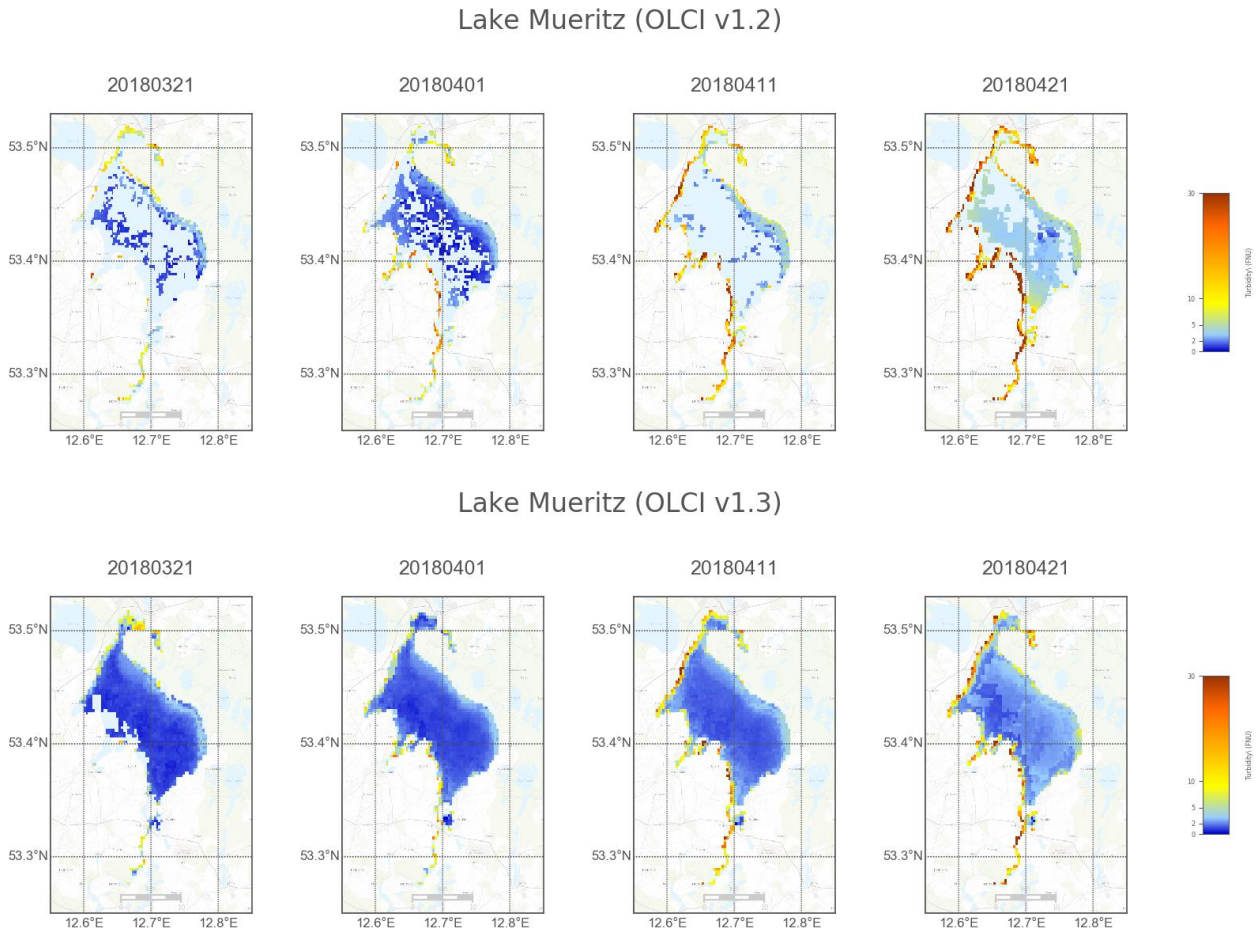


Figure 6: Turbidity during four dekads in Lake Müritz for v1.2 (upper plots) and v1.3.0 (lower plots) from S-3 OLCI

Further improvements have been achieved concerning the identification of shoreline pixels which cause wrong reflectance values. A test has been introduced reducing the number of mixed pixels between land (vegetation) and water in the products. Figure 7Figure 7: Water-leaving reflectance at 620nm in Lake Garda derived from OLCI. Left image: v.1.2 shoreline pixels appear in red, right image: v.1.3.0 spectral brightness threshold test applied to remove shoreline pixels. shows the water-leaving reflectance at 620nm for Lake Garda. While in the left image the shoreline influenced pixels are clearly visible in version 1.2 (red pixels at the western shoreline), their number is reduced by the spectral test applied in version 1.3.0 as shown in the right image. The spectral test identifies pixels where both blue and near infra-red water-leaving reflectance exceed a threshold (see ATBD for details).

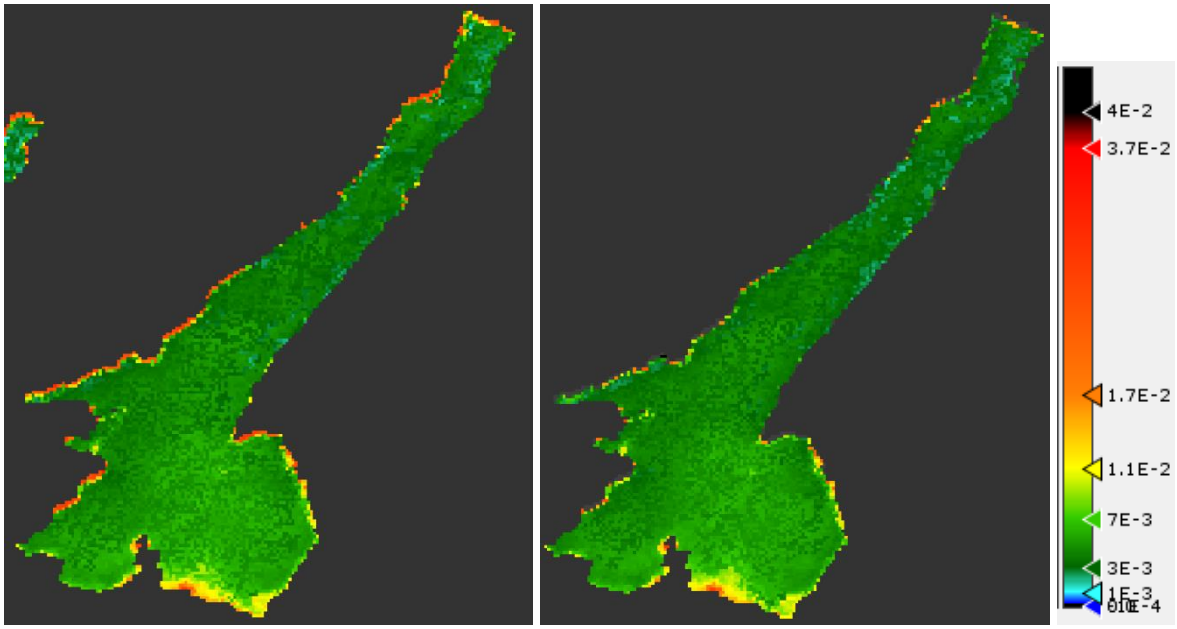


Figure 7: Water-leaving reflectance at 620nm in Lake Garda derived from OLCI. Left image: v.1.2 shoreline pixels appear in red, right image: v.1.3.0 spectral brightness threshold test applied to remove shoreline pixels.

4.1.3 Evolution to version 2.0

4.1.3.1 Polymer Out of Bounds flag

From version 2.0 the Polymer out-of-bounds flag is applied.

In our recent investigation, we have discovered the benefits of the incorporation of an additional mask known as the 'out-of-bounds' (OOB) from Polymer. This mask is designed to identify pixels that exhibit poor atmospheric correction results. The OOB flag is triggered when either the 'logchl' or 'bbs' outputs from the Polymer 'PR05' model (which serves as the default model for the Polymer processing in CGLOPS) exceed the predefined valid range. In the upcoming 2.0 version, the OOB flag will be implemented to eliminate pixels affected by imperfect atmospheric correction. Consequently, in this section, we assess the impact of this flag on the final outcomes in specific lakes of different types, as well as on a global scale.

4.1.3.1.1 Individual lake evaluations

In order to demonstrate how the removal of OOB pixels will influence the lake water products in v2.0 in various types of lakes, four lakes are included in this evaluation, including two extremely turbid lakes (Lake Turkana and Lake Tana) and two clear lakes (Lake Garda and Lake Titicaca).

Turbid lakes: Lakes Turkana and Tana

Figure 8 and Figure 9 display the maps indicating the presence of OOB pixels from Polymer in lakes Turkana and Tana, alongside the corresponding R_w665 lake water products from versions 1.4 and 2.0. In the case of lake Turkana, a total of 249 OOB pixels were identified primarily in the northern part of the lake (Figure 8a). With the implementation of the OOB flag, certain pixels exhibiting extreme values along the lakeshore of the northern region can be effectively eliminated (Figure 8b and c).

In lake Tana, OOB pixels were detected along the lakeshore, specifically in the eastern and southwest areas of the lake (Figure 9a). During the evaluation of the satellite scene in lake Tana, a total of 467 OOB pixels were identified. Notably, a noticeable region displaying visible "artificial boundaries" is observed in the southern eastern section of lake Tana. This occurrence is likely a result of reusing values from adjacent pixels as initial estimates by default in Polymer. Although the application of the OOB mask aids in the removal of certain pixels with disconnected distributions (Figure 9b and c), the region exhibiting visible "artificial boundaries" remains. It is important to mention that an updated version of Polymer is currently being developed to address the issue of "artificial boundaries." This matter is further examined in section 4.1.4 and will be fully resolved in the upcoming v3.0 product through comprehensive reprocessing.

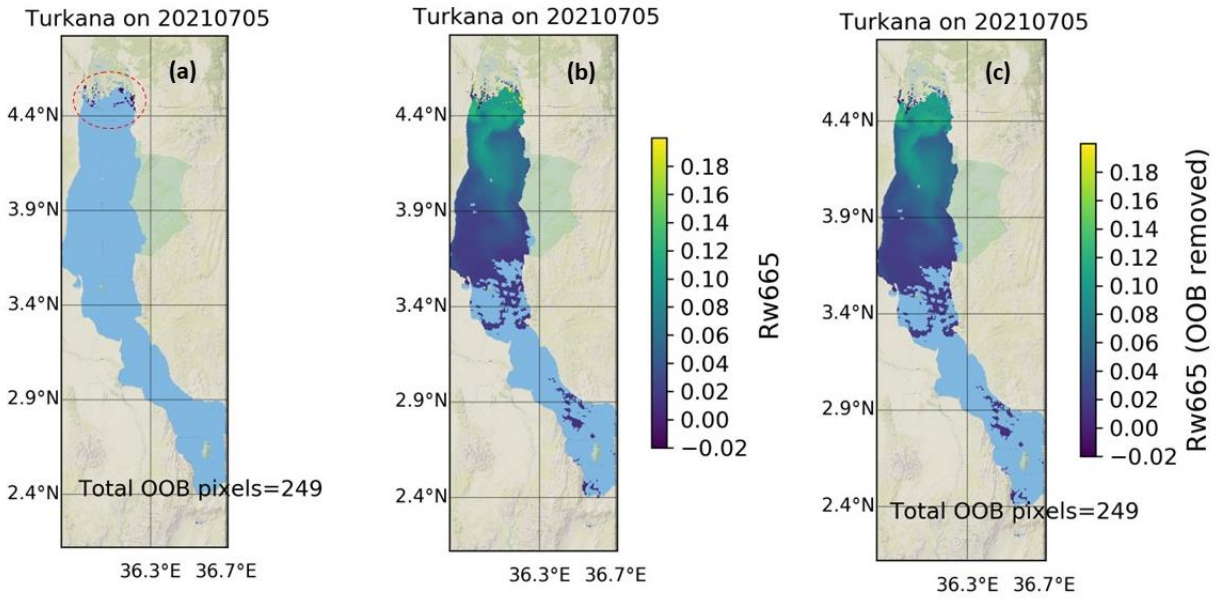


Figure 8 Maps of (a) out-of-bounds from Polymer, (b) R_w665 from lake water products v1.4, and (c) R_w665 v2.0 in Lake Turkana on 5th July 2021

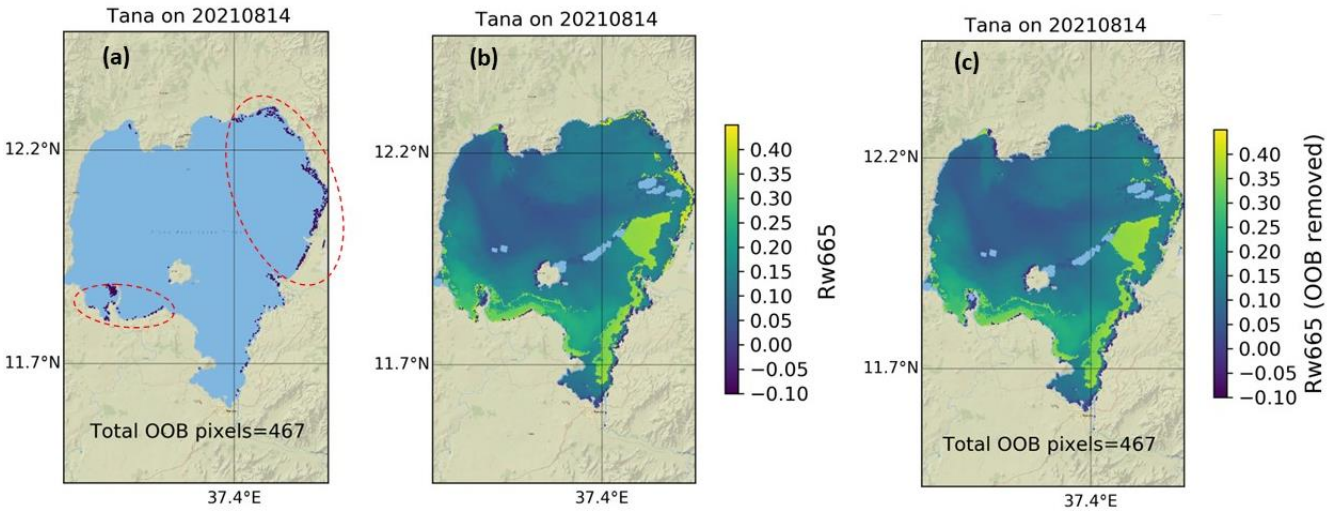


Figure 9 Maps of (a) out-of-bounds from Polymer, (b) R_w665 from lake water products v1.4, and (c) R_w665 v2.0 in Lake Tana on 14th August 2021

Clear lakes: Lakes Garda and Titicaca

During the evaluation of the satellite scene in lake Garda, no OOB pixels were detected (Figure 10). However, in the case of Lake Titicaca, a total of 37 OOB pixels were identified on November 9th, 2018 (Figure 11). The distribution of these identified OOB pixels in Lake Titicaca is primarily concentrated along the lakeshores of the northern and southeastern parts of the lake, although they may not be readily visible on the OOB map (Figure 11a).

In comparison to the findings in turbid lakes (Figure 8 and Figure 9), it is observed that clear lakes exhibit a lower number of identified OOB pixels. Furthermore, it is worth noting that all the OOB pixels identified in the evaluated lakes are characterised by extreme values and are predominantly distributed along the lake shores, which are subject to extreme events or adjacency effects.

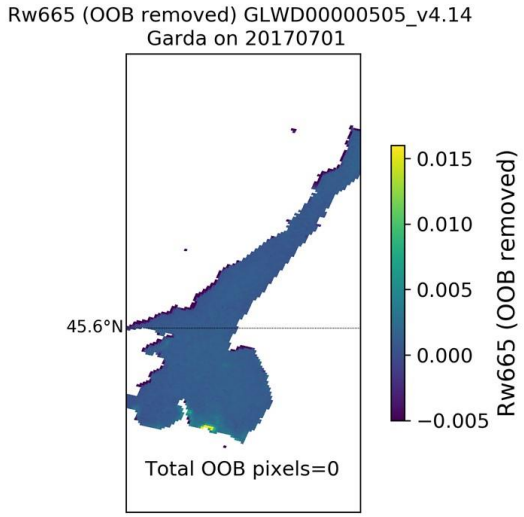


Figure 10: Map of R_w665 v2.0 in Lake Garda on 1st July 2017

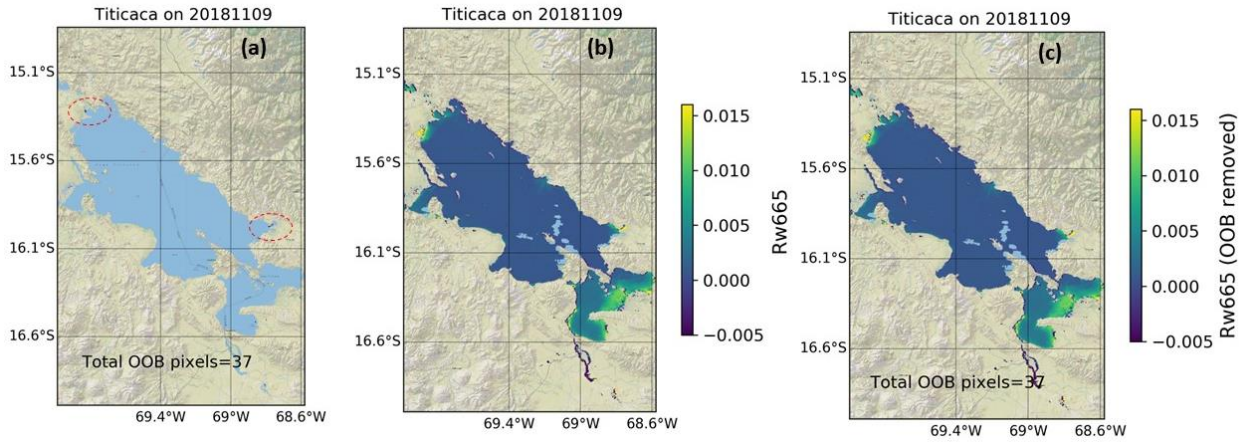


Figure 11: Maps of (a) out-of-bounds from Polymer, (b) R_w665 from lake water products v1.4, and (c) R_w665 v2.0 in Lake Titicaca on 9th November 2018

4.1.3.1.2 Global lake evaluations

To assess the impact of applying the OOB flag on OLCI (300 m resolution) in lake water product v2.0 on a global scale, a comprehensive analysis was conducted, incorporating a total of 4268 lakes from CLMS. The analysis focused on six selected dates in 2021: 15th of February, April, June, August, October, and December, respectively (hereafter referred to by their respective months for brevity).

Figure 12a illustrates the total number of identified OOB pixels across the 4268 lakes. The results indicate that the highest number of OOB pixels was observed in August, followed by June, October, April, February, and December for the six evaluated dates. This pattern can be explained by the relatively lower cloud coverage during summer months, resulting in a higher number of total valid pixels in lakes. In terms of percentage evaluation, the OOB percentages ranged from 0.16% (June and April) to 0.39% (February). This means that, on average, one to four pixels out of every 1000 valid pixels will be removed by the application of the OOB flag in v2.0 when compared to v1.4 products. As for each lake, it is found that half of the evaluated lakes (2175) have no OOB pixels identified in this global evaluation (Figure 13).

Interestingly, the monthly distribution of OOB percentages contrasts with that of pixel numbers (Figure 12). The lowest OOB percentages were found in June and April, followed by August, October, and December, while the highest OOB percentages were found in February. Although the total number of OOB pixels is relatively low during winter months, the high OOB percentages observed can be explained by the reduced number of total valid pixels due to frequent cloud/ice/snow coverage during these periods.

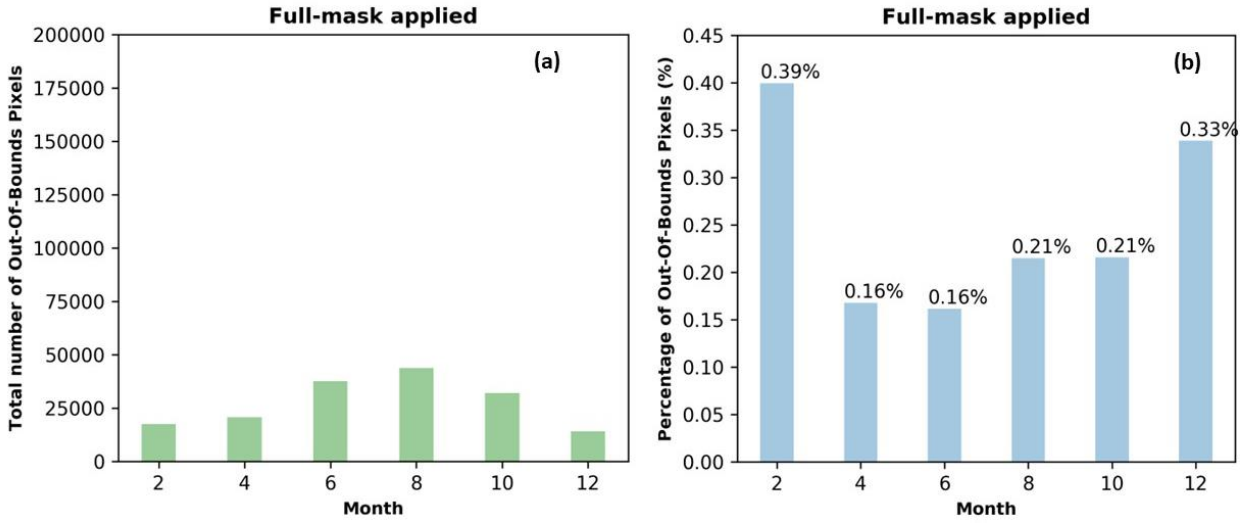


Figure 12: Histograms of (a) total number of out-of-bounds pixels, and (b) percentage of out-of-bounds pixels on each evaluated dates (15th of February, April, June, August, October, and December in 2021) from the 4,268 evaluated lakes globally

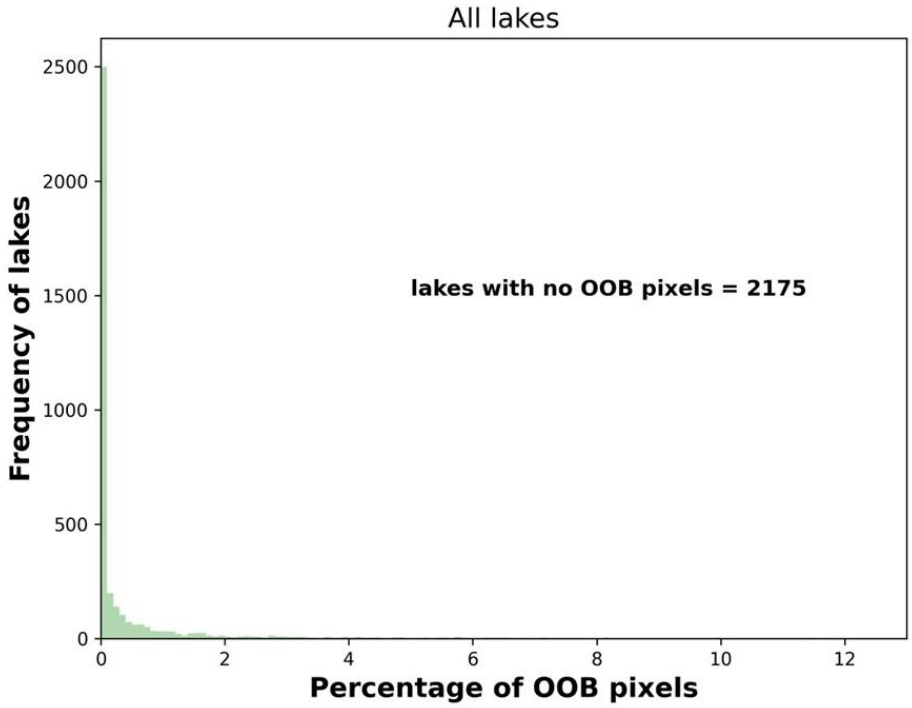


Figure 13: Frequency distribution of lakes globally for the percentage of OOB pixels

4.1.3.2 Floating Cyanobacteria and floating Macrophytes

In a first assessment, a visual inspection was performed. Anecdotal records from articles and social media posts were used to identify occurrences and to assess the ability of the satellite products to represent surface blooms. Visual validation particularly helps to improve the quality of the methodology in terms of omission/commission errors against visual interpretation, to ensure that it is fit for use. In addition, the social media channel CyanoTracker (https://twitter.com/CyanoTracker) was used to identify a number of reported cases of (surface) blooms.

Several CyanoTracker posts included blooms that were visually identified from satellite images across various lakes. These posts were used to locate bloom events and trial the methodology described here. Similarly, a range of published media articles related to these bloom events were studied to specify the type of bloom that took place. Articles linked to in Table 2 included information about the presence of floating macrophytes. Table 2 lists the remote sensing products used to identify and validate the cyanobacteria surface bloom products.

Using observations from CyanoTracker and media articles (Table 2Table 2), we found that all events could be consistently identified. Nevertheless, some uncertainty remains regarding the accuracy of mapping the extent for each floating biomass class. To assess the sensitivity of the class assignments, in situ measurements would be needed for further validation.

Table 2: Articles containing information on the presence of floating macrophytes, used to visually validate the proposed methodology

Journal	Link	Accessed
Mercator Ocean International	https://www.mercator-ocean.eu/en/news/mercator-ocean-arcan36sargassesmte_en/	28-03-2023
The New Humanitarian	https://www.thenewhumanitarian.org/report/74270/kenya-livelihoods-hit-water-hyacinth-takes-over-lake-victoria	28-03-2023
The Scientific World Journal	https://www.researchgate.net/publication/257938835_Gichuki_et_al_Water_hyacinth_succession_in_Lake_Victoria	28-03-2023

Table 3: EO products for Sentinel-3 OLCI selected to test the surface bloom product

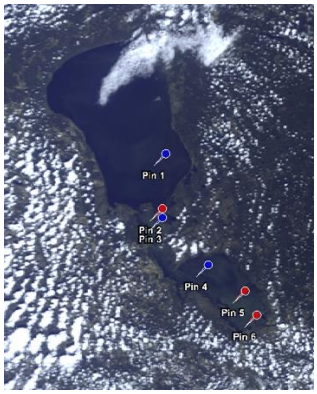
Date	Sensor	Lake
2016-09-16	Sentinel-3 OLCI	Curonian lagoon
2017-07-22	Sentinel-3 OLCI	Baltic Sea
2019-07-24	Sentinel-3 OLCI	Lake Erie
2020-07-18	Sentinel-3 OLCI	Baltic Sea
2020-08-08	Sentinel-3 OLCI	Lake Pepsi
2022-04-11	Sentinel-3 OLCI	Lake Suchitlán

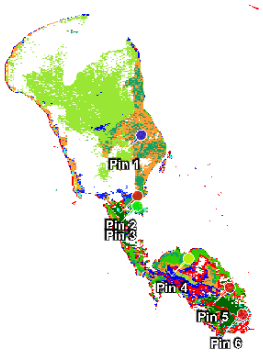
In a second step, the spectral information within the detected bloom and in the surrounding were compared and assessed. We assume that the water around surface blooms of cyanobacteria is affected by immersed cyanobacteria. Therefore, we should expect the typical light absorption features of cyanobacteria to be present in atmospherically corrected reflectance spectra. In severe bloom conditions, these features may also show clearly in TOA spectra. Examples of likely omission errors would be the detection of a macrophyte patch surrounded by water in which cyanobacteria are evident (which is possible, but less likely to occur), whereas a commission error example would be a floating cyanobacteria detection event without evidence of cyanobacteria in surrounding waters, which is very unlikely to occur.

Lake Peipsi

Lake Peipsi is known to exhibit cyanobacteria bloom due to agricultural nutrient inputs from its catchment, and internal loading. The blooms particularly manifest in the south basin and are known to form buoyant layers during calm weather. On 8th August 2020, the probability of occurrence of floating cyanobacteria is found to be high at locations where clear green features are present in the top-of-atmosphere reflectance (Figure 14). TOA spectra show elevated near infrared reflectance at

all selected locations, with pin 1 in the northern basin having the lowest reflectance signal, and the most realistic spectrum following atmospheric correction. Atmospheric correction (Polymer v4.13) has a tendency to over-correct highly turbid situations, and this effect is strongly seen in the region affected by bloom, which also shows instability in the assignment of Optical Water Types. All atmospherically corrected results for the selected pins also raise the 'inconsistency flag', which is raised, if at any band, the sum of the atmospheric path radiance and water-leaving radiance exceeds the TOA signal corrected for Rayleigh scattering, indicating that the optimised fit of these components is less likely to be physically sound. This flag was originally developed for SeaWiFS so may not be highly relevant here. From the similarity of the spectral shapes in the TOA imagery, it seems likely that some of the locations which are not flagged as floating cyanobacteria either have sub-pixel floating matter, or that cyanobacteria were aggregated near the water surface, which would logically result in poor atmospheric correction because bio-optical models for water assume a well-mixed condition. The TOA and atmospherically corrected spectra all show clear presence of absorption in the 620-nm region indicative of the phycocyanin pigment, strongly supporting that there is a strong cyanobacteria component even where it is not near or at the water surface. The floating vegetation probability was low across the lake. The floating cyanobacteria mask therefore seems likely clear of commission error.





Pin	cyano	macroph	Dominant OWT	AC quality flags
1	0.0	0.0	9	Inconsistency
2	0.95	0.0	7	Inconsistency
3	0.0	0.0	1	Inconsistency
4	0.0	0.0	9	Inconsistency
5	0.94	0.0	13	Inconsistency
6	0.82	0.0	3	Inconsistency

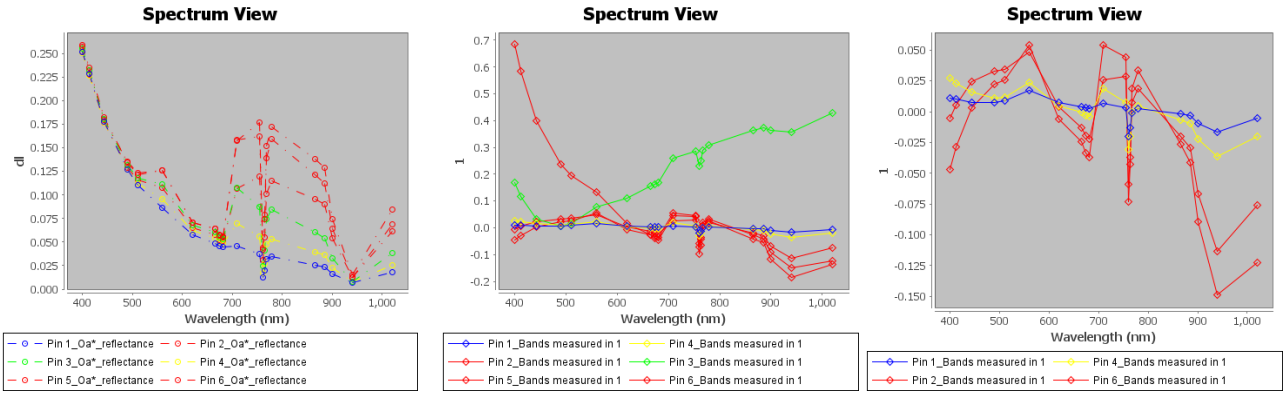


Figure 14: Detection of floating cyanobacteria in Lake Peipsi on 8 Aug 2020 (Sentinel 3A). (top-left) RGB image generated from top-of-atmosphere reflectance and (top-centre) the per-pixel dominant Optical Water Type. (top-right) Probabilities for floating cyanobacteria and macrophytes derived from the TOA reflectance, and the dominant optical water type and atmospheric correction quality flags generated during L2 processing. (bottom, left-to-right), TOA spectra, atmospherically corrected spectra, and atmospherically corrected spectra with outlier results removed

Lake Erie - OLCI

Similar to the previous case, Lake Erie is known to exhibit surface blooms of cyanobacteria in summer, particularly in western parts. Figure 15 shows the spectra along a transect from open waters to those influenced by riverine inputs. Two locations within the floating cyanobacteria mask correspond to highly elevated near infra-red TOA radiance, whereas surrounding waters are representative of mixed conditions. In the visible domain, the spectra have similar shapes including a sharp drop from 560 nm to 620nm, which is often associated with phycocyanin absorption. Elevated reflectance is visible at all locations within the bloom, which shows a clear front between pins 5 and 6. Atmospheric correction is again shown to fail in the area affected by floating cyanobacteria, and the results are also not stable, with similar TOA spectra resulting in large difference in shape and amplitude of the corrected spectra.

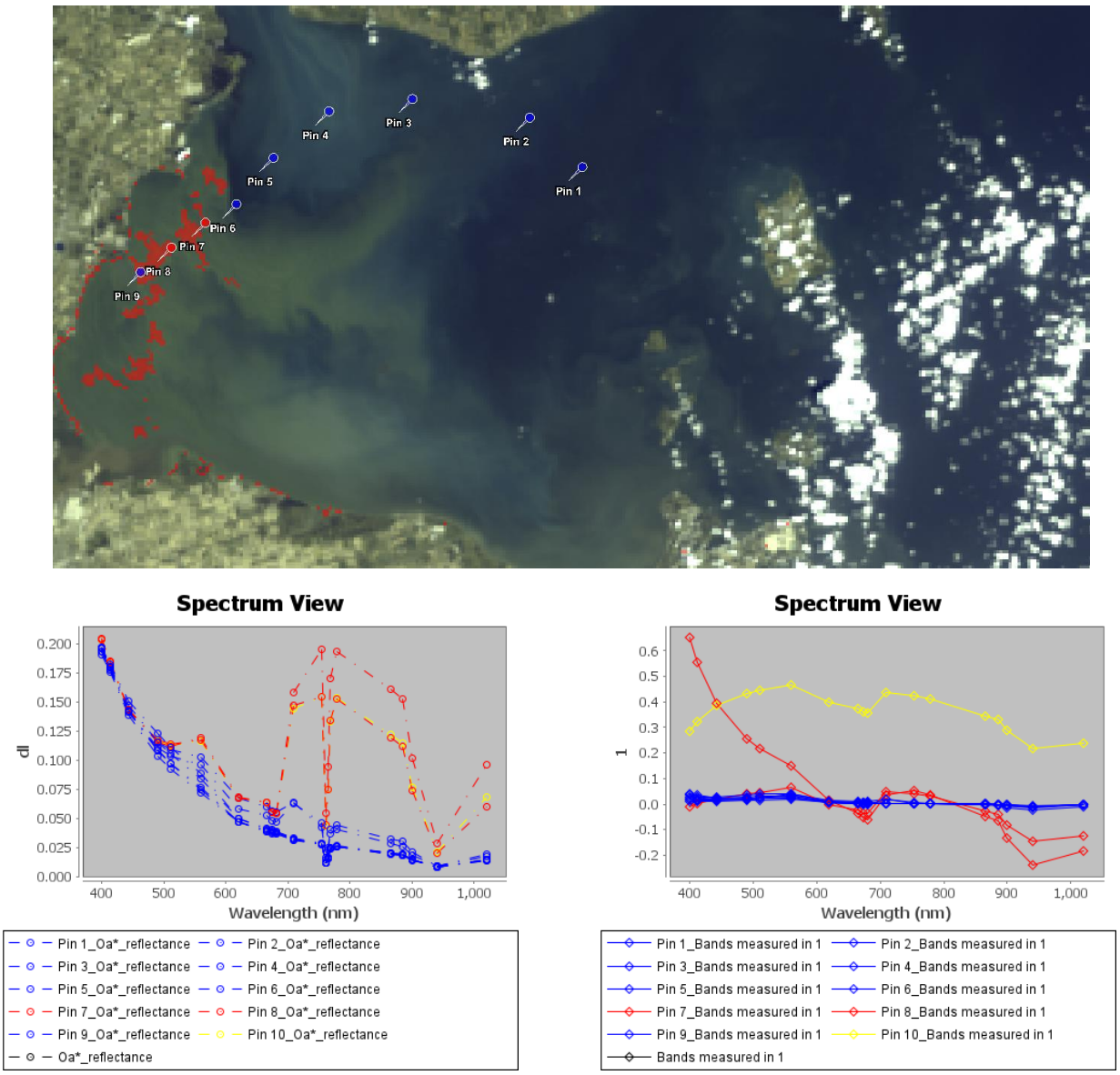


Figure 15: Floating cyanobacteria in western Lake Erie. (top) RGB image from TOA reflectance with floating cyanobacteria mask overlaid in red, and the locations for which spectra are plotted in the bottom graphs. (bottom-left) TOA reflectance spectra, (bottom-right) atmospherically corrected spectra

The differentiation between floating cyanobacteria and floating macrophytes can be demonstrated in Winam Gulf. Winam Gulf is a large bay in the north-eastern part of Lake Victoria. It is approximately 130km long and 30km wide, located in Kenya. Winam Gulf is known to have abundant floating macrophytes and cyanobacteria. Floating macrophytes provide habitat for various species of fish, insects, and other aquatic organisms. In Winam Gulf, some of the common

floating macrophytes include water hyacinth, water lettuce, and papyrus. Blooms of cyanobacteria have been observed in the Winam Gulf as well, and they can cause problems for local fishing communities who rely on the lake for their livelihoods.

Figure 16 and Figure 17 Figure 16show an example of the detection of floating material, with MSI and OLCI. Both images show similar spatial pattern for the abundance of floating macrophytes and floating cyanobacteria in the same bay. The EO images were taken three days apart and it seems that the macrophytes drifted west over that time. The cyanobacteria surface blooms seemed to be locally stable in the south-western part of the bay. The visual inspection showed that the methodology worked well, based on the spatial pattern of each type of floating blooms. The spatial pattern of floating cyanobacteria shows wind-swept filaments whereas the floating macrophytes are more strictly clustered. Several articles confirmed the presence of water hyacinth and cyanobacteria blooms in the Winam Gulf.

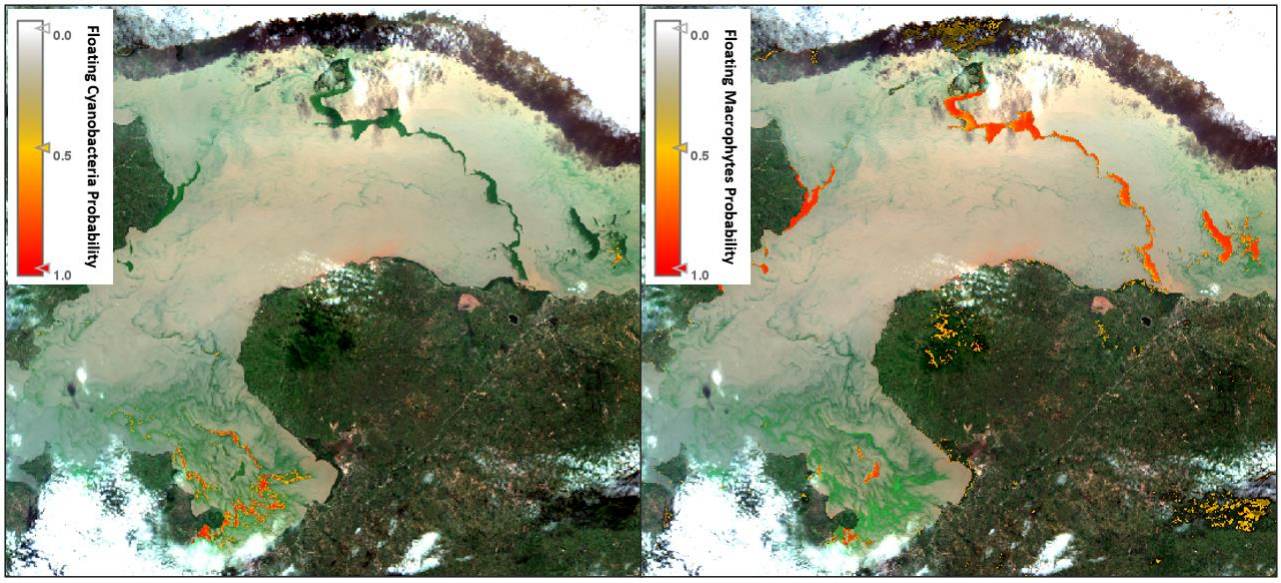


Figure 16: Detection of floating cyanobacteria (left) and floating macrophytes (right) in the Winam Gulf with Sentinel-2 MSI on 26th June 2018

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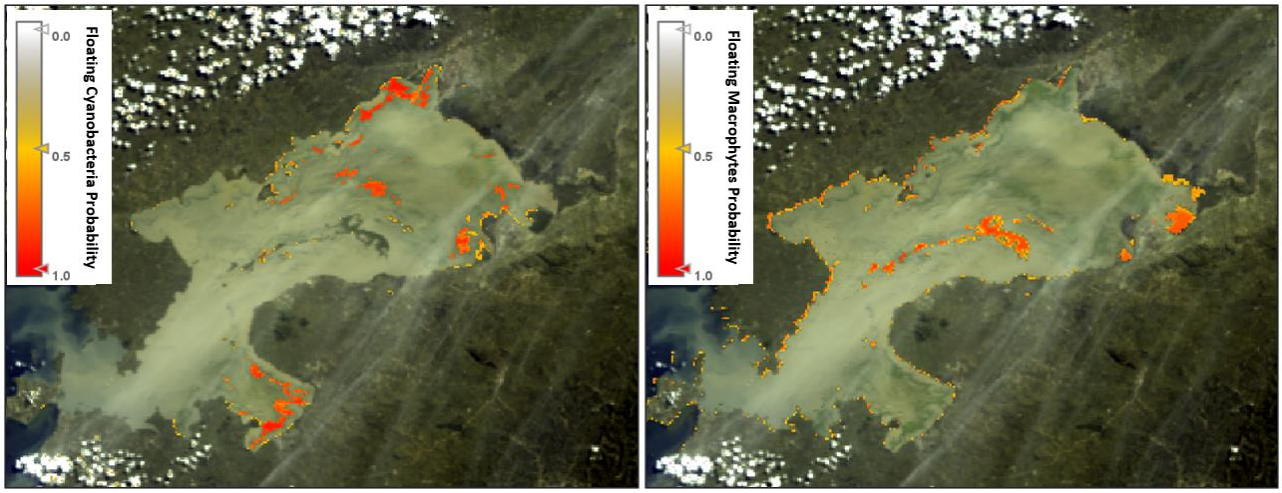


Figure 17: Detection of floating cyanobacteria (left) and floating macrophytes (right) in the Winam Gulf with Sentinel-3 OLCI on 23rd June 2018

4.1.4 Evolution to version 3.0

Version 3.0 will use a new version of Polymer, v4.15, which incorporates experimental (optional) processing modes specifically designed for OLCI in complex coastal and inland waters. The key upgraded modes include: (1) the optional parameterisation of mineral absorption switching for Chl>10 (option min_abs=-2), (2) updated first guess to test several points across the cost function and not reuse the result of the previous pixel (option initial_points), and (3) the inclusion of SWIR bands of 1020 for OLCI for bands_corr and bands_oc.

The upgraded version of Polymer v4.15 presents a potential solution to the problem of unrealistic spectral shapes with visible artificial boundaries on maps, which had been a significant issue in extremely turbid regions with previous versions. This session aims to conduct a comprehensive evaluation between Polymer versions 4.14 and 4.15, including both satellite-image-based and in situ matchup-based analysis. Following shows the evaluation results of satellite-image-based comparison, for the detailed results on in situ matchup-based analysis please refer to Section 4.4.1.

To demonstrate the performance of different Polymer versions in various types of lakes, four lakes are included in this evaluation, including two extremely turbid lakes (Lake Turkana and Lake Tana) and two clear lakes (Lake Garda and Lake Titicaca).

Turbid lakes: Lakes Turkana and Tana

Figure 18 and Figure 19 illustrate a comparison of lake water-leaving reflectance between Polymer v4.14 and v4.15 in Lake Turkana and Lake Tana, with OLCI bands ranging from 443 to 945 nm being included. The comparison includes all valid pixels on the atmospheric-corrected images. For Lake Turkana, the outputs of the two versions show good agreement for most bands, particularly for bands between 560 and 779nm, with scatters lying close to the unity line and R-values above 0.6 (Figure 18). Although rare, some disagreements are observed between the two versions, exhibiting extremely high values (>0.4), which can be potentially removed in stages later by filtering out bright pixels. On the other hand, for Lake Tana, two distinct patterns are observed between the two versions. Specifically, v4.15 increases concerning v4.14 in low-value sessions (e.g., when $R_w560<0.25$ in v4.14) for all evaluated bands, whereas in high-value sessions (of v4.14), v4.15 outputs are dramatically depressed. The unique behaviours in the two evaluated lakes could be related to the fact that the image studied in Lake Turkana is less 'bright/turbid' ($R_w<0.2$ for all bands in v4.14) than that in Lake Tana. Additionally, in Lake Tana, the extremely 'bright' pixels in v4.14 are shown to be dramatically depressed in v4.15.

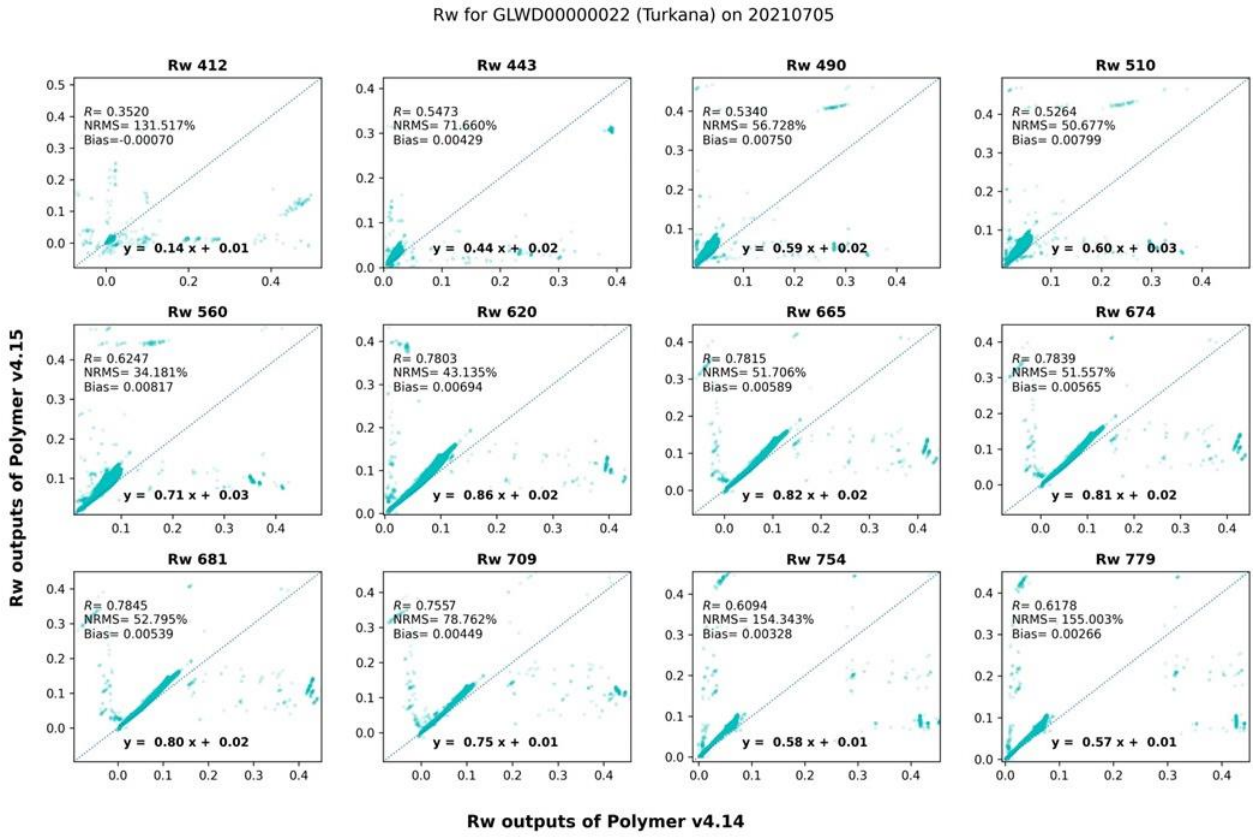


Figure 18: Comparison of lake water-leaving reflectance between v4.14 and v4.15 in Lake Turkana

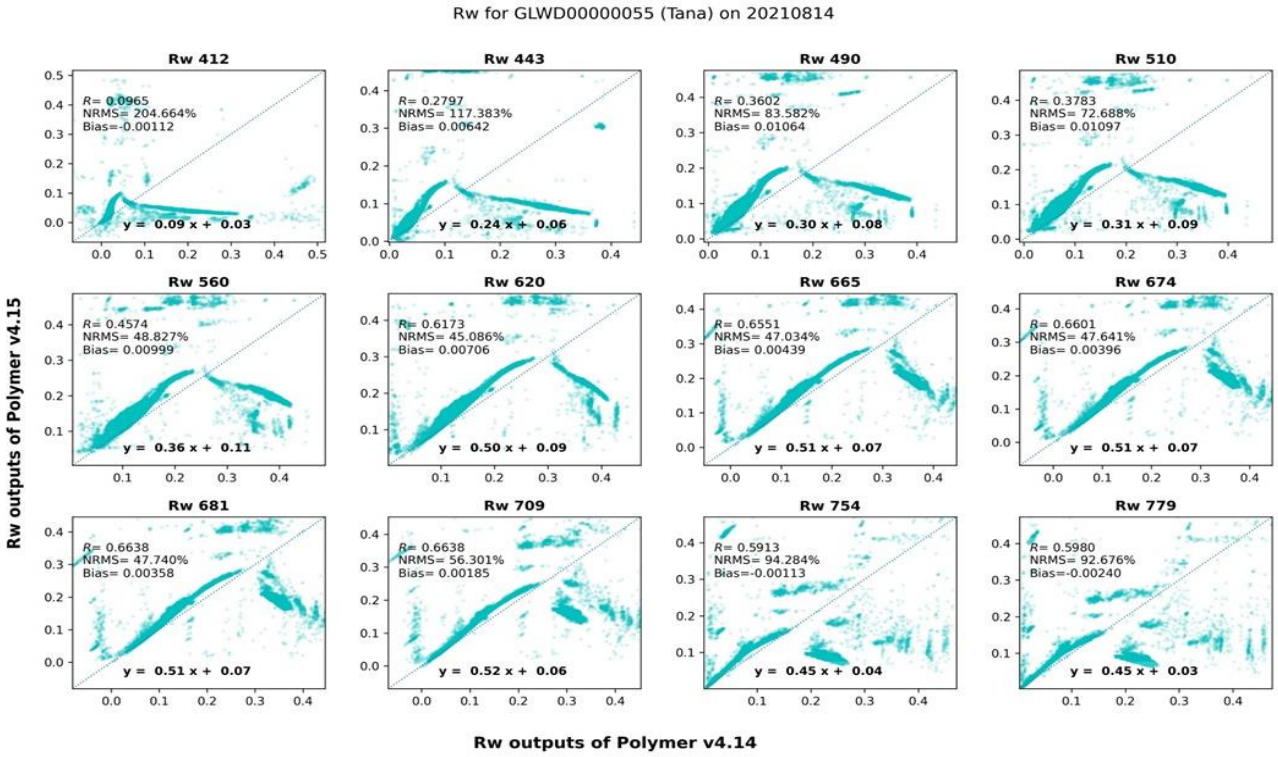


Figure 19: Comparison of lake water-leaving reflectance between v4.14 and v4.15 in Lake Tana

To provide further validation for Polymer versions v4.14 and v4.15, Figure 20 displays histograms of the distribution of R_w for the two versions in lakes Turkana (a-c) and Tana (d-f) at bands 443, 560, and 665nm. The median values of the three evaluated bands in lake Turkana are close to each other, consistent with the observations on the scatter plots (Figure 18). Moreover, the v4.15 bars are slightly shifted to larger values compared to v4.14. On the other hand, the differences between the median values of the two versions for each band are more significant in lake Tana compared to those in lake Turkana. In the low-value sections (<0.25), v4.15 is generally higher than v4.14, whereas in the high-value sections (>0.25), v4.14 shows more high-value outliers. As mentioned previously, this phenomenon is not observed in lake Turkana mainly due to the fact that the studied scene in lake Turkana is less 'bright' than that in lake Tana, in which more pixels are distributed in the 'high-value' section.

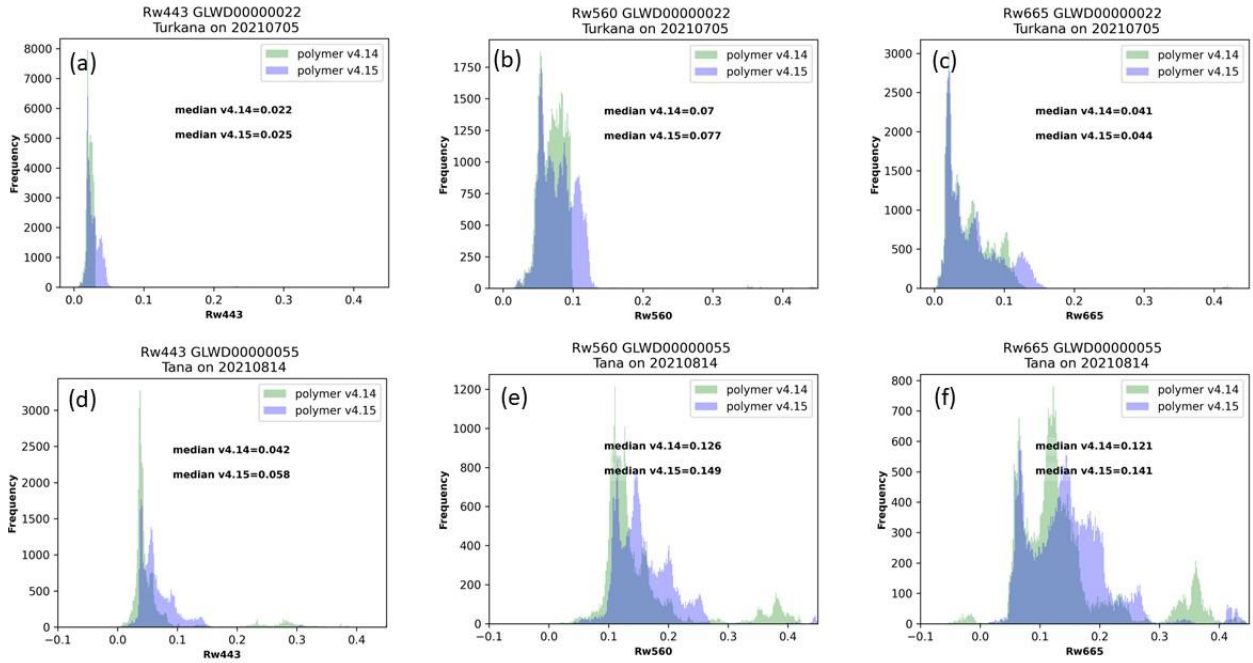


Figure 20: Histogram distributions of R_w Polymer outputs (v4.14 and v4.15) for (a-c) Lake Turkana and (d-f) Lake Tana, at bands 443, 560 and 665nm

Regarding spectral characteristics, the median spectra of both versions in both lakes exhibit similar shapes (Figure 21). However, the median spectra of v4.15 are slightly higher than those of v4.14 for most of the evaluated bands. In Lake Turkana, the two versions present almost identical median spectra and 30-70th percentiles span. Conversely, in Lake Tana, the differences between the two versions are more significant, with v4.15 showing higher median spectra and span of 30-70th percentiles than v4.14. These results are consistent with the findings in Figure 20.

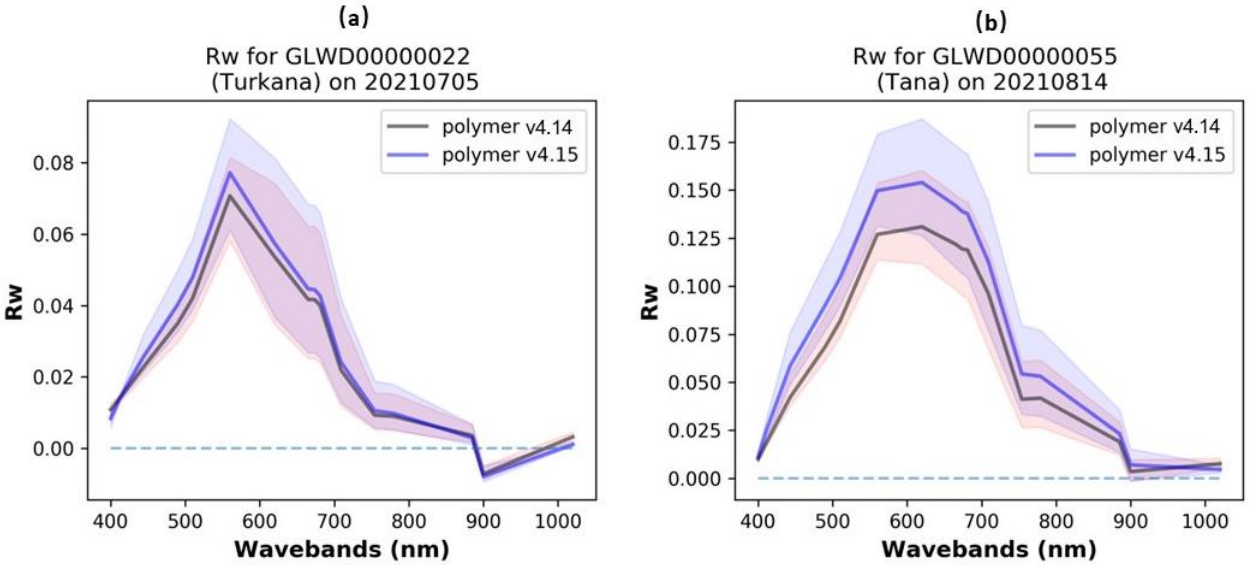


Figure 21: Spectral comparison of Polymer v4.14 and v4.15 in (a) Lake Turkana and (b) Lake Tana. Lines in the centre indicate median values, and the filled regions represent 30th to 70th percentiles

To compare the spatial differences in the outputs of Polymer v4.14 and v4.15, Figure 22 displays the maps of the two versions at R_w665 for lakes Turkana and Tana. In Lake Turkana, Polymer v4.14 and v4.15 exhibit similar distribution patterns, with v4.15 producing higher retrievals than v4.14 in the northern regions with high R_w665 values. In lake Tana, more noticeable differences between the two versions are found, particularly in the regions adjacent to the southeast and southwest lakeshores. A region with visible "artificial boundaries" is observed in the southern east of lake Tana in v4.14, likely due to re-using values from adjacent pixels as initial guesses by default in previous versions of Polymer. However, this phenomenon is not present in v4.15, which has several initial points enabled for processing. In the southwest of Lake Tana along part of the lakeshores, v4.15 generates generally higher retrievals than v4.14.

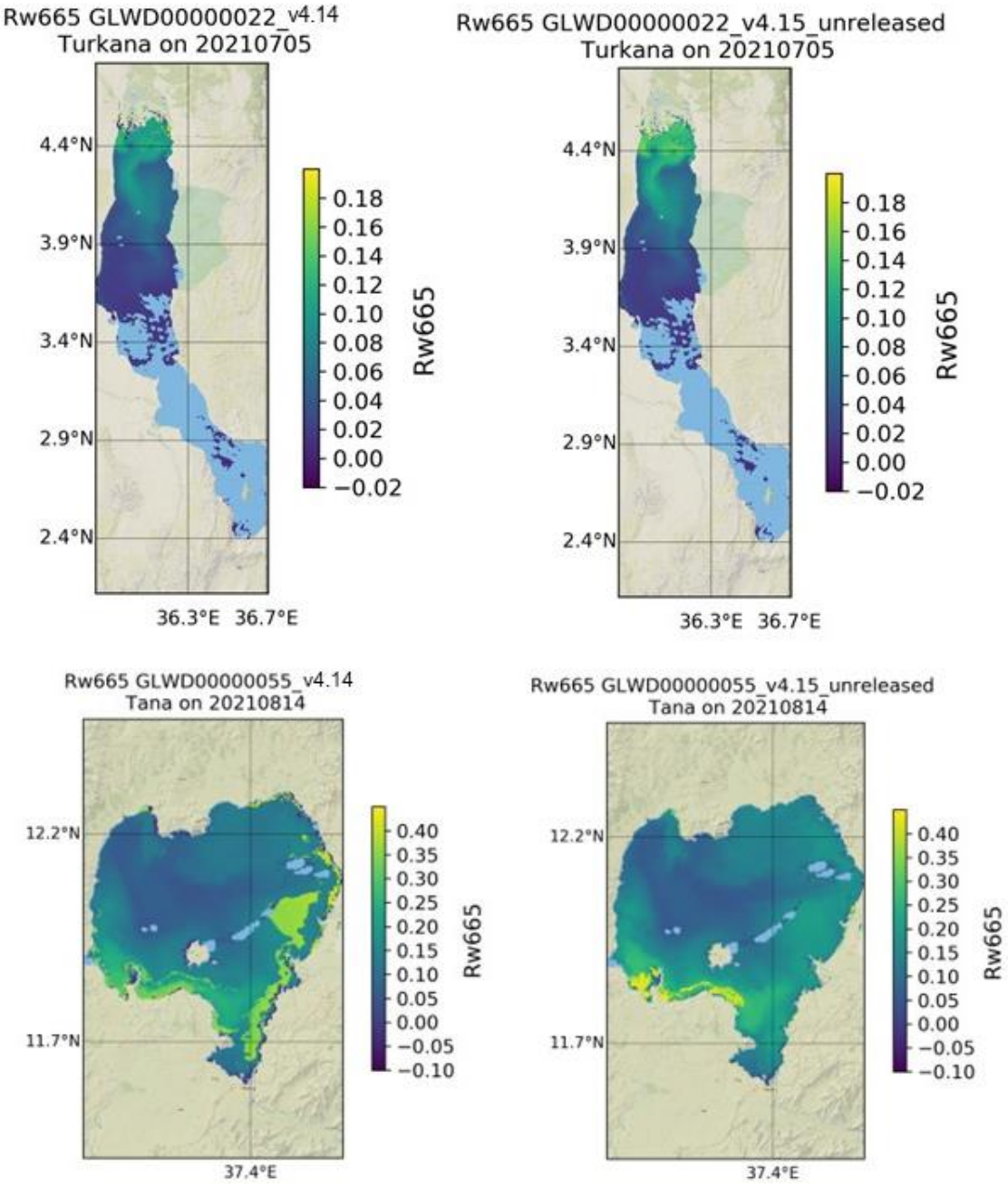


Figure 22. Maps of R_w665 from Polymer v4.14 and v4.15 in lakes Turkana and Tana

Clear lakes: Lakes Garda and Titicaca

The spatial distribution of R_w665 from the two versions of Polymer for two clear lakes, Lakes Garda and Titicaca, is illustrated in Figure 23. The comparison of the two versions shows good

consistency in the majority of the regions in both lakes. However, it is interesting to note that v4.14 and v4.15 behave differently along the lakeshores of both lakes. Specifically, v4.14 shows extremely low (or even negative) values in the lake shores, while v4.15 shows significantly elevated values for the same regions. This phenomenon is particularly noticeable in the northwest shore of Lake Garda and the southern shores of Lake Titicaca.

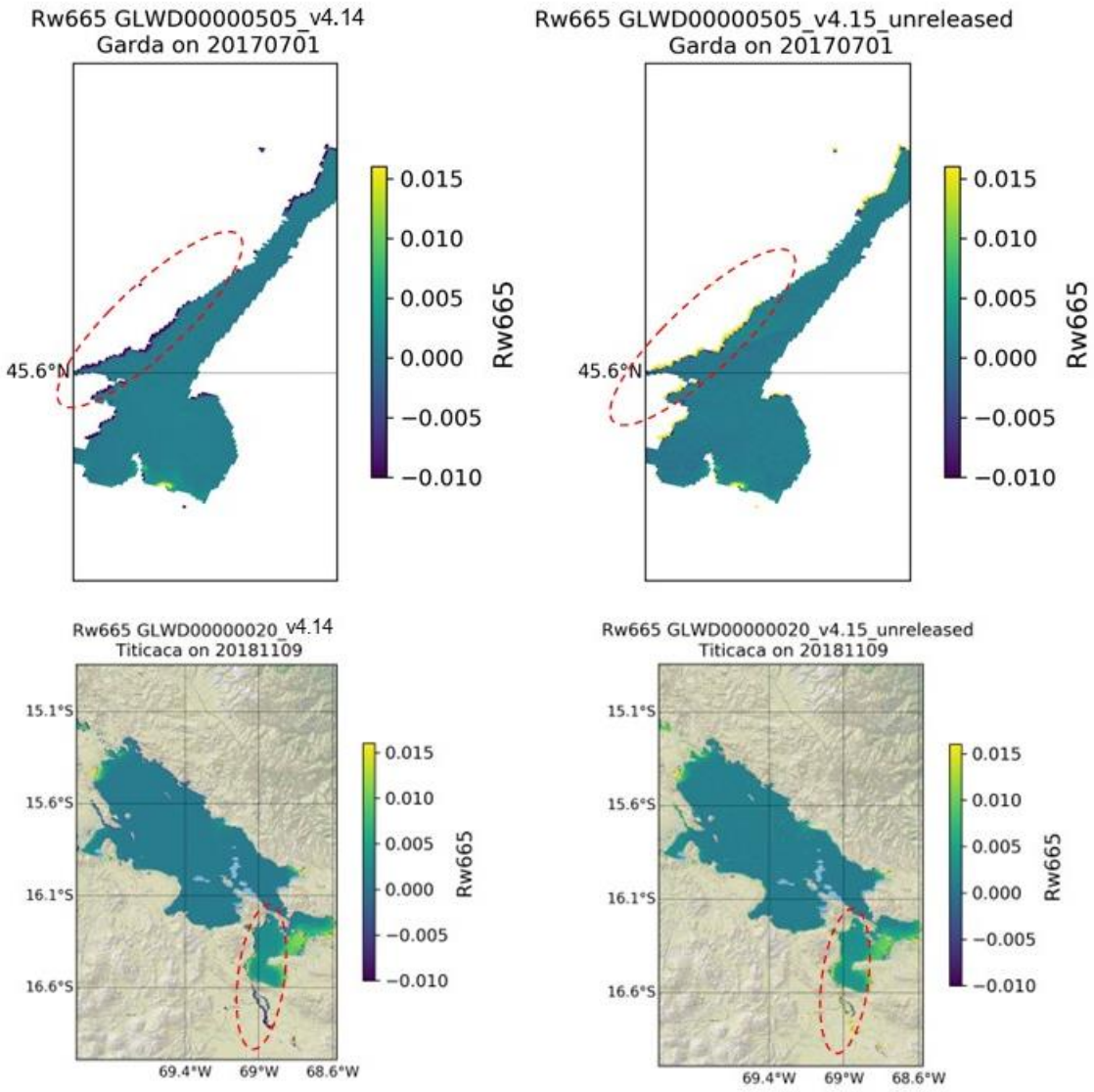


Figure 23: Maps of R_w665 from Polymer v4.14 and v4.15 in Lake Garda (upper panel) and Lake Titicaca (lower panel)

To better illustrate the performance of the two Polymer versions in clear lakes, two histogram figures are presented here: one focusing on the low R_w value range that includes the main lake bodies (Figure 24), and the other focusing on the high R_w range, which mainly includes the outliers identified along the lake shores (Figure 25). As shown in Figure 24, the two versions exhibit perfect agreement in both lakes, with similar distribution shapes and nearly identical median values for all three evaluated bands (443, 560, and 665nm). Figure 25 reveals that both Polymer versions exhibit extremely high outliers at 443nm, while for bands 560 and 665nm, only v4.15 shows high-value outliers.

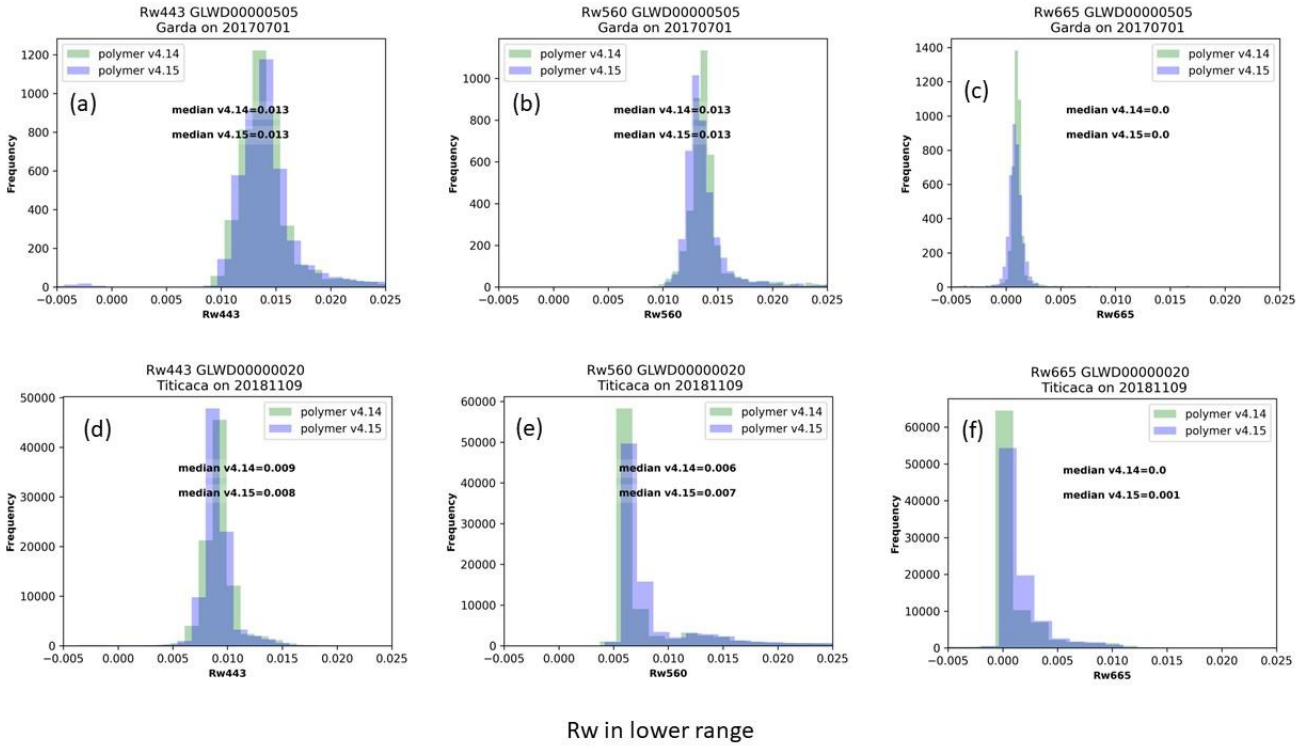


Figure 24: Histogram distributions of R_w Polymer outputs (v4.14 and v4.15) for (a-c) Lake Garda and (d-f) Lake Titicaca, at bands 443, 560 and 665nm. For better demonstration of the results in both low and high value ranges, only R_w in low range of 0-0.025 are shown

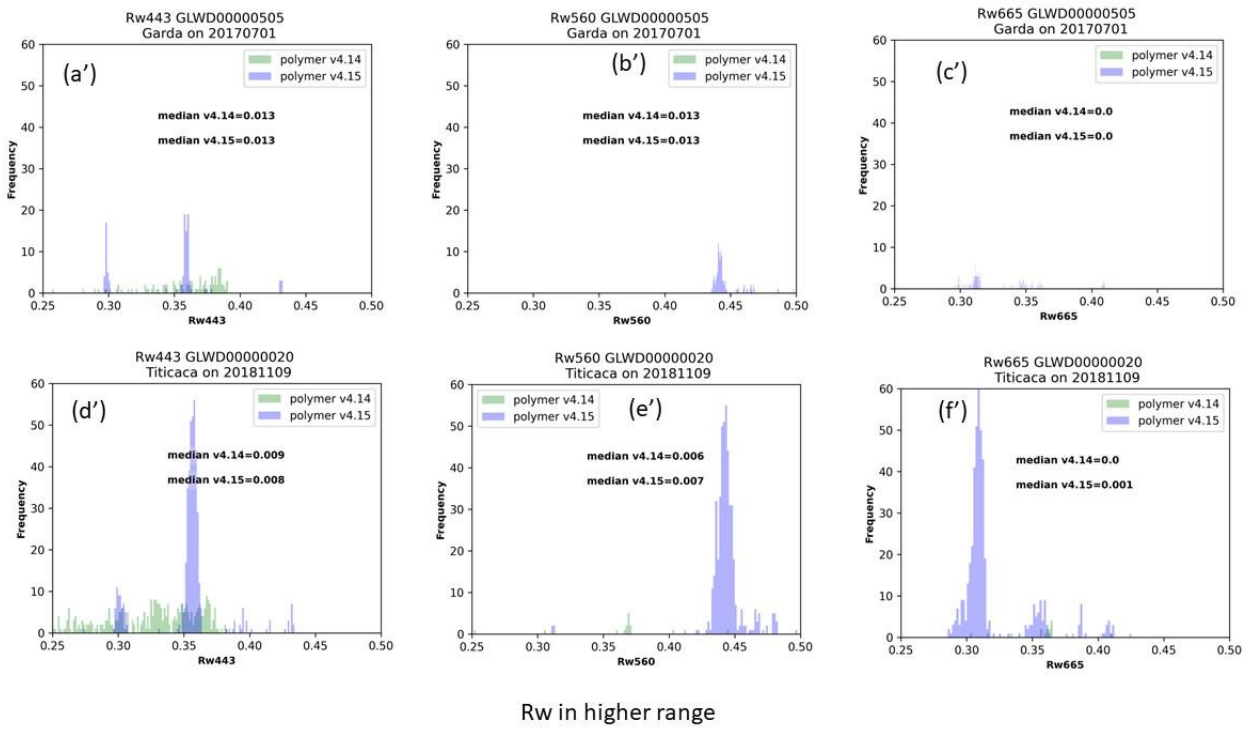


Figure 25: Histogram distributions of R_w Polymer outputs (v4.14 and v4.15) for (a'-c') Lake Garda and (d'-f') Lake Titicaca. As supplementary of Figure 7, only R_w ranging from 0.25 to 0.5 are shown in this figure

In terms of spectral characteristics, Figure 26 shows the median and 30-70th percentile spectra for both versions in clear lakes Garda and Titicaca. The median spectra show perfect agreement between the two versions in lake Garda (Figure 26a). However, it is worth noting that in this lake, both versions show negative values of R_w709, likely due to over-correction of atmospheric effects. In Lake Titicaca, the two versions generally show the same shape for bands after 510nm (Figure 26b). However, for the blue bands, the outputs of v4.15 are relatively lower than those of v4.14 in Lake Titicaca (Figure 26b).

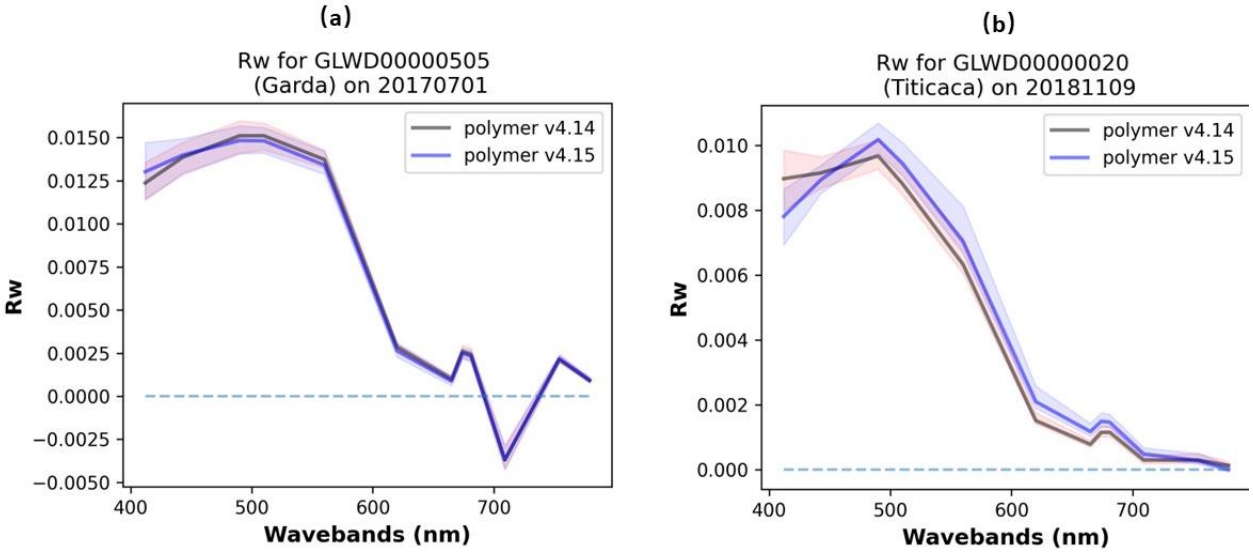


Figure 26: Spectral comparison of Polymer v4.14 and v4.15 in (a) Lake Garda and (b) Lake Titicaca. Lines in the centre indicate median values, and the filled regions represent 30th to 70th percentiles

4.2 CONSISTENCY BETWEEN S3-OLCI 300M PRODUCT V1.3.0 AND S2-MSI 100M PRODUCT V1.3.0

The following four figures show the consistency of temporal evolution between 300 m and 100 m lake water products during four years (2016/05 – running for 300 m) and one year (2019 – running for 100 m) of **available** OLCI and MSI observations, respectively. The horizontal axis represents the day of year and vertical axes give the respective product. The first six sub-plots show the reflectance for 443nm, 490nm, 560nm, 665nm, 705nm and 865nm. The last two sub-plots show turbidity and trophic state index. The plots are shown for Lake Turkana, Kenya (Figure 27), Lake Vänern, Sweden (Figure 9, Lake Balaton, Hungary (Figure 10) and Lake Garda, Italy (Figure 11).

In general, consistency between Sentinel-3 OLCI observations is high between years (2016-2019) for all shown parameters (reflectance bands, turbidity and trophic state index), whereas consistency in reflectance retrieval is relatively poor between Sentinel-2 MSI and Sentinel-3 OLCI in the blue bands. More detailed descriptions are given below the individual figures.

Note that axis scaling differs between plots of reflectance bands within and between the lakes.

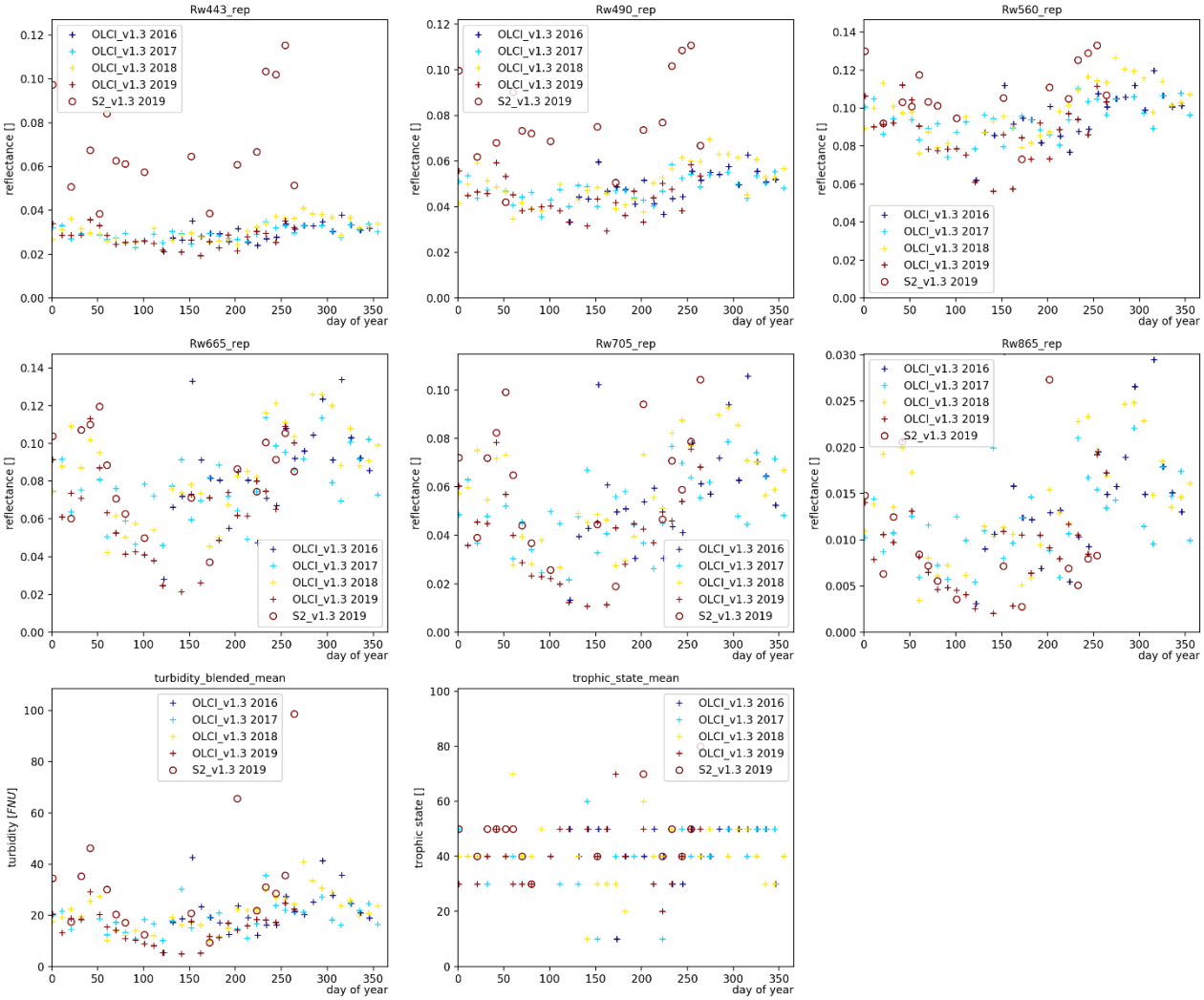


Figure 27: Lake Turkana. Yearly time series of water leaving reflectance at 443, 490, 560, 665, 705 and 865nm, Turbidity and Trophic State Index in the Northern part of the lake. Each color (blue 2016 to red 2019) represents a year. S3-OLCI products are marked by crosses, the S2-MSI products by circles

Lake Turkana as turbid lake shows high reflectance values over all bands and in turbidity (note the different scaling of the y-axis compared to other plots). A minimum around day of year DOY 150 and maxima at DOY 50 and 280 are visible in green, red and nIR bands, and in turbidity. This trend is seen for all years, while 2019 shows the lowest absolute values.

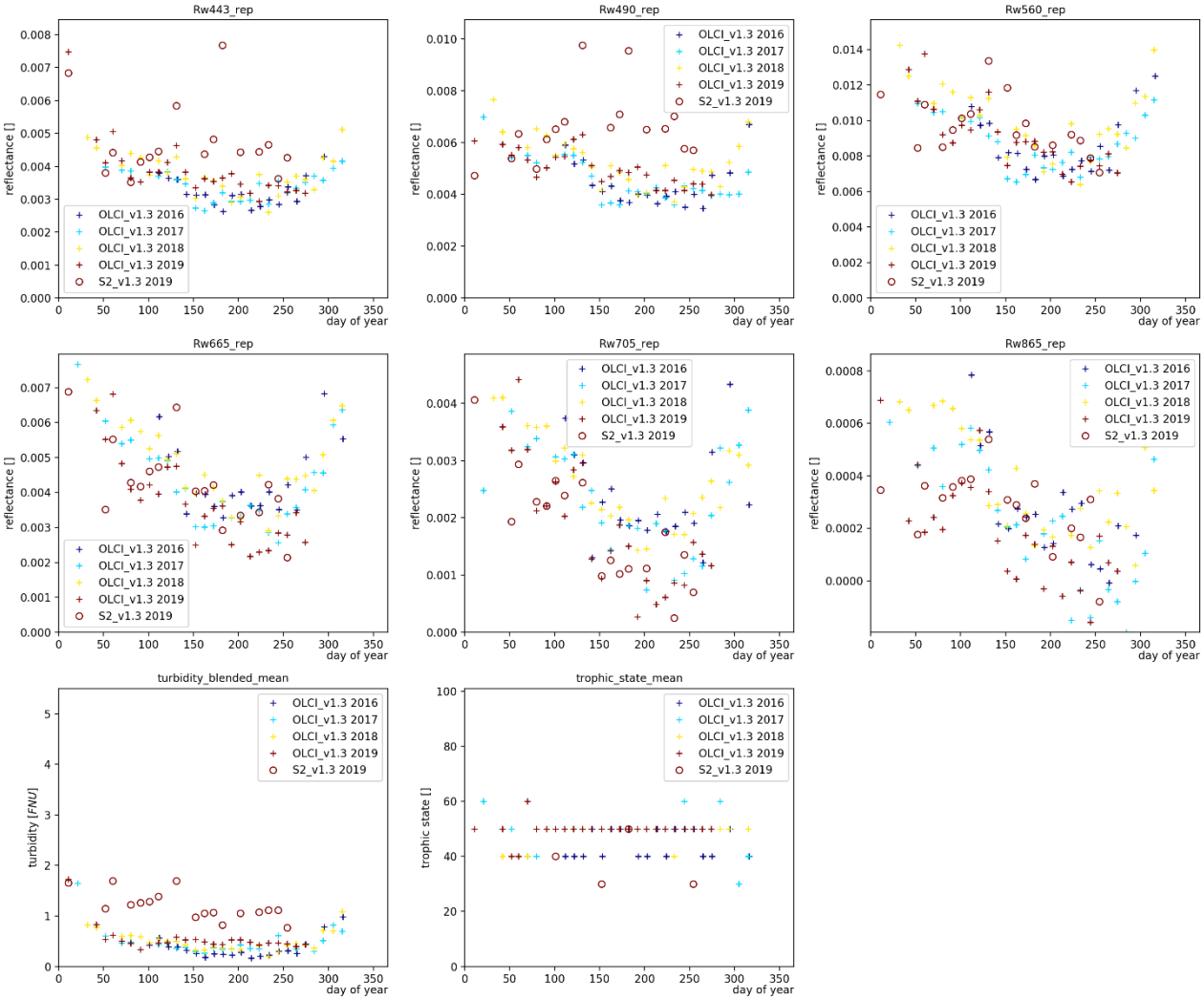


Figure 28: Lake Vänern. Yearly time series of water leaving reflectance at 443, 490, 560, 665, 705 and 865nm, turbidity and trophic state in the Eastern basin. Each colour (blue 2016 to red 2019) represents a year. S3-OLCI products are marked by crosses, the S2-MSI products by circles

Lake Vänern shows very similar behaviour in all years, with low variability over the year and across years. The minimum of reflectance and turbidity is in late summer, which is the clear water phase of the lake. High values during wintertime indicate that the flagging of ice pixels is not sufficient. It is expected that no information is available in winter months due to ice coverage. The trophic state index in 2016 is with a value of 40 lower than for the years 2018 and 2019 (TSI = 50). Both years were characterised by high temperatures and high algal blooms.

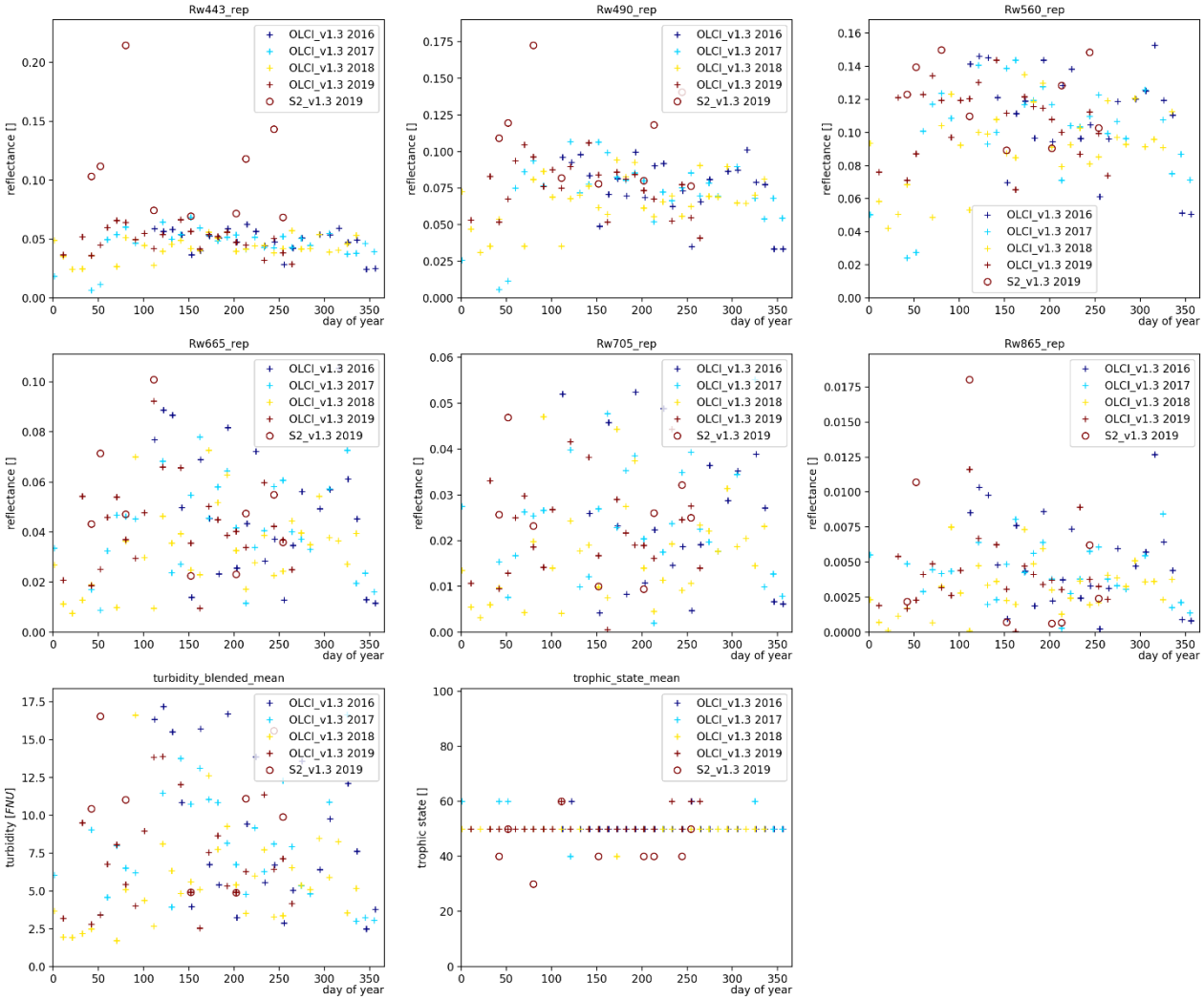


Figure 29: Lake Balaton. Yearly time series of water leaving reflectance at 443, 490, 560, 665, 705 and 865nm, turbidity and trophic state in the middle of the lake. Each colour (blue 2016 to red 2019) represents a year. S3-OLCI products are marked by crosses, the S2-MSI products by circles

Lake Balaton shows stable reflectance in the blue bands and large scatter in the green, red and nIR bands. The scatter is similar between the different years. Turbidity is variable over the whole year, with maximum towards the early summer months. TSI is stable at 50 (mesotrophic) in most of the dekads.

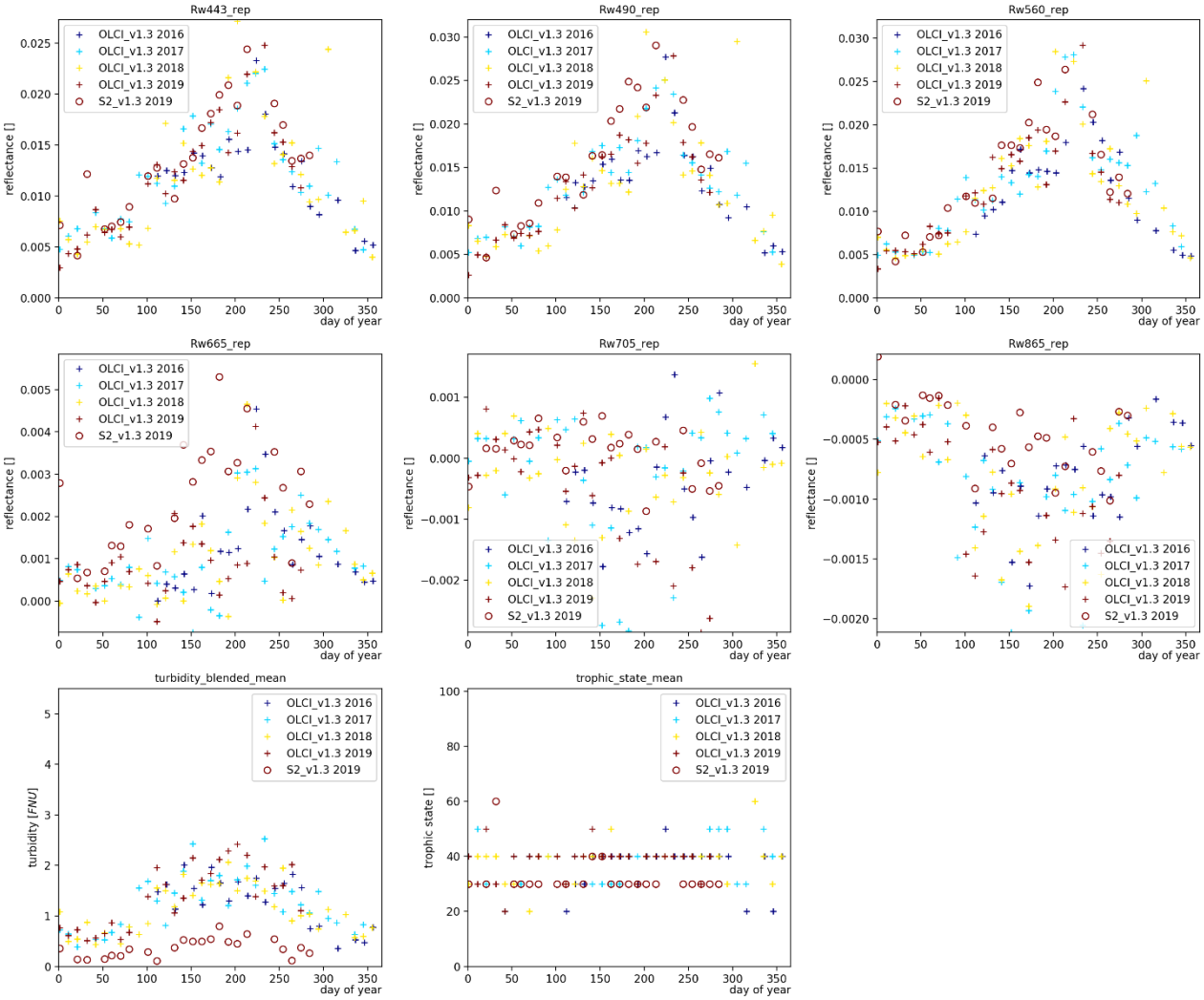


Figure 30: Lake Garda. Yearly time series of water leaving reflectance at 443, 490, 560, 665, 705 and 865nm, turbidity and trophic state in the centre of the Southern part. Each colour (blue 2016 to red 2019) represents a year. S3-OLCI products are marked by crosses, the S2-MSI products by circles

For Lake Garda, a very good agreement between years and between Sentinel-2 100m and OLCI 300m products is achieved for all bands except for 665nm. The turbidity is similar in all OLCI datasets but lower for the 100m products. A clear seasonal trend of the reflectance in the short wavelengths and in turbidity is visible. A larger scatter is shown in band 705 and 865nm, while the range for scatter is the same for all years.

The same comparison will be performed for version 3.0 when production is ready to start.

4.3 VISUAL INSPECTION - CONSISTENCY OF TIME SERIES AND MAPS

4.3.1 Consistency individual lakes

4.3.1.1 Lake Vänern



Lake Vänern is the largest lake of Sweden and the fourth largest in Europe. It is situated in a region with Archaean rocks, but the deposits in the drainage area mainly consist of moraines that are poor in nutrients. A rift zone of north-south extension divides the lake into two major basins. Numerous small islands form a considerable archipelago in relation to the open water areas. Good water exchange between the archipelago and the open water is decisive for the generally extensive spreading of pollutants (ILEC World Lake Database). Chl-a concentration in pelagic Lake Vänern varies between 0.6 and 5.9 µg L^{-1} and water quality is consequently classified as moderately nutrient rich. Total biomass nevertheless corresponds to that of oligotrophic lakes (Philipson et al. 2015, Willén et al. 2001).

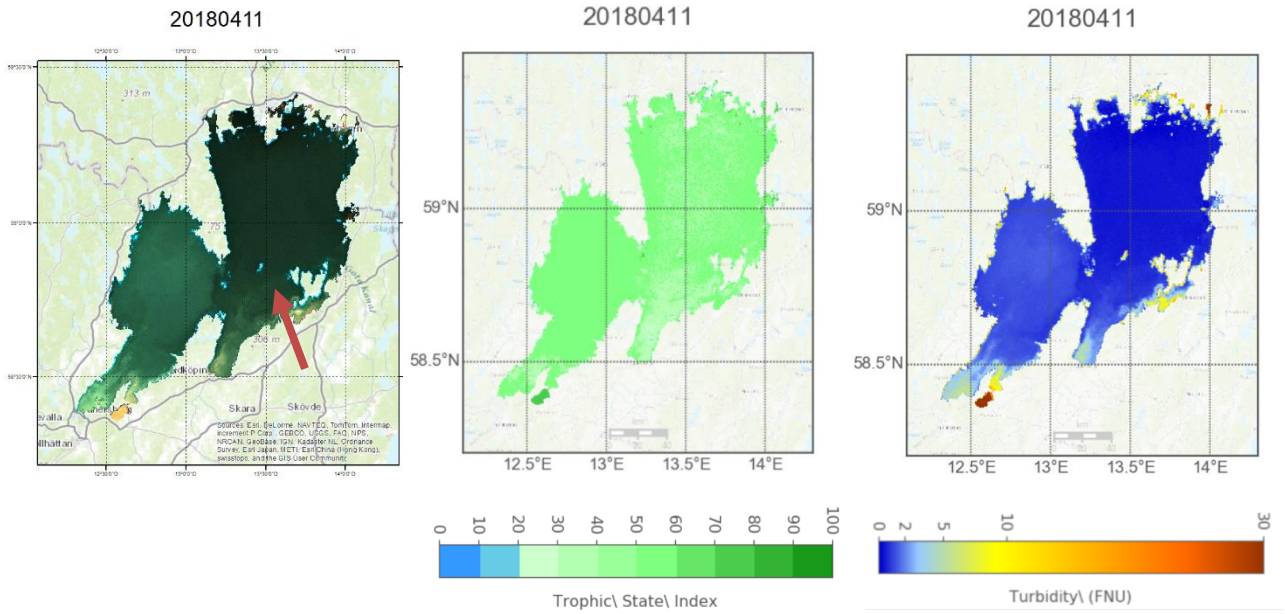


Figure 31: Lake Vänern - Left: RGB derived from water-leaving reflectance (R_w665_rep, R_w560_rep, R_w490_rep). The arrow in the left images marks the position of time series extraction. Middle: Trophic State Index. Right: Turbidity during the second dekad in April 2018 from S-3 OLCI

Figure 31 shows the spatial distribution of all products for Lake Vänern. Clear water covers most of the lake while higher water constituent concentrations occur in bights in the southern, eastern and northern parts of the lake. Figure 32 provides the dekad turbidity maps over a full year (2018). The winter dekads show missing values due to ice cover and probably also due to low light levels. In early spring artefacts are visible in the maps, likely caused by mixed water and ice during the melting period.

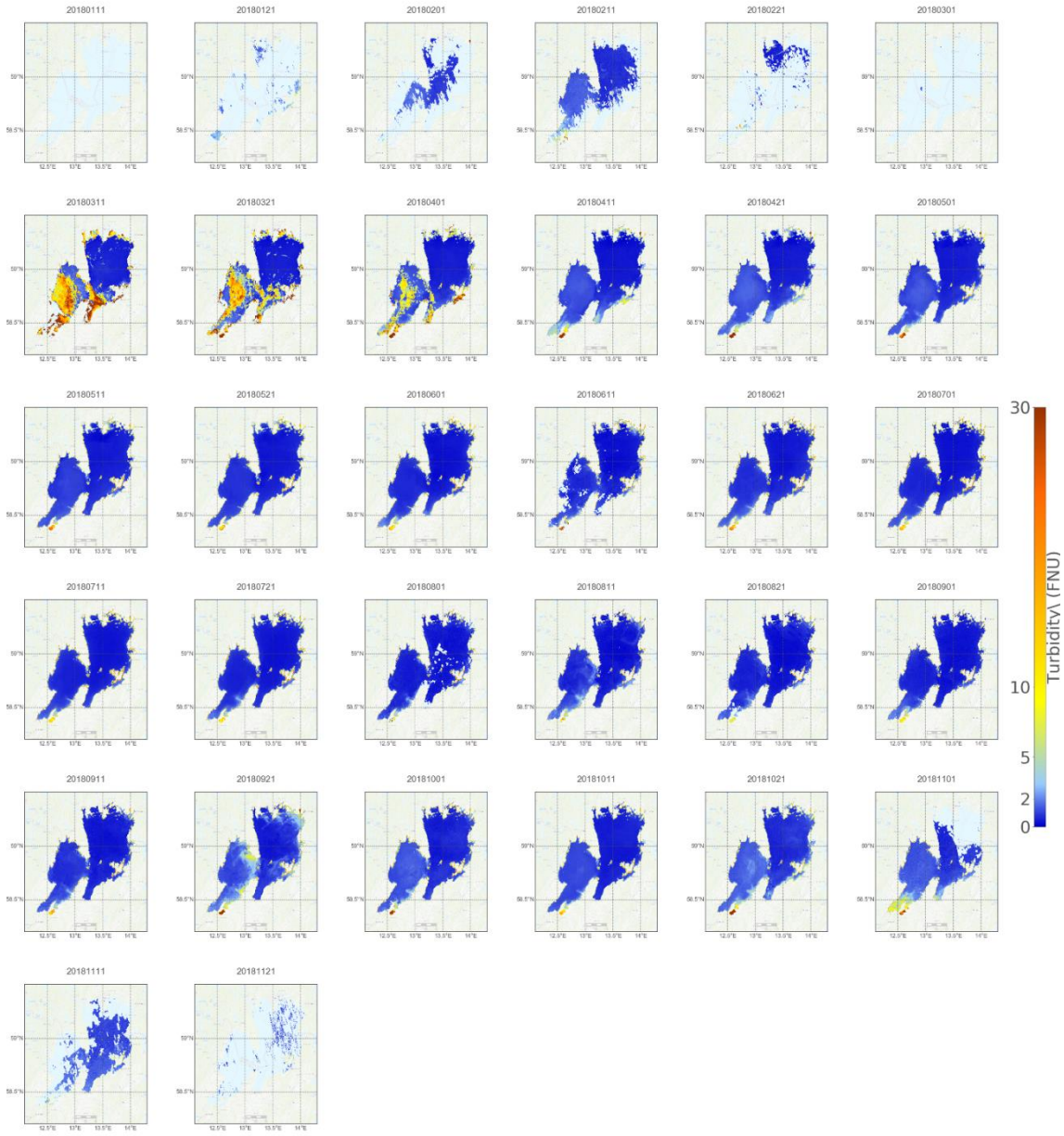


Figure 32: Maps showing Turbidity and its consistency over the year in Lake Vänern per dekad in 2018 from Sentinel-3 OLCI observations

The time series shown in Figure 33 for LWQ at selected locations show low turbidity values during spring, summer and autumn, with slightly higher values in early spring and later autumn. The data gap in winter is due to ice cover and low light availability. Higher values occur in early spring, probably due to melting ice, as suggested above, but not triggering the bright pixel or ice flags. The time series for the trophic state index have been complemented with in situ data which is collected within the framework of the coordinated Swedish environmental monitoring program. The in situ data were categorised according to Carlson (1977), i.e. in the same way as the satellite derived chlorophyll-a concentration. Both datasets show a TSI around 40 to 50, whereas the majority of satellite derived products are in category 50 compared to 40 in the in situ measurements.

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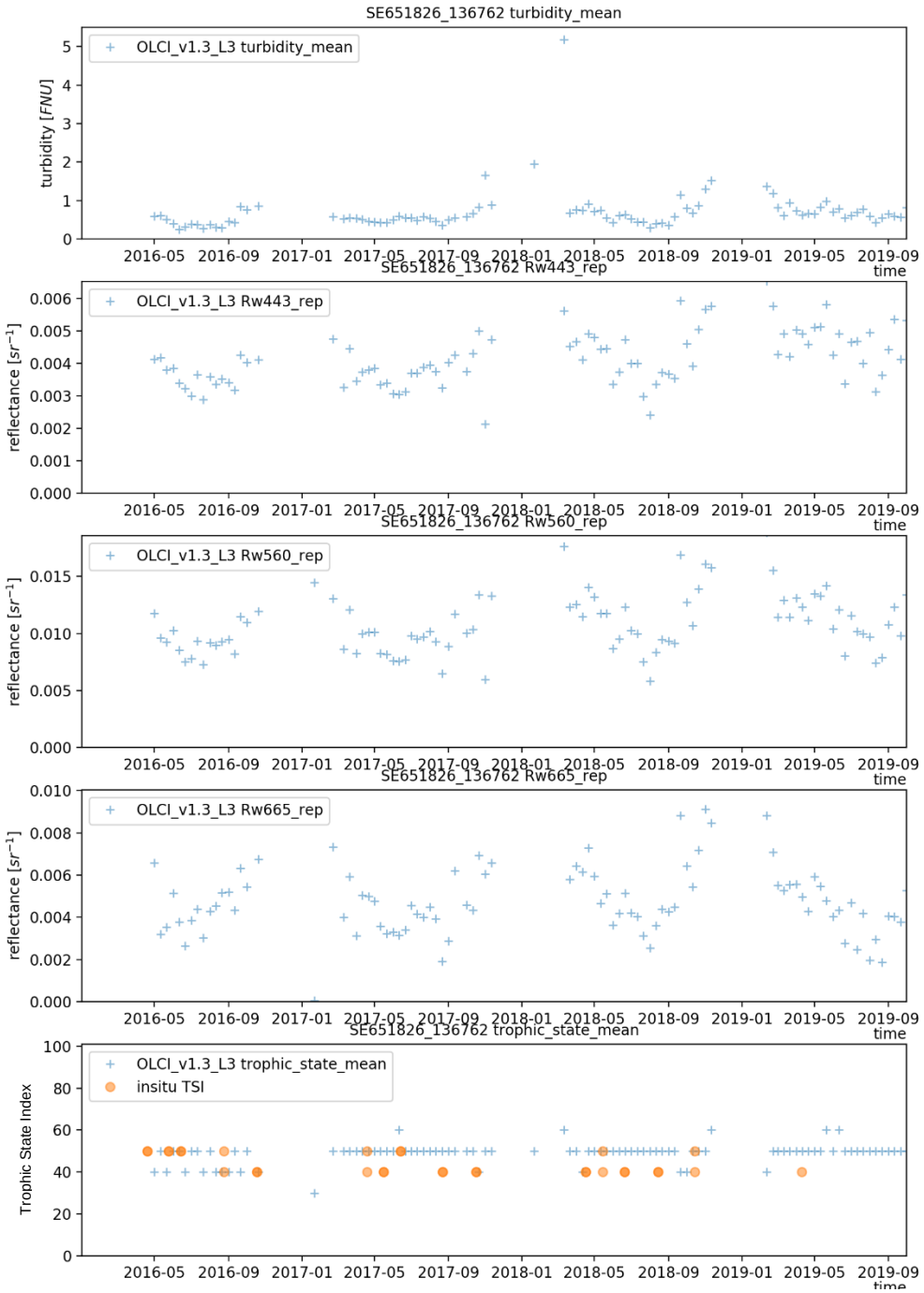


Figure 33: Time series in Lake Vänern showing consistency of the Turbidity, R_w443, R_w560, R_w665 and trophic state index products. Trophic state classes range from 0 to 100, separated in 10 categories corresponding to chlorophyll-a concentration following Carlson (1977)

4.3.1.2 Lake Müritz



Lake Müritz is the largest lake in Germany that is completely located within Germany. It has been strongly influenced by wastewater, which is reduced since the 1970s. Improved water quality is observed since the 1990s, reflected in reduced turbidity of the lake. The lake has large inter-annual changes in phyto- and zooplankton succession. It is categorised as mesotrophic.

(https://www.umweltbundesamt.de/themen/wasser/seen/zustand#textpart-1)

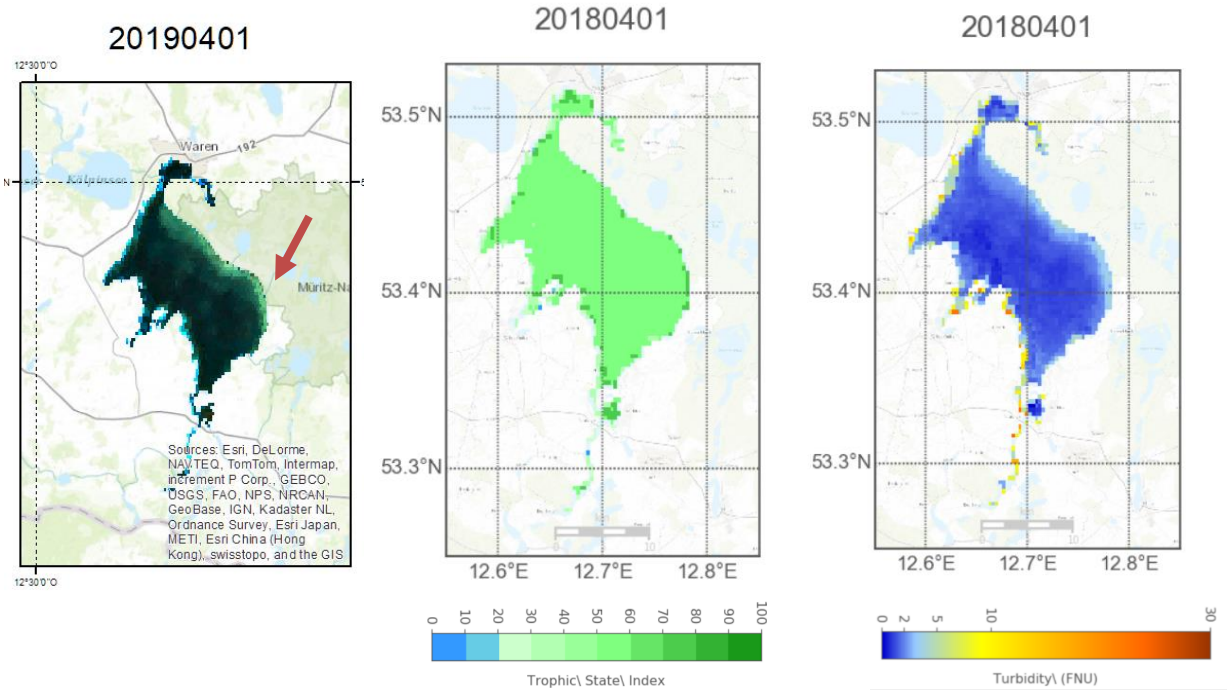


Figure 34: Lake Müritz - Left: RGB derived from water leaving reflectance (R_w665_rep, R_w560_rep, R_w490_rep). The arrow marks the position of time series extraction (Figure 36). Middle: Trophic State Index. Right: Turbidity during the first dekad in April 2018

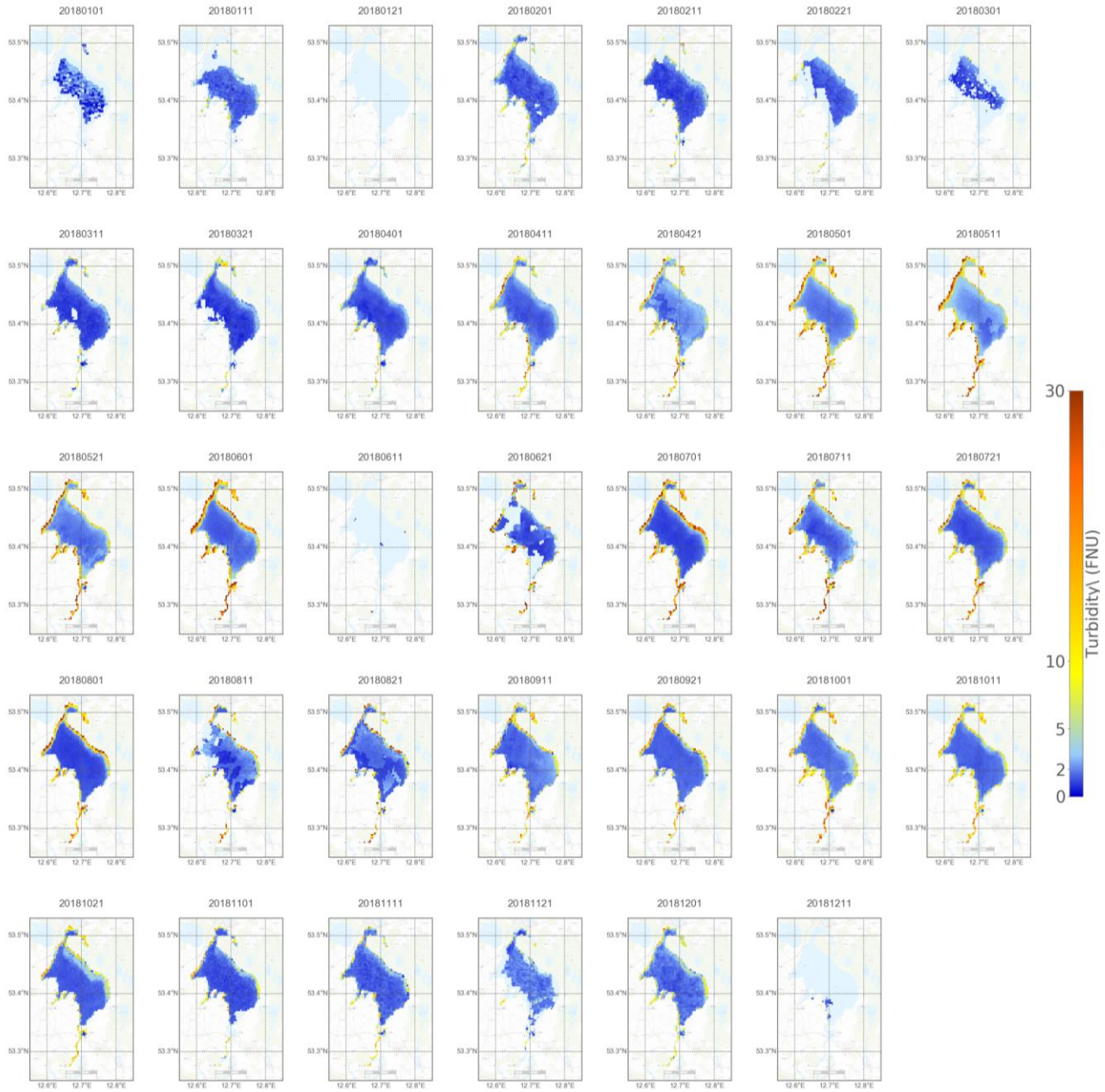


Figure 35: Maps showing Turbidity and its consistency over the year in Lake Müritz, Germany, in 2018 (dekad products).

Lake Müritz is categorised as mesotrophic lake which corresponds to Carlson TSI classes 40 and 50. The lake has low turbidity in most of its parts with lowest values during summer months due to the clear water period between spring bloom and autumn blooms. Very high values along the shoreline indicate misclassification of mixed pixels and non-flagged bottom reflection which cause artificially high turbidity values. Some of the maps show artificial structures which are mainly

caused by suboptimal cloud detection in one of the input products. Figure 36 shows the time series of turbidity, selected water reflectance bands and the trophic state index. Areas close to the north-eastern shoreline often show higher values in reflectance and turbidity. The region is characterised by shallow water covering bright sands so that bottom reflection likely contributes to water-leaving reflectance.

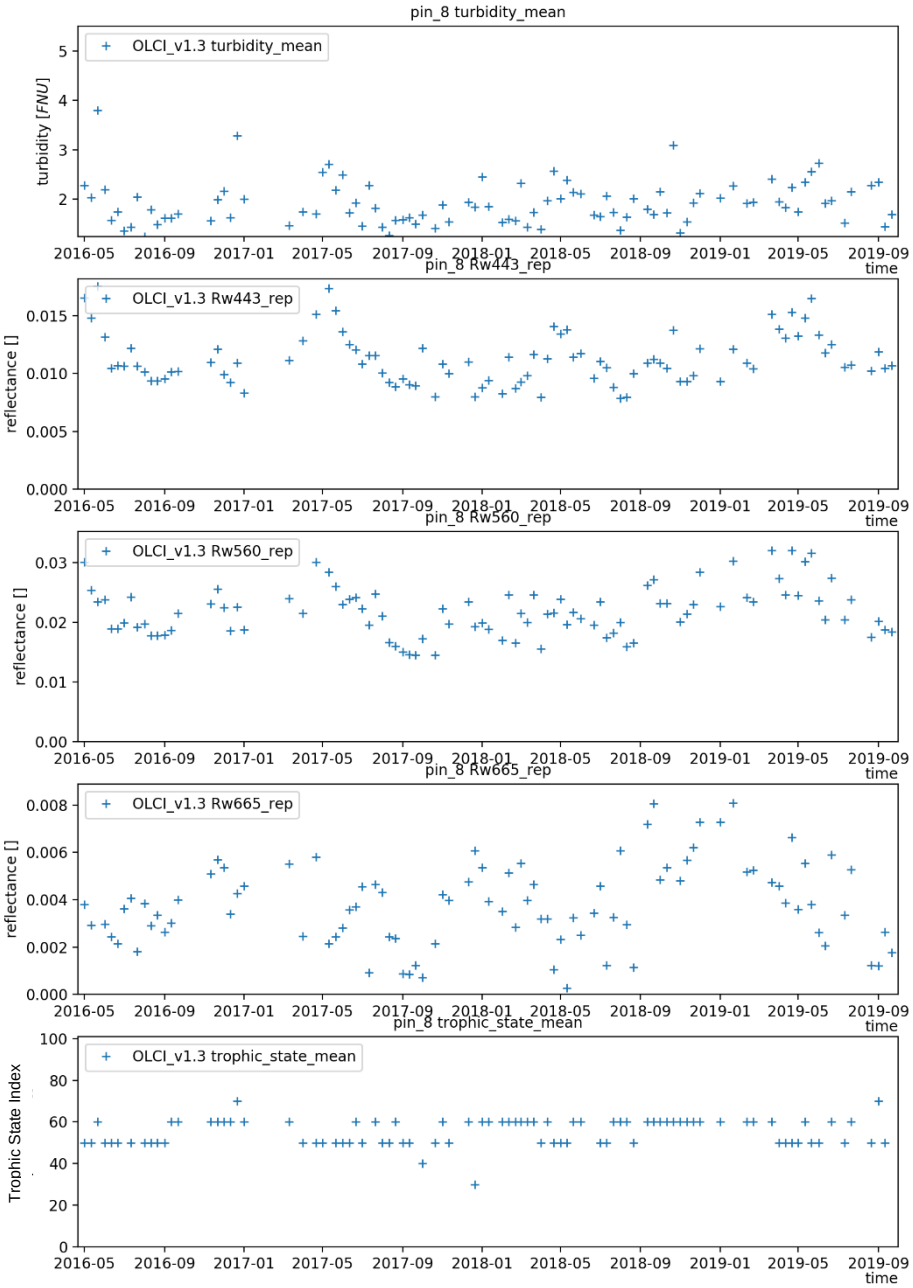


Figure 36: Time series of Lake Müritz showing consistency of the Turbidity, R_w443, R_w560, R_w665 and trophic state index products. Trophic state classes range from 0 to 100, separated in 10 categories corresponding to chlorophyll-a concentration following Carlson (1977)

4.3.1.3 Lake Turkana



Lake Turkana is the world´s largest permanent desert lake, Kenya's largest lake and the largest lake in the eastern rift. Lake Turkana is popularly known as the 'Jade Sea' due to its remarkable algal colouration.

It is filled with semi saline waters, well mixed and has constant temperatures. 90% of its freshwater are received via the Omo River from Ethiopia. The regulation of the Omo River will impact the lake water regime and probably water quality. The Omo River will permanently capture sediments and nutrients that would otherwise be transported to feed the lake.

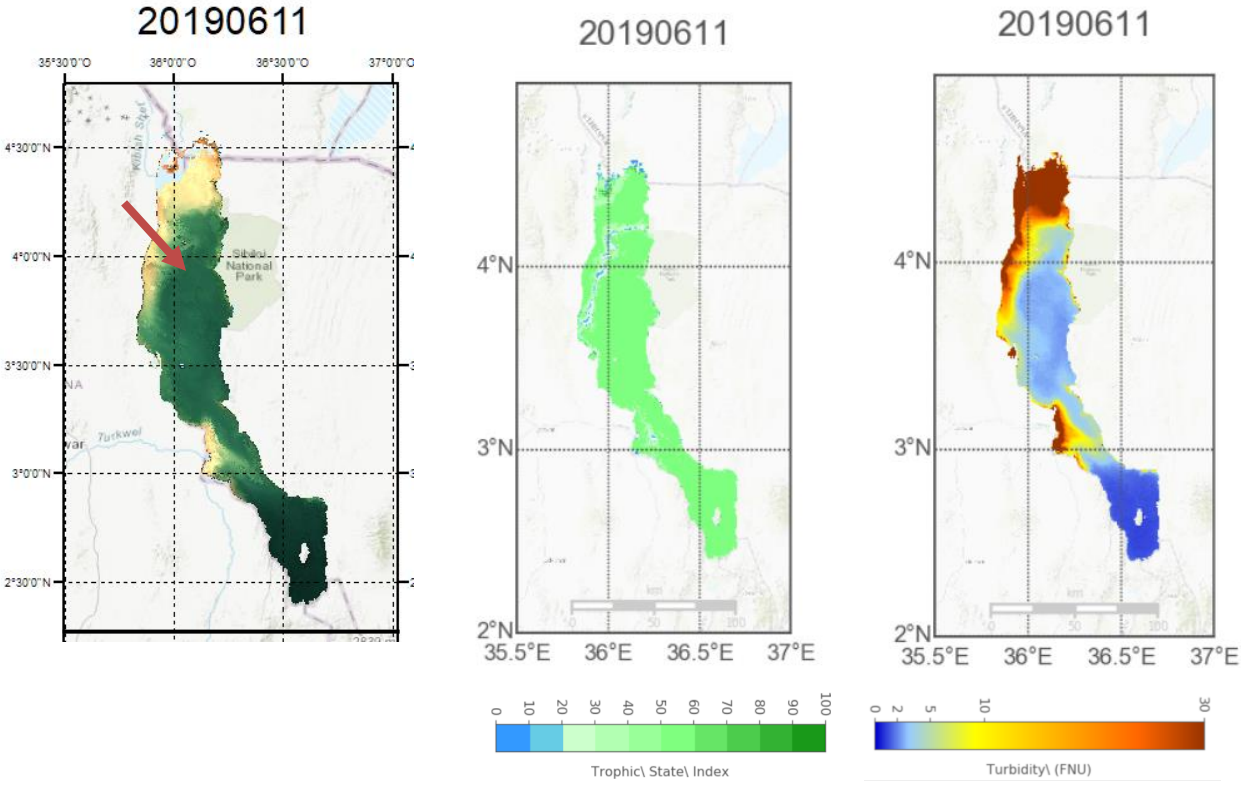


Figure 37: Lake Turkana - Left: RGB derived from water leaving reflectance (R_w665_rep, R_w560_rep, R_w490_rep). The arrow marks position of time series extraction. Middle: Trophic State Index. Right: Turbidity during the second dekad in June 2019

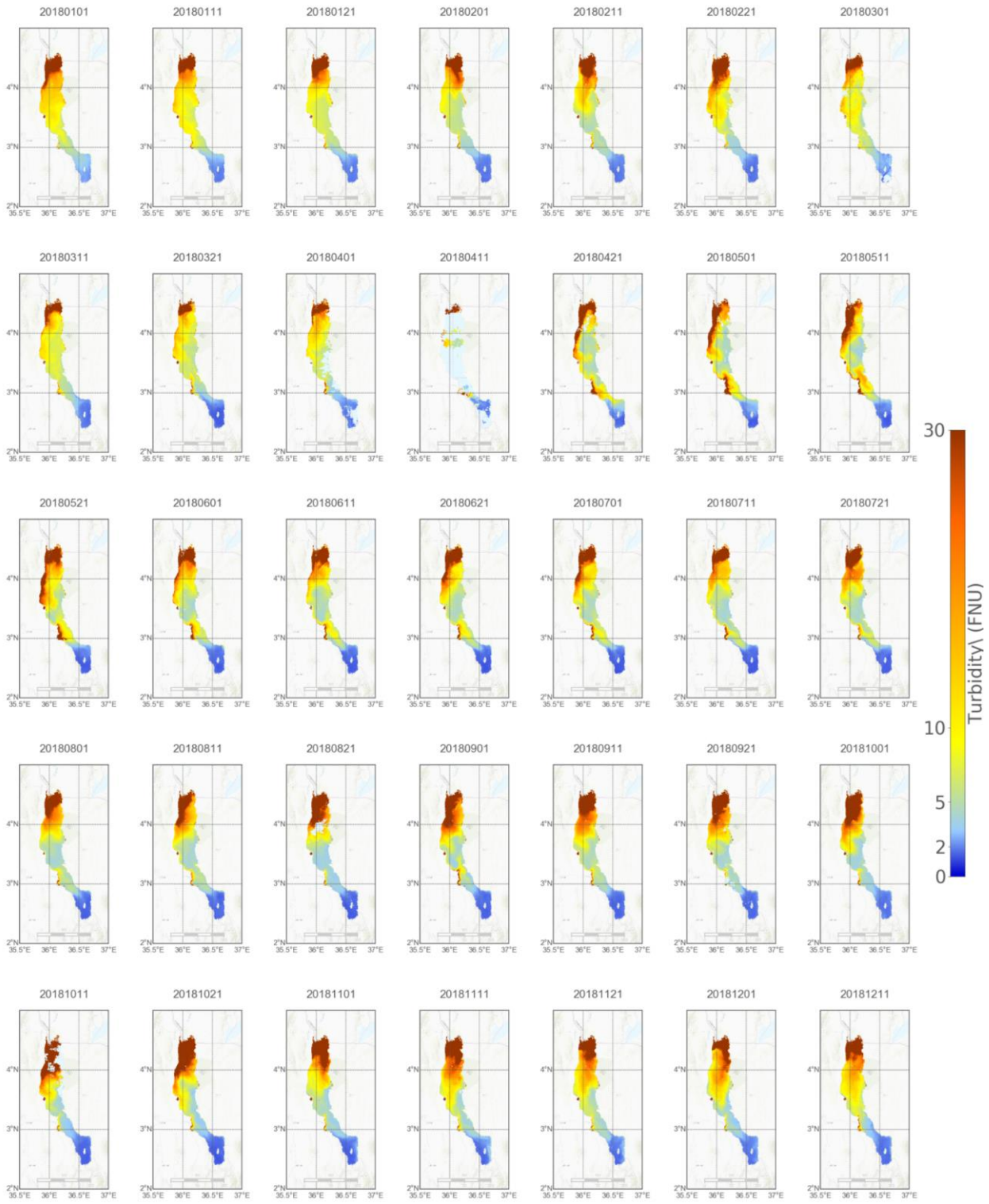


Figure 38: Maps showing turbidity and its consistency over the year of Lake Turkana, Kenya, in 2018 (dekad products).

Lake Turkana is characterised by the strong gradient of Turbidity from North to South, which can be seen in each of the above 10-days turbidity maps in Figure 38. The distribution of suspended sediment a strong seasonal trend, with higher and larger concentration during and shortly after the raining season. This effect is demonstrated in the time series plots in Figure 39 from a position in the Northern part of the lake.

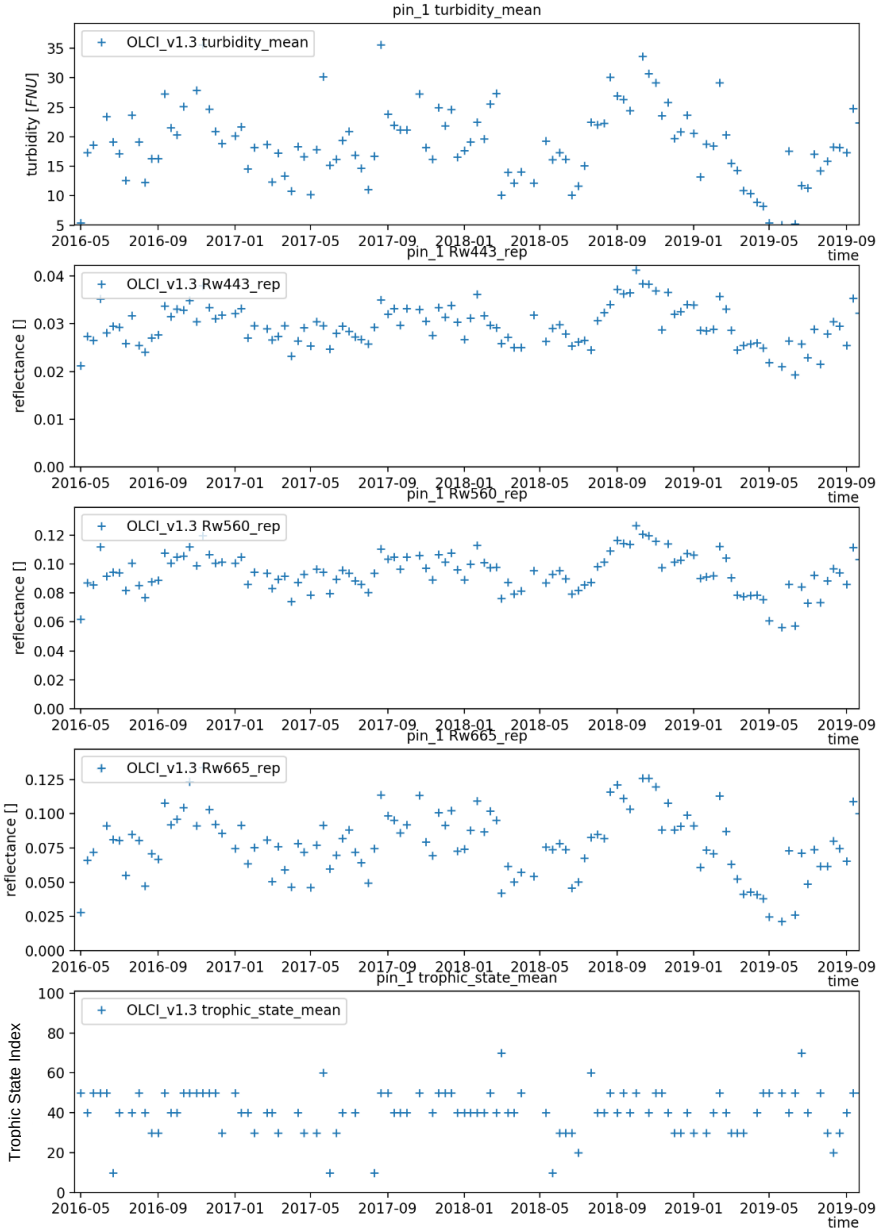


Figure 39: Time series of Lake Turkana showing consistency of the Turbidity, R_w443, R_w560, R_w665 and trophic state index products. Trophic state classes range from 0 to 100, separated in 10 categories corresponding to chlorophyll-a concentration following Carlson (1977)

4.3.1.4 Lake Balaton



Lake Balaton is the largest lake in Central Europe, with a surface area of 593km^2. The lake is very shallow at an average depth of 3.2m. The main inflow, the Zala River, feeds into the southwestern end of the lake, while the Sio-canal drains the lake in the eastern basin into the River Danube. Wind resuspension of sediments can render transparency low and in wintertime the lake is ice covered (ILEC World Lake Data Base). Lake Balaton is characterised by its turquoise colour. The Eutrophic state has changed in the past years, with more eutrophic situation in the west and mesotrophic in the eastern part of the lake.

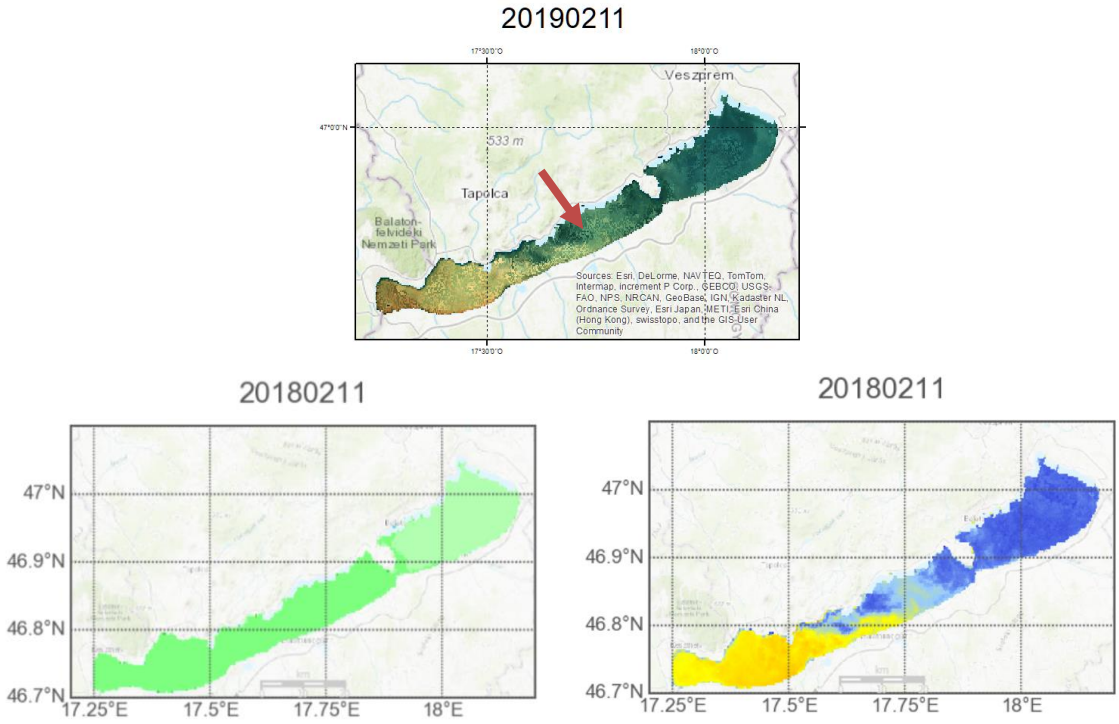


Figure 40: Lake Balaton - Left: RGB derived from water leaving reflectance (R_w665_rep, R_w560_rep, R_w490_rep). The arrow marks the position of the time series extraction. Middle: Trophic State Index. Right: Turbidity during the second dekad in February 2018

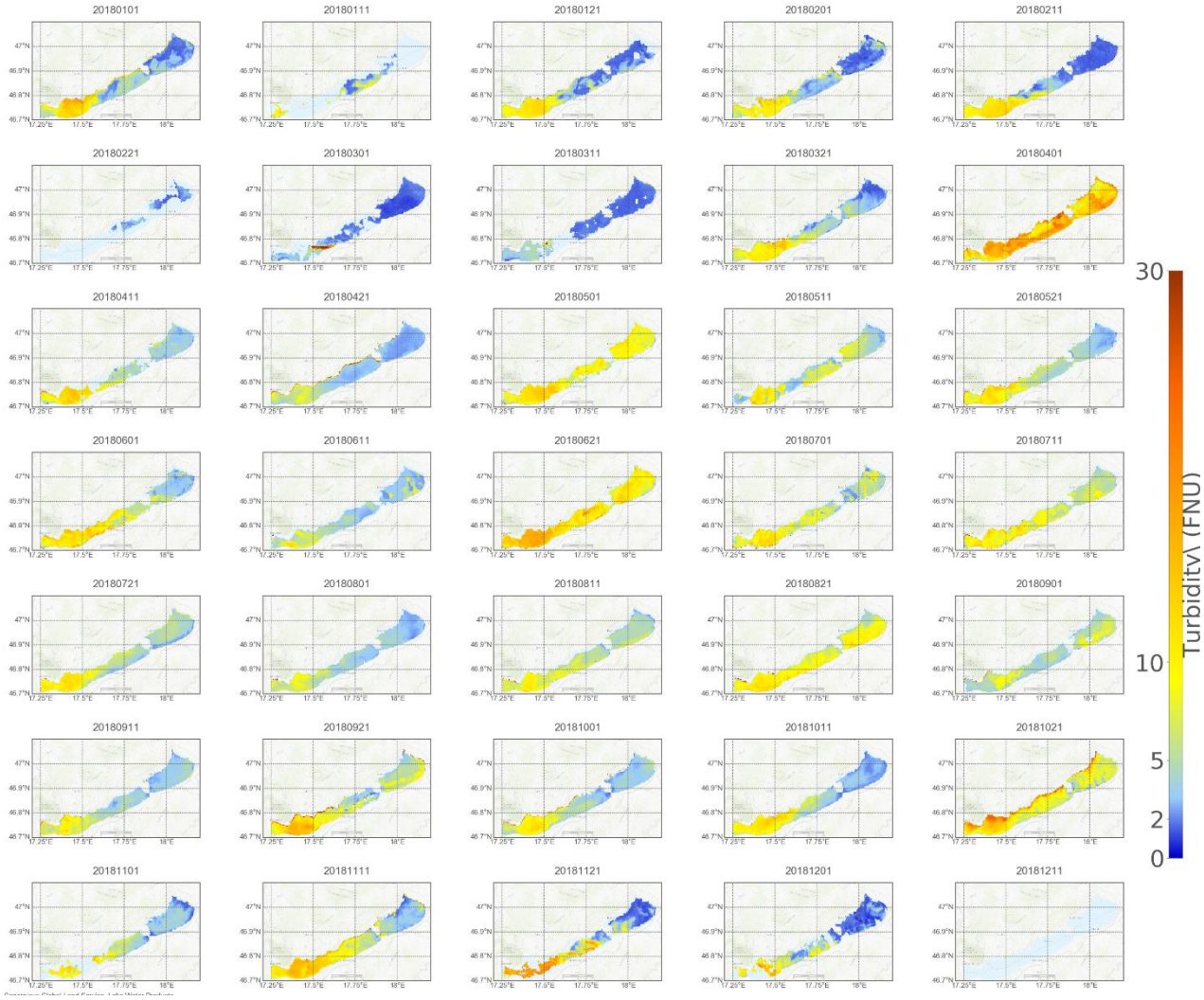


Figure 41: Maps showing turbidity and its consistency over the year of Lake Balaton, Hungary, in 2018 (dekad products)

Lake Balaton typically has more turbid waters in the western basin and clearer waters in the eastern basin. Total suspended matter in the lake is dominated by inorganic particles with concentrations typically in the order of 18–28 g m^{-3} (2010 annual mean) but can exceed 100 g m^{-3} during windy periods (Riddick et al. 2015, Riddick et al. 2019). The maps in Figure 41 clearly show the different water types in the western and eastern parts of the lake. The patterns are consistent between subsequent dekads with occasional lake-wide resuspension events (e.g. 20180401, 20180501 and 20180621).

A gradient is also present in Chl-a concentration and therefore TSI, with higher values in the west (fed directly by the Zala river) and lower concentration in the east. In the western Keszthely basin, concentrations range from 3 to 45 mg m^{-3} and in the eastern Siófok basin from 3 to 20 mg m^{-3}.

This corresponds to TSI categories 50 - 70 in the western part and 50 - 60 in the eastern part following the categorization applied to the CGLOPS products. Figure 42 shows the time series in a position in the centre of the lake (see red arrow in Figure 40) with TSI classes mainly at 50 and turbidity between 5 and 20 NTU. Turbidity is variable without clear seasonal trends.

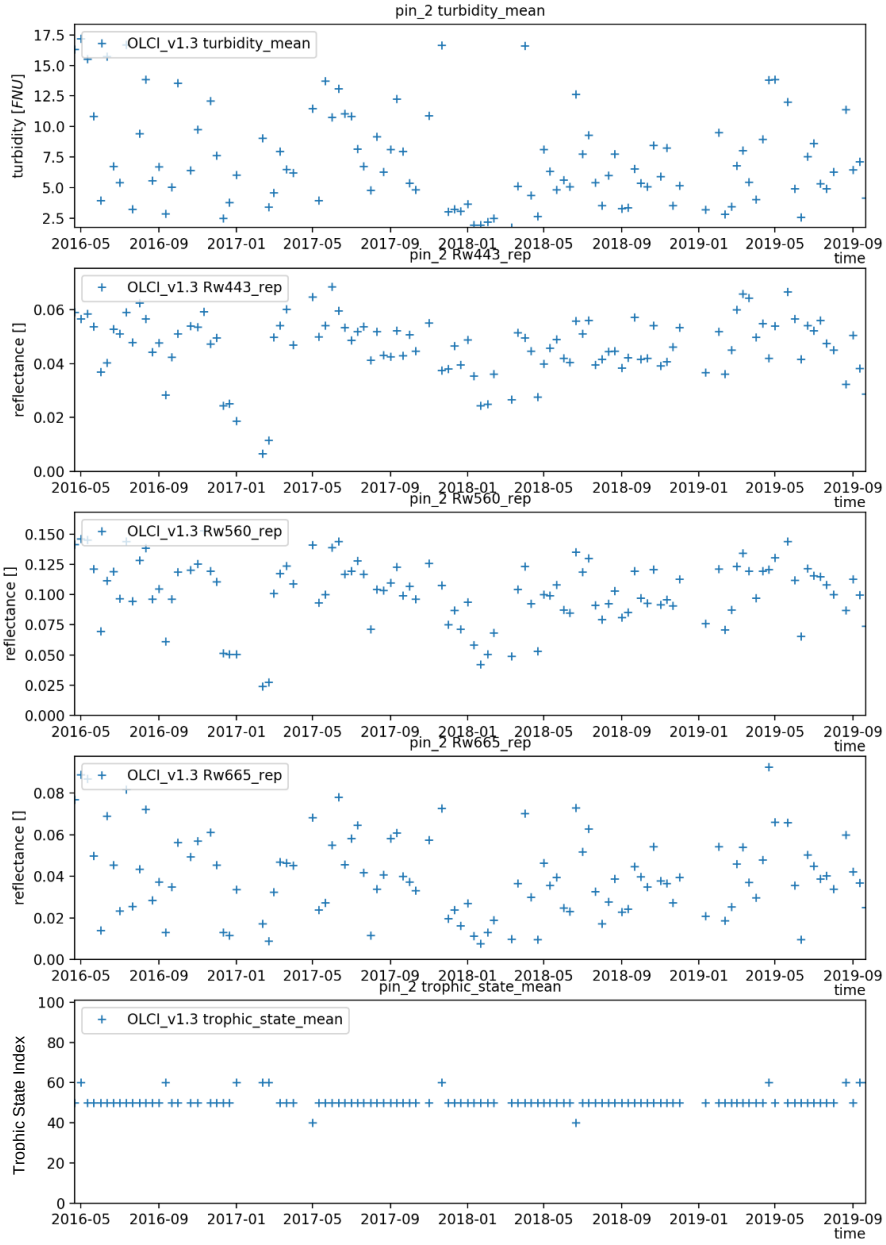


Figure 42: Time series of Lake Balaton showing consistency of the Turbidity, R_w443, R_w560, R_w665 and trophic state index products. Trophic state classes range from 0 to 100, separated in 10 categories corresponding to chlorophyll-a concentration following Carlson (1977)

4.3.1.5 Lake Garda



Lake Garda is the largest lake in Italy. The northern part of the lake is narrower and surrounded by mountains. The lake is oligo-mesotrophic with mild seasonal phytoplankton blooms and occasionally a deeper chlorophyll-a maximum. Lake Garda is a major tourist destination, and its water is used in agriculture and as drinking water. Lake Garda is mesotrophic (Giardino et al. 2007).

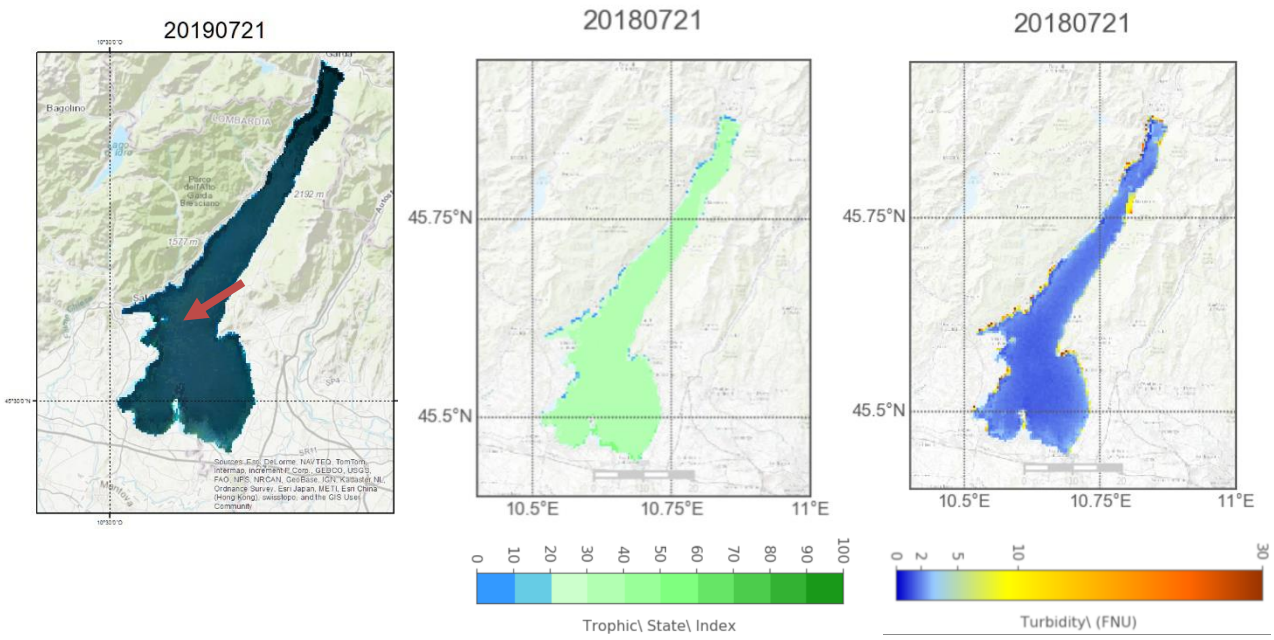


Figure 43: Lake Garda - Left: RGB derived from water leaving reflectance (R_w665_rep, R_w560_rep, R_w490_rep). The arrow marks the position of the time series extraction. Middle: Trophic State Index. Right: Turbidity during the third dekad in July 2018

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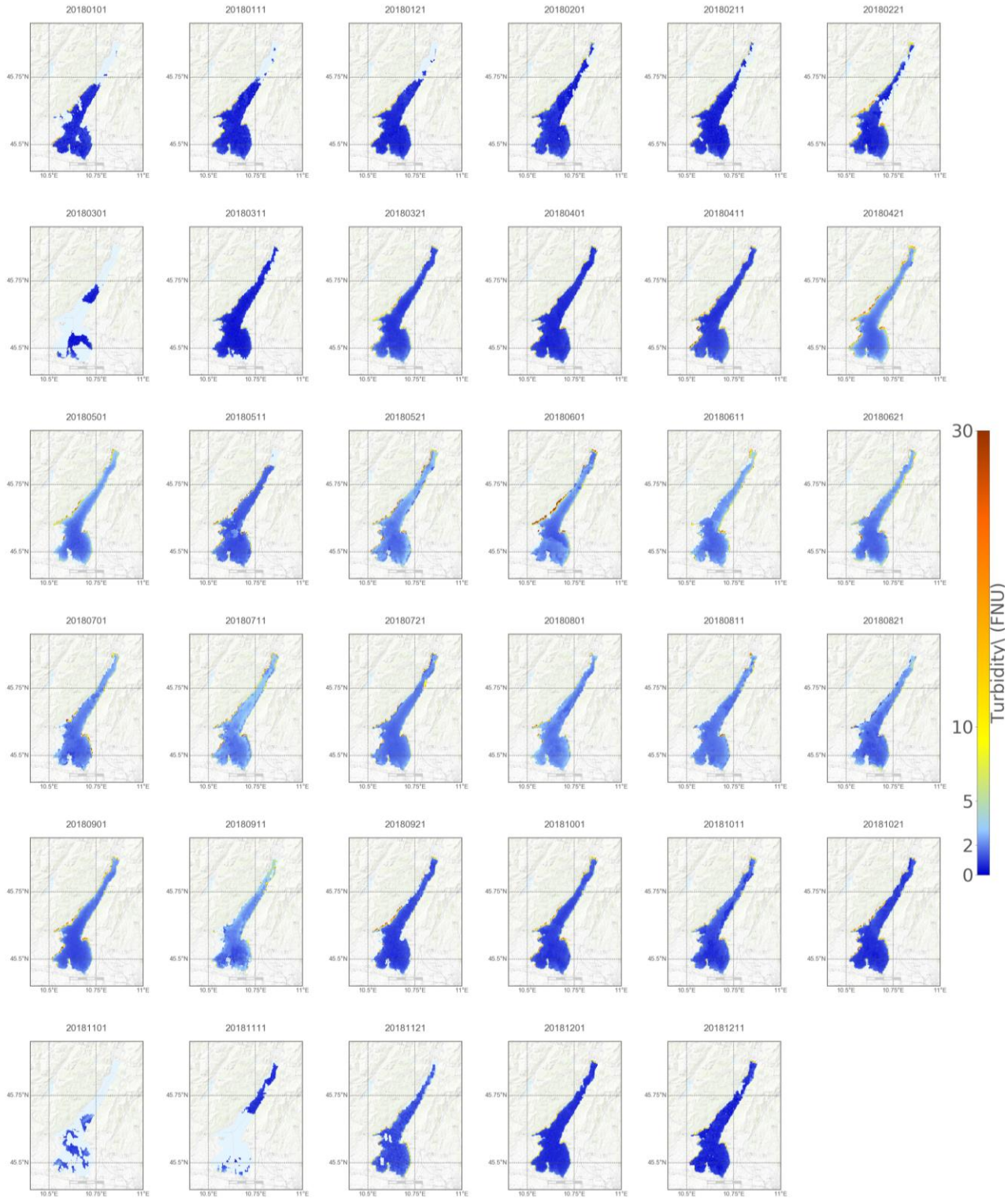


Figure 44: Maps showing turbidity and its consistency over the year of Lake Garda, Italy, in 2018 (dekad products).

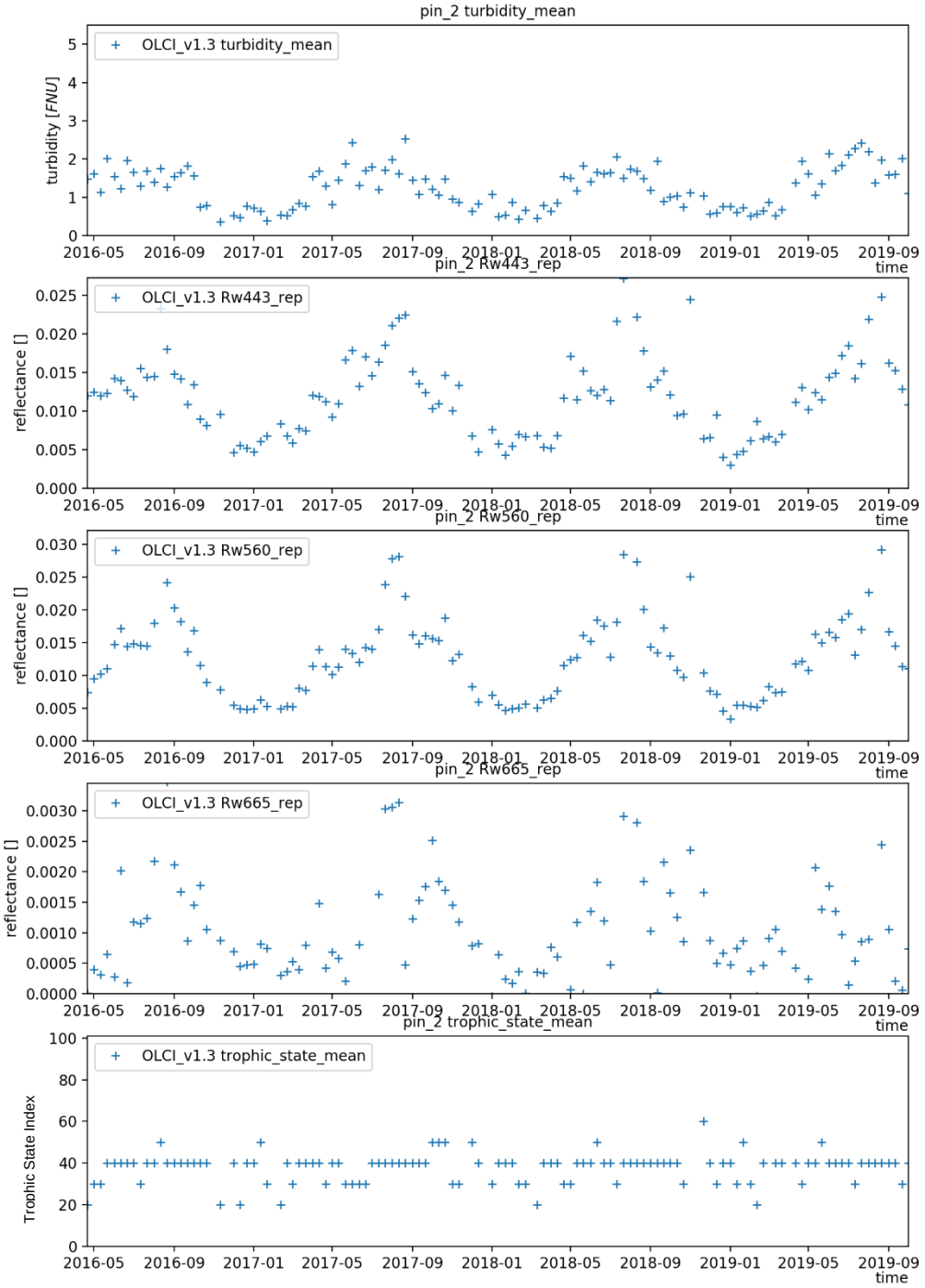


Figure 45: Time series in Lake Garda showing consistency of the Turbidity, R_w443, R_w560, R_w665 and trophic state index products. Trophic state classes range from 0 to 100, separated in 10 categories corresponding to chlorophyll-a concentration following Carlson (1977)

The maps of the dekads for Lake Garda show very consistent turbidity within the lake and over the year. It starts with very clear water in January until March followed by an increase of turbidity in the last dekad in April. During summer, higher turbidity is visible in the whole lake, decreasing again end of August to reach clear water end of September. This clear seasonal trend is also showing the time series of turbidity and reflectance in Figure 45. Turbidity is changing between 0 and 2 FNU but following a clear seasonal cycle with higher values in late spring and summer months. The trophic state index is mainly 40 with variations between 20 and 50. TSI of 40 corresponds to chlorophyll-a concentrations between 0.9 and 2.6 mg m^{-3}. Giardino et al. (2007) provided the following concentration ranges for Lake Garda: Chlorophyll-a and TSM concentrations range from 0.5 to 12 mg m^{-3} and from 0.1 to 5.5 g m^{-3}, respectively (average concentrations of Chl-a, TSM are around 2.7 mg m^{-3} and 2.5 g, respectively).

4.3.1 Performance of mapping water constituents per optical water type

The optical water type itself is not a parameter within the final LWQ products delivered to the users. However, some examples of their distribution and the influence they have on the blending of algorithms is demonstrated here. Maps of the OWT in two lakes as well as spatial transects of trophic status and turbidity parameters are provided to show whether class boundaries introduce unrealistic spatial discontinuities in the products.

The first example of Lake Huron shows no clear transitions at class borders neither in the map (Figure 27) nor in the transects (Figure 28). The turbidity algorithm that is mapped here, however, is the same for the three identified dominant optical water types: 3, 9 and 13 (Binding et al 2010). The potential influence of blending other algorithms is therefore expected to be negligible.

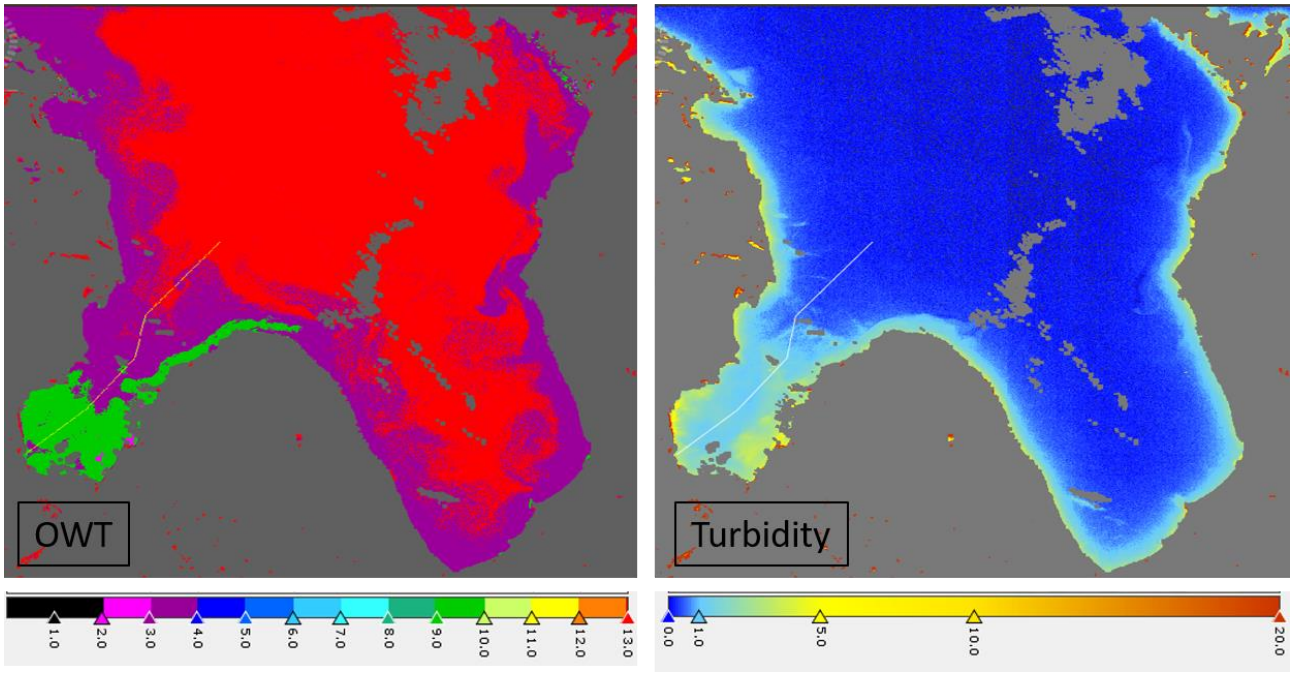


Figure 46: Lake Huron OWT (left) and Turbidity in NTU (right) for OLCI acquisition 26 August 2017. The line marks the transect shown in Figure 47

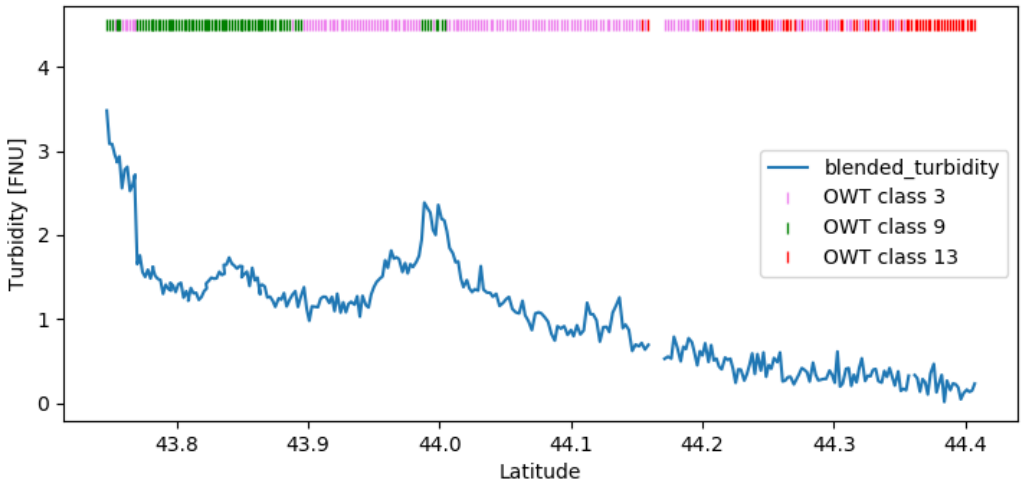


Figure 47: Blended Turbidity and OWT classes along a transect in Lake Huron (transect location shown in Figure 46)

The following example of Lake Turkana (Figure 29) blends between water types 2, 4, 9, 11 and 13. Water types 9 and 11 share the same tuned algorithm (Binding et al. 2010) whereas 2 and 4 are derived from a separate algorithm (Vantrepotte et al., 2011). The result is nevertheless seamless (Figure 30).

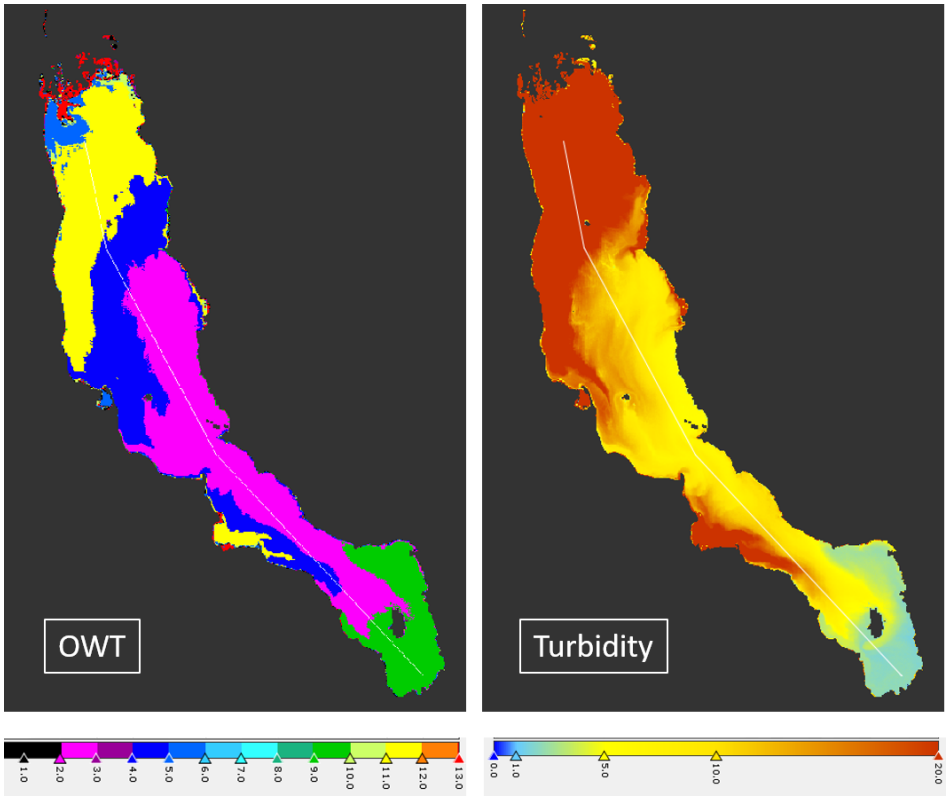


Figure 48: Lake Turkana OWT (left) and Turbidity (right) for OLCI acquisition 19 September 2017. The line marks the transect shown in Figure 49

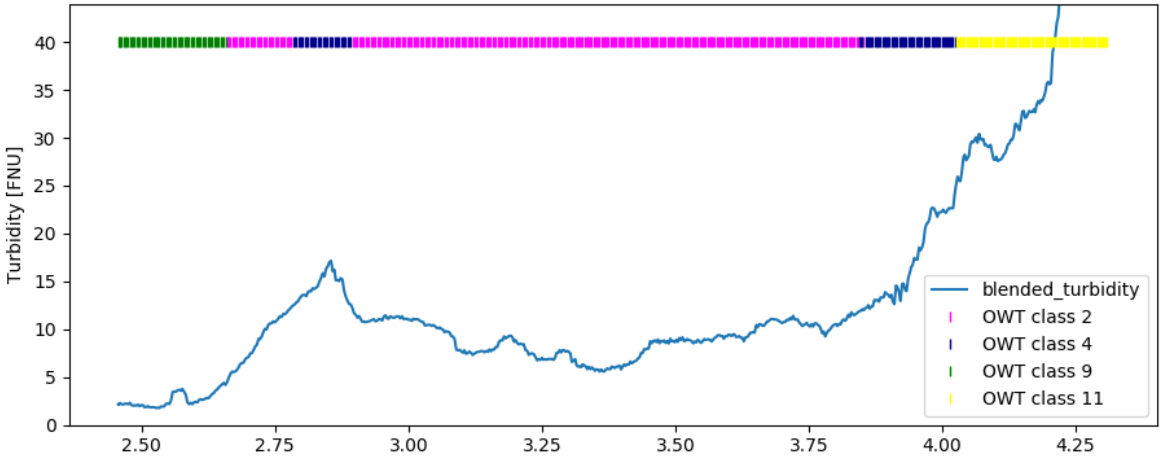


Figure 49: Blended Turbidity and OWT classes along a transect in Lake Turkana (position of transect marked in Figure 48).

Results shown in Figure 44 are from a recent uncertainty characterization analysis of the chlorophyll-*a* product in the *Calimnos* v1.04 dataset. This result is included here to demonstrate that applying the weighted average of the two algorithms associated with the highest OWT

similarity for each observed pixel provides a marked improvement over selecting, for each lake and each observation day, the algorithm that is most suitable for the lake-wide predominant optical water type. In this case, when a single algorithm is applied per lake, algorithms for extremely high concentrations are not commonly selected, causing a saturation effect. This is likely associated with patchy phytoplankton blooms. The dynamic algorithm selection approach is therefore preferred as it is better equipped to deal with optical gradients within individual lakes.

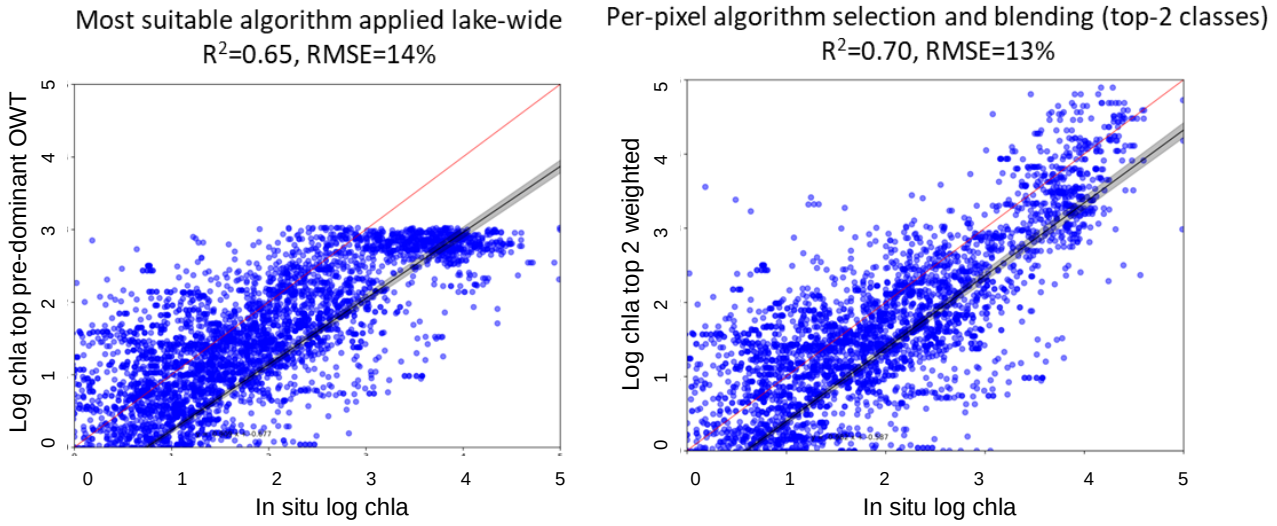


Figure 50: Comparison of the performance of (left) the best chlorophyll-a algorithm for the predominant optical water type of each lake and (right) per-pixel selection and blending of the two highest ranking algorithms based on optical water type membership scores

4.4 COMPARISON WITH IN SITU DATA

4.4.1 Comparison with in situ data – Lake Water Reflectance – Matchup Analysis

Polymer v3.5 was validated against in situ reflectance data contained in Limnades within the GloboLakes project. Polymer gave the best performance (unbiased error) in each waveband, although a consistent underestimation is apparent (Figure 7). This systematic error is cancelled out in the whole-chain calibration of TSM and chlorophyll-a retrieval. A correction is not attempted for the LSR because there are not enough in situ validation data available to inspect the performance per Optical Water Type. Further improvement of the Polymer atmospheric correction will be based on results of ongoing investigations, which have thus far shown that the mineral absorption model should not be universally applied, as was previously considered a solution. It should be noted that underestimation of reflectance has a minimal effect on water constituent retrieval algorithms that operate primarily on the shape of the reflectance spectrum. This is the case for turbid water chlorophyll-*a* algorithms and some TSM algorithms.

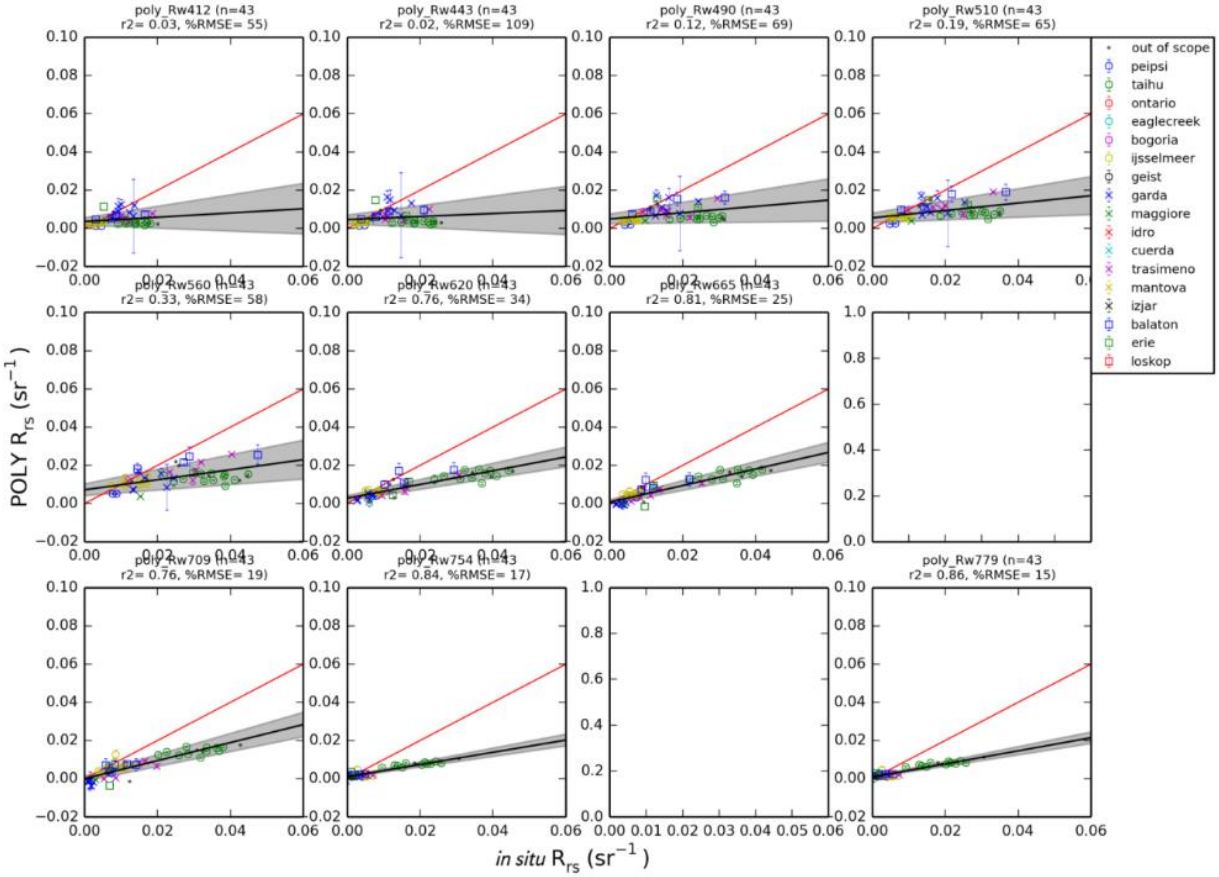


Figure 51: Polymer v3.5 validated against in situ reflectance data contained in Limnades (source: UK NERC GloboLakes). Here, results for a 3x3 pixel window with a 3-day difference between satellite and in situ observation, are shown. Linear regression fits are shown in black with the shaded area marking the 95% confidence interval. The red line marks unity

Hyperspectral water-leaving reflectance spectra ranging from 400 to 900 nm between in situ and S3 OLCI data are shown in Figure 52. The in situ observation data were derived by ASD Fieldspec Spectro-radiometer and have been provided by IREA within CNR.

Both examples show good agreement in the spectral shape, while water-leaving reflectance is significantly underestimated from the satellite data, especially in the second example. The scatterplot in Figure 53 shows the results from all available in situ points in Lake Garda, also underlying the underestimation of reflectance in all bands.

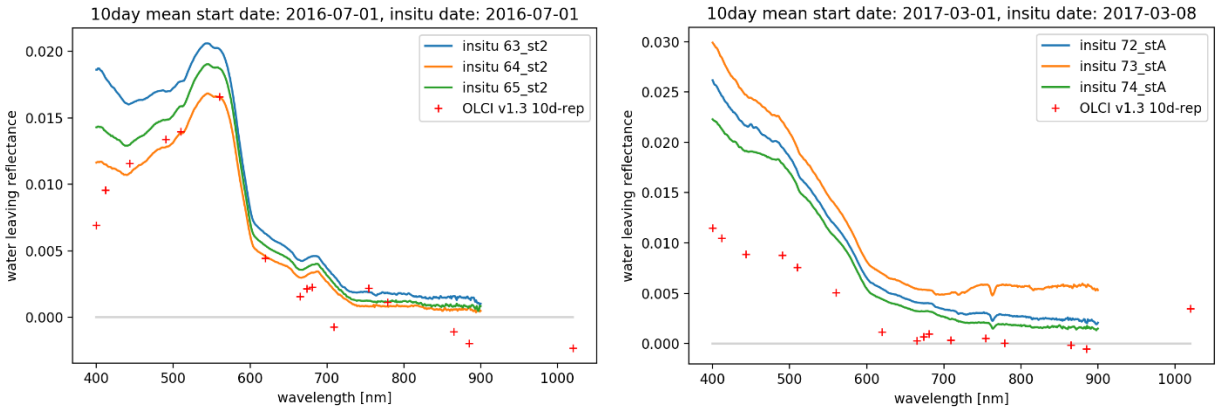


Figure 52: Representative water-leaving reflectance spectrum of S3-OLCI for a ten-day period, compared to in situ measurements (ASD spectro-radiometer) in Lake Garda. Left: measurement in the Southern part, right: in the Northern part (see red arrows in map below). In situ data: CNR

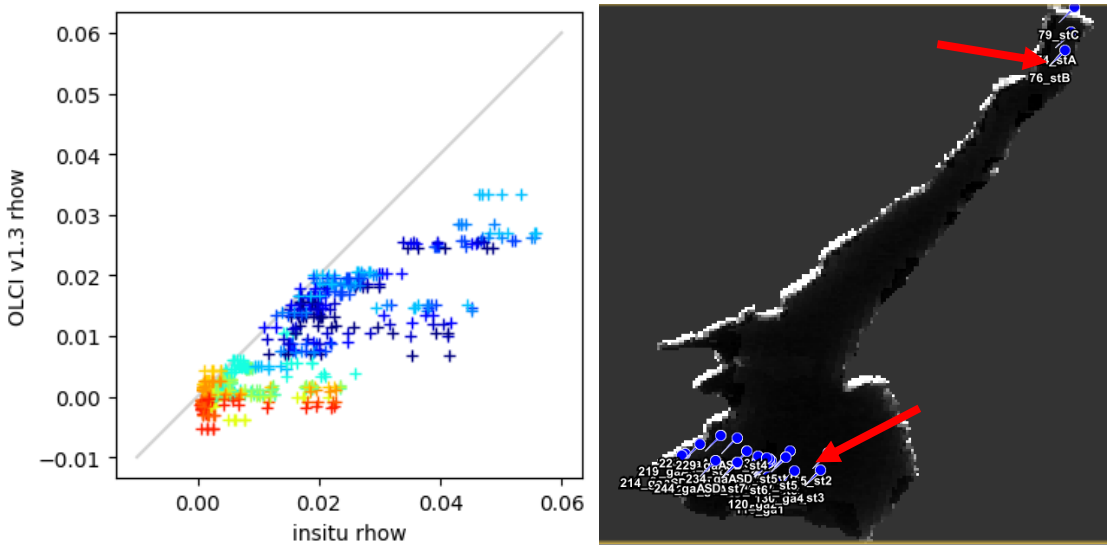


Figure 53: S3-OLCI Level 3 water leaving reflectance compared to in situ reflectance from all in situ measurements. The spectra presented in Figure 52 have been taken from the locations indicated by the red arrows. Hyperspectral data were integrated over the spectral response function of Sentinel-3A. The colour of the markers corresponds to S3-OLCI central wavelengths from 400nm (dark blue) to 900nm (red). In situ data: CNR

4.4.2 Updated comparison with in situ data – Lake Water Reflectance – Matchup Analysis

A comparison between Polymer v4.14 and v4.15 has been performed using in situ measurements of R_{rs}. We have used the Gloria dataset (Lehmann et al, 2023) and WISP stations to generate a set of matchups with the OLCI sensors. Each matchup is constructed from a 3x3 macro-pixel of data centred at the location of the in situ point. There must be at least 5 non-NaN values in the 9 (3x3) pixels and there must be no LAND flagged within the macro-pixel. In addition, pixels were masked out using a selection of flags from IDEPIX (IDEPIX_INVALID, IDEPIX_CLOUD, IDEPIX_CLOUD_AMBIGUOUS, IDEPIX_CLOUD_SURE, IDEPIX_CLOUD_BUFFER, IDEPIX_CLOUD_SHADOW, IDEPIX_LAND, IDEPIX_SNOW_ICE), OLCI quality flags (saturated_Oa01 – saturated_Oa21) and the Polymer OUT_OF_BOUNDS flag. Finally, a homogeneity test was performed on the macro-pixel where the coefficient of variation at 560 nm, CV560 (standard deviation / mean at R_w560), must be less than 0.2.

In addition to the above filtering on the macro-pixels, the in situ data were also filtered to remove suspect points. The criterion used was to discard any where the R_{rs} was greater than 0.5 or less than -0.01 for any equivalent satellite band. Data were also filtered on database specific columns: for Gloria this used the “flagged” column to remove any points that had been flagged, and for the WISPCloud database the “rrs_quality” was used to remove any points labelled as “suspect”. The data were also offset corrected in the NIR to remove biases. In situ data were integrated over the sensor response function to match the equivalent satellite bands. If there were multiple in situ data at the same location on the same day then these were aggregated to use the median spectra.

When making the comparison between the two Polymer versions a common set of matchups has been used. As not all in situ data contain R_{rs} up to 865 nm and higher, these higher bands have fewer matchups.

A comparison between the absolute residual error and distance to land showed that the points with highest residual error are within 5 km of land. It should be noted that a static distance to land map was used (Carrea et al, 2015) meaning that data points’ distance to land is referenced against the land / water boundaries of that era and not of the date of the satellite image. It can be seen (Figure 54) that close to land Polymer v4.15 has a higher error compared to Polymer v4.14 processed data. This is likely due to adjacency effects. Elsewhere the residuals imply a similar performance between the two Polymer versions.

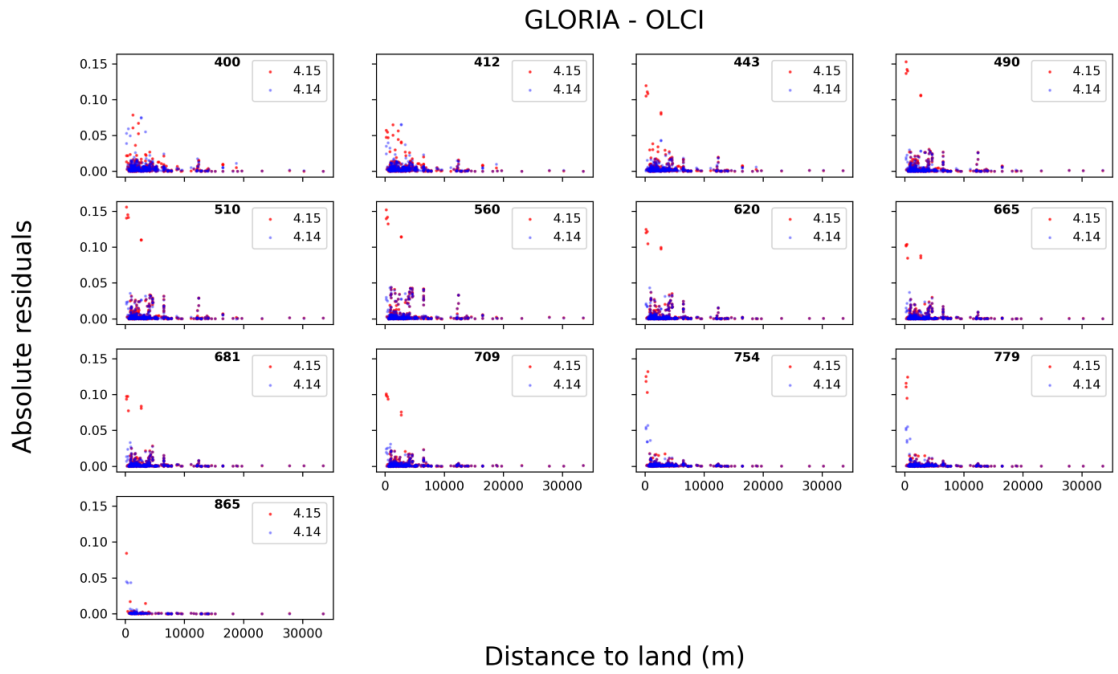


Figure 54: Plot showing absolute residual error of matchups vs distance to land for OLCI processed bands

GLORIA database

The bias from Polymer v4.15 is lower than that of v4.14 but the performance in terms of MAE, RMSE and R has degraded a small extent in Polymer v4.15 (Figure 55, Figure 56, Figure 57), but we do still include some matchups with high standard deviation within the macro-pixel which could increase the magnitude of these error metrics. Looking at the OLCI spectra of matchups whose standard deviation in the 3x3 macro-pixel of the 560nm band > 0.01 for each lake (Figure 55) shows that largely the spectra have similar shape with elevated R_w in the blue and NIR. The median spectra of the remaining matchups for each lake (i.e. those with standard deviation < 0.01 in the 560nm band) tends to show lower R_w. It's expected that this elevated R_w is due to adjacency effect and/or heavy bloom state of the water.

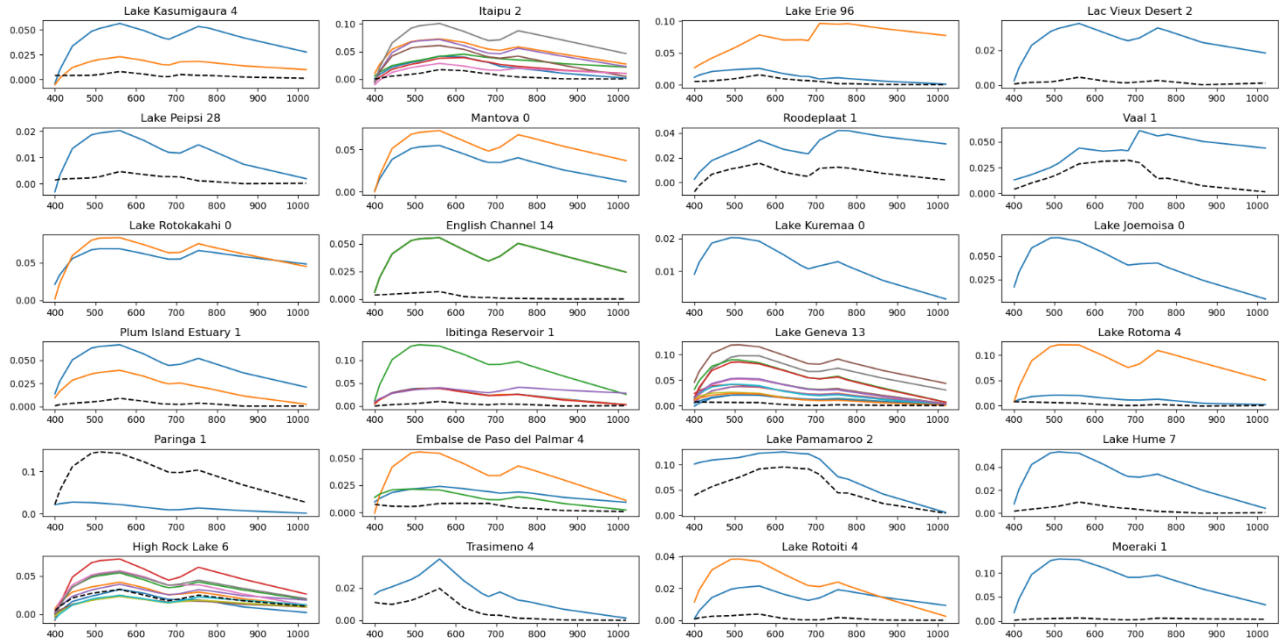


Figure 55: Solid lines show OLCI R_w spectra where the standard deviation of the 3x3 macro-pixel in the 560nm band is > 0.01. Dashed lines show the median OLCI spectra of matchups with standard deviation below 0.01

Table 4: Statistics of OLCI Gloria matchups

	R_w (n)												
	400 (395)	**412 (395)**	**443 (395)**	**490 (395)**	**510 (395)**	**560 (395)**	**620 (395)**	**665 (395)**	**681 (395)**	**709 (395)**	**754 (385)**	**779 (385)**	**865 (135)**
bias_v14	0.0030	0.0026	0.0011	-0.0011	-0.0020	-0.0033	-0.0021	-0.0015	-0.0012	-0.0013	0.0021	0.0019	0.0010
bias_v15	0.0012	0.0016	0.0014	0.0001	-0.0007	-0.0019	-0.0008	-0.0003	0.0000	-0.0003	0.0027	0.0024	0.0011
rmse_v14	0.0100	0.0082	0.0060	0.0074	0.0086	0.0121	0.0081	0.0060	0.0056	0.0055	0.0064	0.0063	0.0066
rmse_v15	0.0093	0.0100	0.0142	0.0178	0.0186	0.0201	0.0158	0.0129	0.0121	0.0122	0.0126	0.0118	0.0076
mae_v14	0.0055	0.0048	0.0036	0.0040	0.0045	0.0060	0.0039	0.0030	0.0028	0.0027	0.0024	0.0022	0.0016
mae_v15	0.0051	0.0051	0.0052	0.0059	0.0064	0.0077	0.0054	0.0043	0.0040	0.0039	0.0029	0.0027	0.0014
r_v14	0.4421	0.4886	0.5993	0.6036	0.5982	0.6320	0.7674	0.8112	0.8163	0.7976	0.3483	0.3742	0.3709
r_v15	0.0852	0.1320	0.1874	0.2238	0.2357	0.3089	0.4197	0.4754	0.4980	0.4515	0.2005	0.2229	0.3213

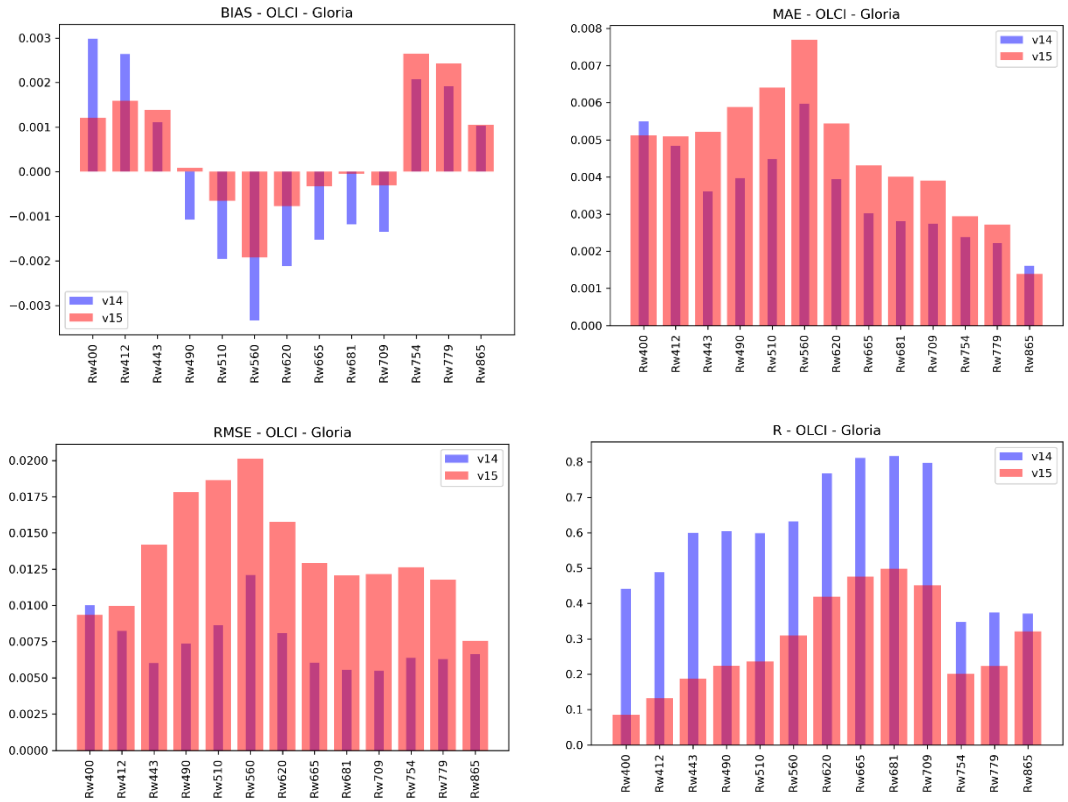
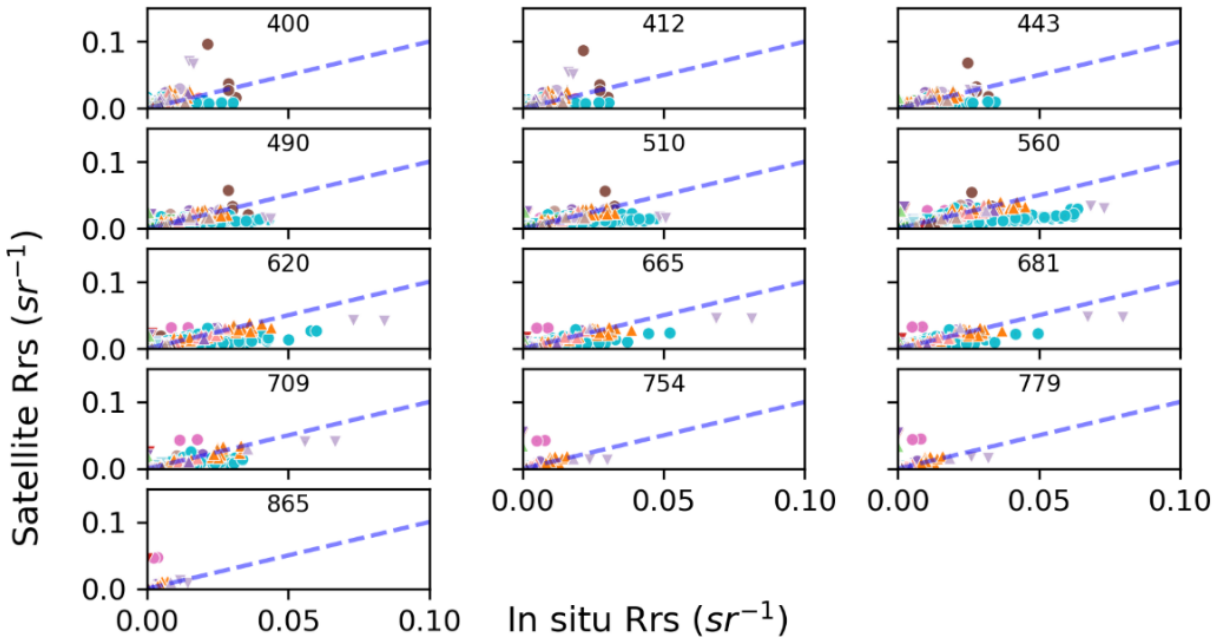


Figure 56: Plots of OLCI Gloria statistics



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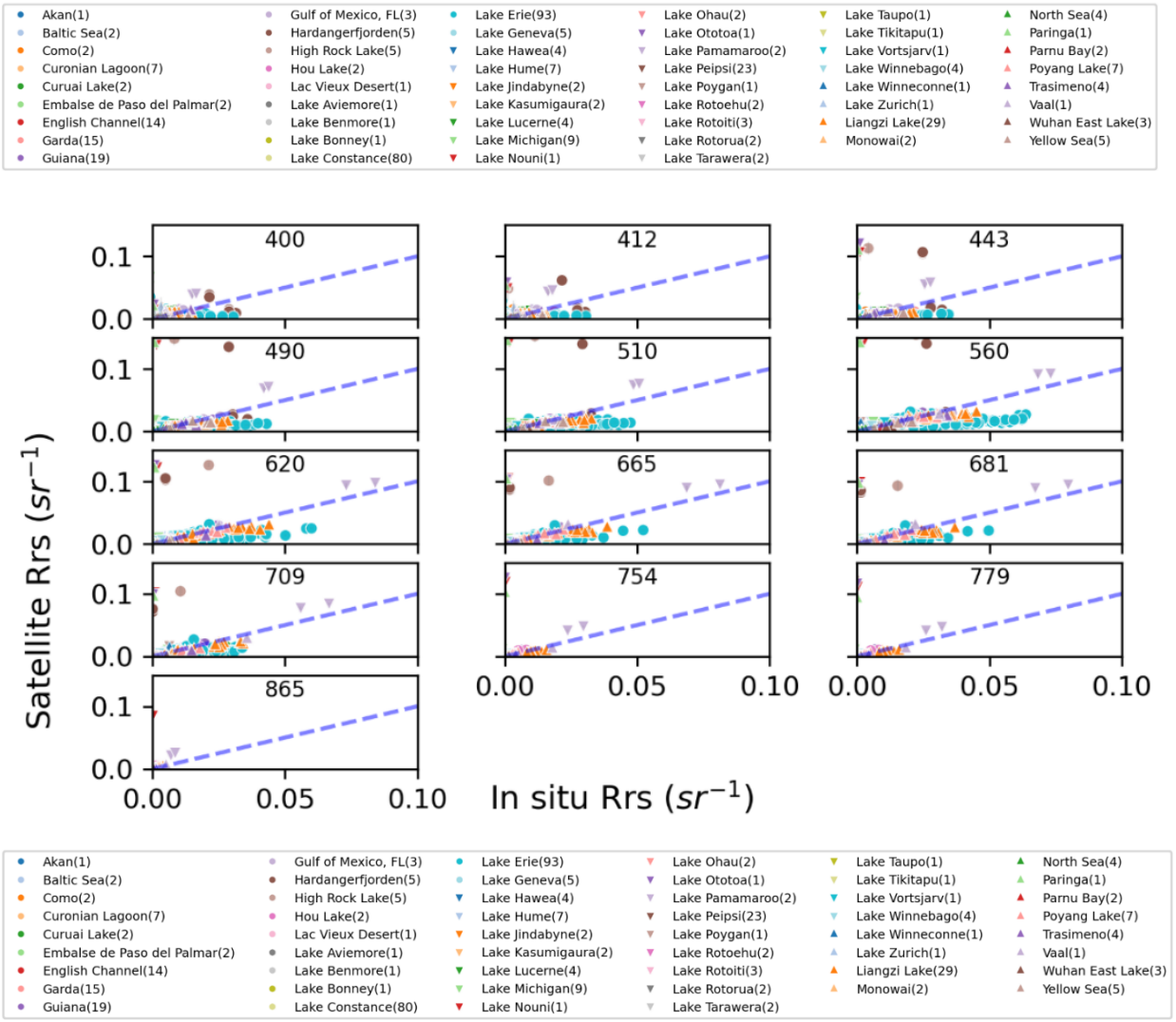


Figure 57: GLORIA Matchups: Scatter plots for R_w bands from (top) Polymer v4.14 and (bottom) Polymer v4.15

WISPCloud database

The statistical analysis (Table 5, Figure 58, Figure 59) of the WISPCloud data suggests that the two Polymer versions perform comparatively well with small improvements noticed in the Polymer v4.15 dataset.

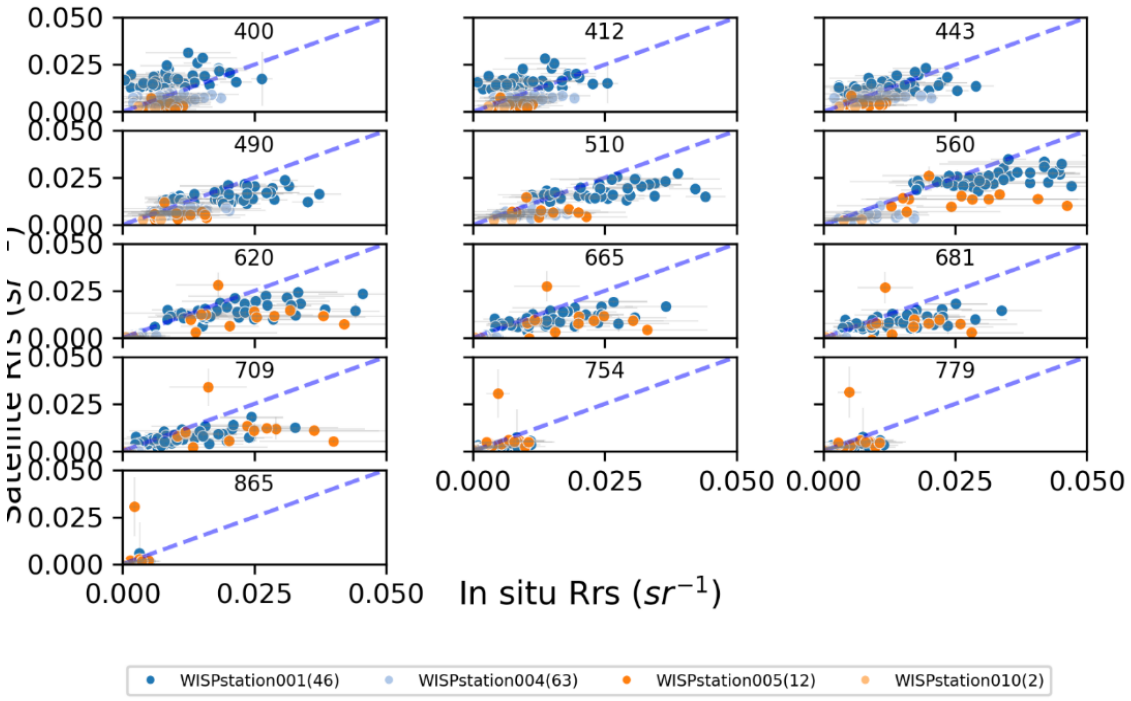
Table 5: Statistics of OLCI WISPCloud matchups

	R_w (n)												
	400	412	443	490	510	560	620	665	681	709	754	779	865

	(123)	(123)	(123)	(123)	(123)	(123)	(123)	(123)	(123)	(123)	(123)	(123)	(123)
bias_v14	0.001	0.0002	-0.0017	-0.0035	-0.0043	-0.0059	-0.0045	-0.0035	-0.0032	-0.0037	-0.0005	-0.0007	-0.0004
bias_v15	-0.0017	-0.002	-0.0028	-0.0039	-0.0046	-0.006	-0.0044	-0.0034	-0.0031	-0.0037	-0.0001	-0.0008	0.0007
rmse_v14	0.0067	0.0058	0.0048	0.006	0.007	0.0101	0.0083	0.0067	0.0061	0.0071	0.0031	0.0032	0.0028
rmse_v15	0.0053	0.005	0.0053	0.0065	0.0074	0.0101	0.0081	0.0064	0.0058	0.0067	0.0021	0.0021	0.0016
mae_v14	0.0053	0.0045	0.0038	0.0043	0.0049	0.0063	0.005	0.0041	0.0037	0.0043	0.0014	0.0014	0.001
mae_v15	0.0041	0.0039	0.0039	0.0046	0.005	0.0063	0.0048	0.0039	0.0035	0.004	0.0013	0.0013	0.0008
r_v14	0.3171	0.3393	0.5105	0.7393	0.8274	0.8740	0.8395	0.7926	0.7744	0.7574	0.4607	0.4771	0.3085
r_v15	0.3169	0.3212	0.4148	0.7240	0.8254	0.8802	0.8610	0.8411	0.8378	0.8309	0.6856	0.7583	0.3844



Figure 58: Plots of statistics for S-3 OLCI WISPCloud for Polymer v4.14 and v4.15



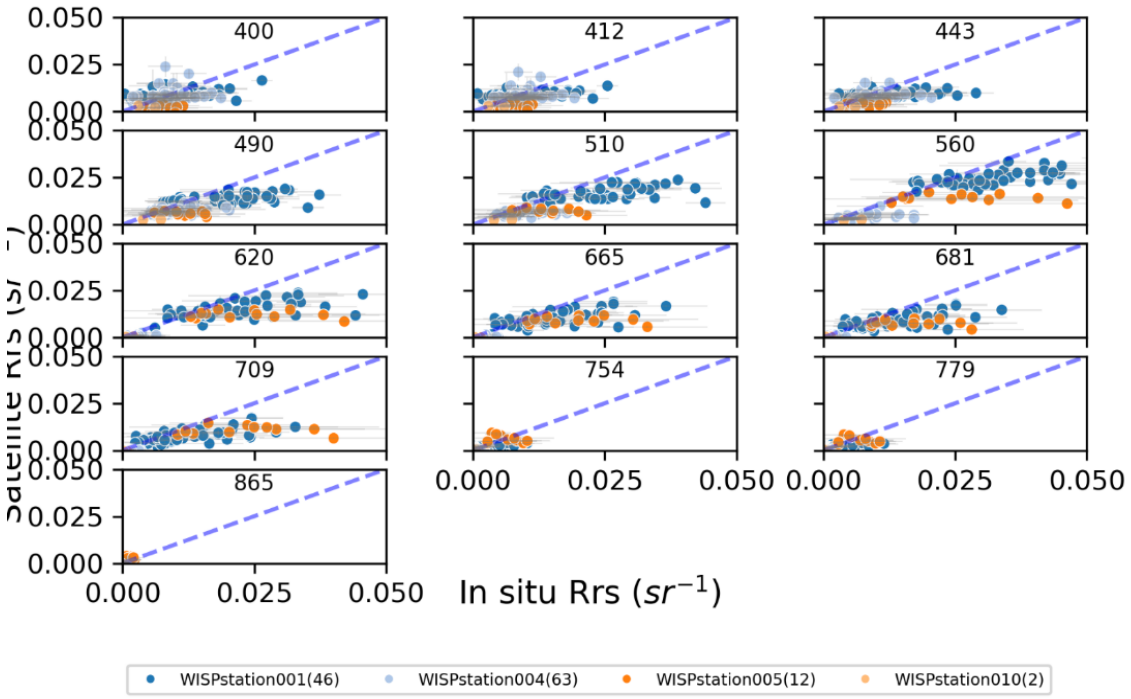


Figure 59: WISPCloud Matchups: Scatterplots for R_w bands from (top) Polymer v4.14 and (bottom) Polymer v4.15

SORAD database

The statistical analysis (Table 6**Error! Reference source not found.**, Figure 60 and Figure 61) of the SORAD data suggests that the two Polymer versions perform comparatively well but with increased RMSE for Polymer v4.15.

Table 6: Statistics of OLCI SORAD matchups

	R_w												
	400 (14177)	**412 (14177)**	**443 (14177)**	**490 (14177)**	**510 (14177)**	**560 (14177)**	**620 (14177)**	**665 (14177)**	**681 (14177)**	**709 (14177)**	**754 (14177)**	**779 (14177)**	**865 (14177)**
bias_v14	-0.0017	-0.0013	-0.0010	-0.0011	-0.0012	-0.0013	-0.0015	-0.0015	-0.0010	-0.0016	0.0001	0.0000	-0.0001
bias_v15	-0.0026	-0.0019	-0.0011	-0.0009	-0.0009	-0.0010	-0.0012	-0.0013	-0.0008	-0.0015	0.0001	0.0001	0.0003
rmse_v14	0.0155	0.0137	0.0109	0.0087	0.0079	0.0073	0.0059	0.0052	0.0048	0.0049	0.0053	0.0051	0.0043
rmse_v15	0.0145	0.0134	0.0130	0.0130	0.0127	0.0118	0.0096	0.0080	0.0074	0.0077	0.0100	0.0095	0.0078
mae_v14	0.0053	0.0048	0.0038	0.0031	0.0029	0.0028	0.0024	0.0023	0.0020	0.0023	0.0014	0.0013	0.0012
mae_v15	0.0053	0.0047	0.0038	0.0032	0.0032	0.0034	0.0030	0.0027	0.0023	0.0027	0.0017	0.0016	0.0012
r_v14	-0.0117	-0.0038	0.0428	0.2287	0.3682	0.7569	0.7499	0.7081	0.6288	0.7965	0.3729	0.3997	0.2908
r_v15	-0.0500	-0.0161	0.0273	0.1008	0.1741	0.5213	0.4951	0.4661	0.4099	0.5841	0.1917	0.1979	0.0853

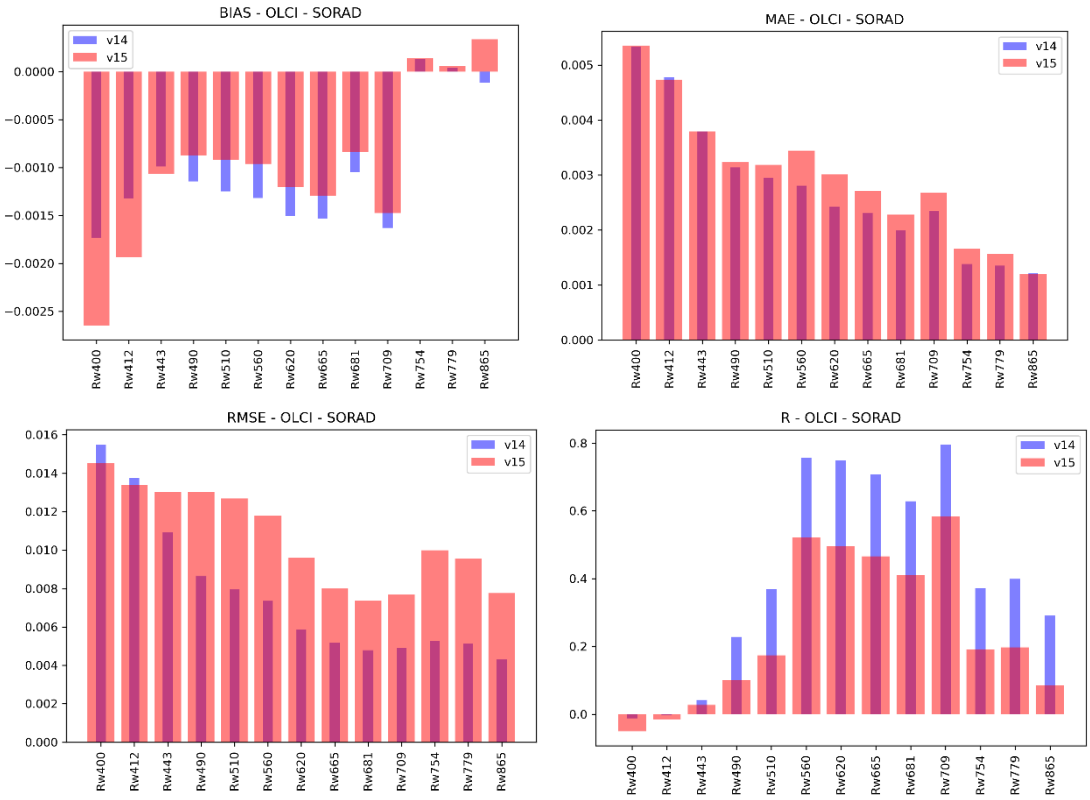
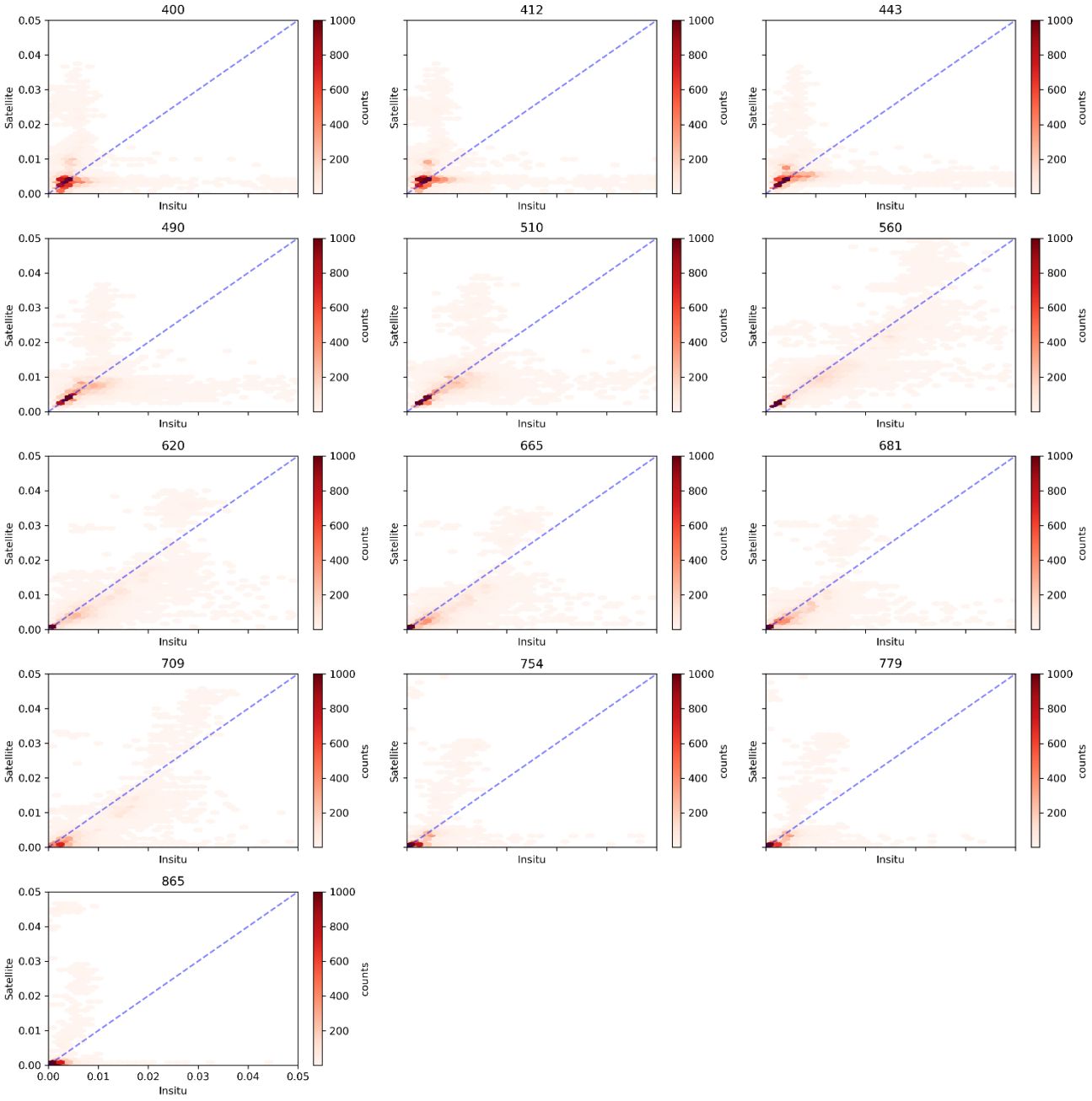


Figure 60: Plots of statistics for OLCI SORAD matchups for Polymer v4.14 and v4.15

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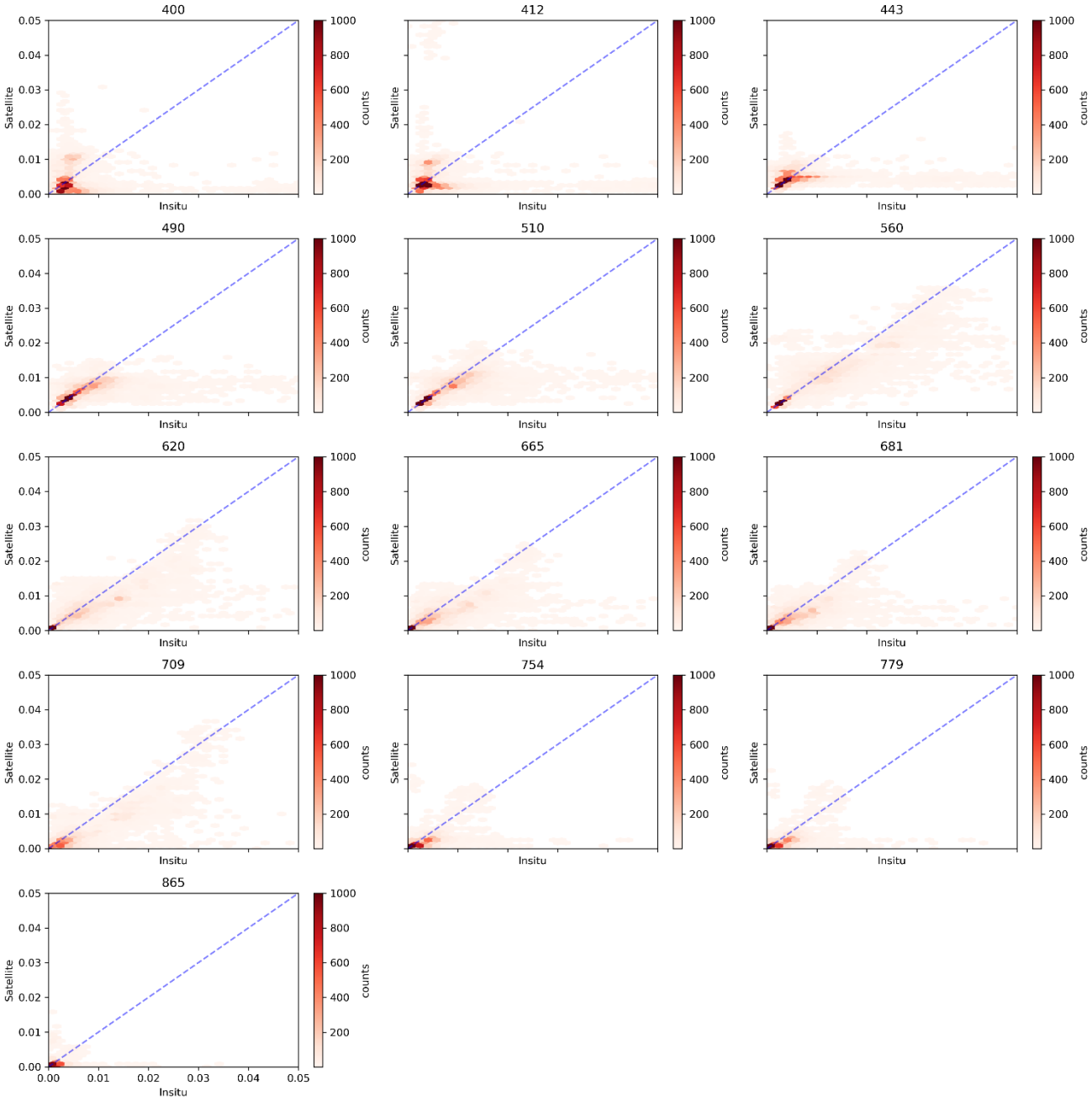


Figure 61: SORAD Matchups: Density heatmaps for R_w bands from (top) Polymer v4.14 and (bottom) Polymer v4.15

4.4.3 Comparison with in situ data – Turbidity - Time series at sampling stations (v1.3)

Time series comparison between in situ data and EO derived parameters show the behaviour of both measurement techniques over time. The focus is on the consistency of the time series on the one hand and on the comparability of the datasets on the other hand. The order of magnitude and seasonal patterns are investigated. For Lake Michigan a number of monitoring stations both in the open parts of the lake and in the coastal area of the City of Milwaukee with river inflow provide in-situ measurements, which were extracted from the US data bases STORET (https://www3.epa.gov/storet/) and WQP (https://www.waterqualitydata.us/).

Lake Michigan is dominated by clear waters falling into two optical water types. Only close to the coast more turbid waters can be found. For demonstration, two time series are shown for a station in the clear lake water and for one station close to the shore. They are marked in Figure 62. The respective time series are shown in Figure 63. The upper plot shows the time series in the central part of the lake with turbidity below 0.8 FNU, the lower plot for the coastal station shows turbidity of up to 5 NTU. In situ data (orange points) and CGLOPS v1.3 Turbidity product (blue crosses) correspond in their magnitude and in some parts in the seasonal cycle. As the CGLOPS turbidity product is a 10-days average, differences are within expectation in the more dynamic area like the coastal station near the river inflow.

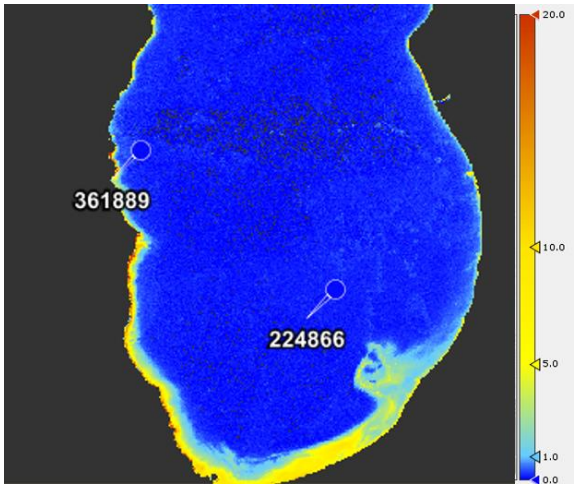


Figure 62: Turbidity (NTU) in Lake Michigan (10D average 20170401-20170410, OLCI). Arrows indicate the positions shown in the time series plots

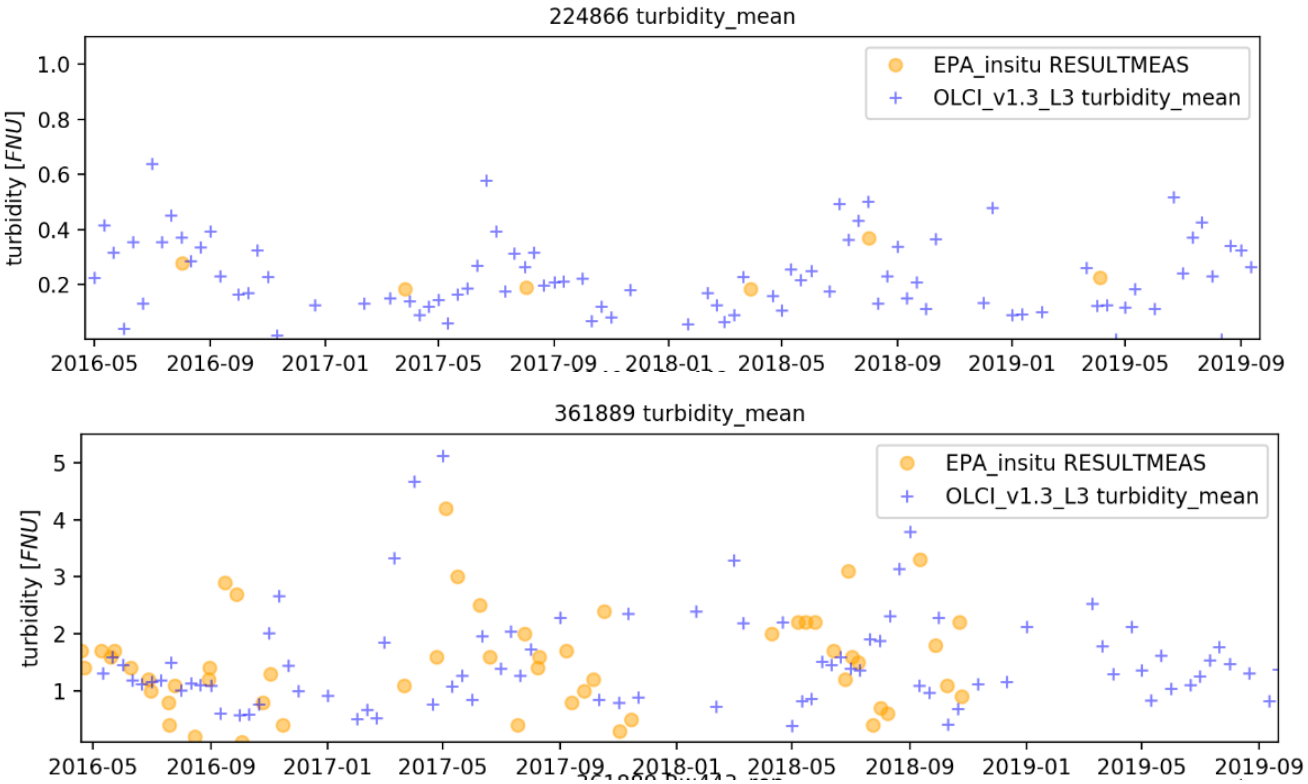


Figure 63: Turbidity time series at two stations in Lake Michigan. Clear waters in central part of the lake (upper panel) and more turbid water close to the coast (lower panel). In situ data: US Data bases STORET (http://www3.epa.gov/storet/) and WQP (http://waterqualitydata.us/portal/)

4.4.4 Comparison with in-situ data – Turbidity (v2.0)

For version 2.0 onward turbidity is calculated using the single band algorithm of Nechad et al. (2010, 2015, 2016), with the input wavelength varied per optical water type. There were insufficient in situ turbidity data available for algorithm tuning. Rather, tuned TSM algorithms are used and the result converted to turbidity using a constant scaling factor of 1.04 (Figure 65). For this same reason there are insufficient in situ data to validate the blended turbidity product directly, so the blended TSM is compared with the GLORIA in situ database before conversion to turbidity (**Figure** 64). Performance of the blended product is stable across the TSM value range.

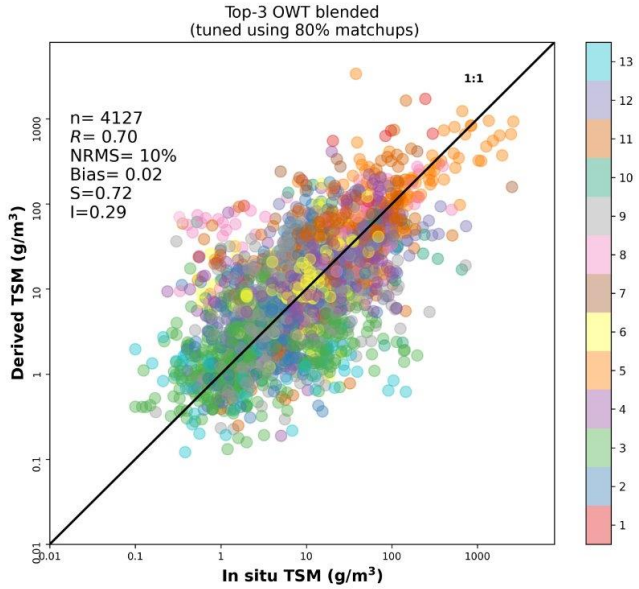


Figure 64. In situ vs blended satellite-derived TSM. Colours indicate dominant OWT number

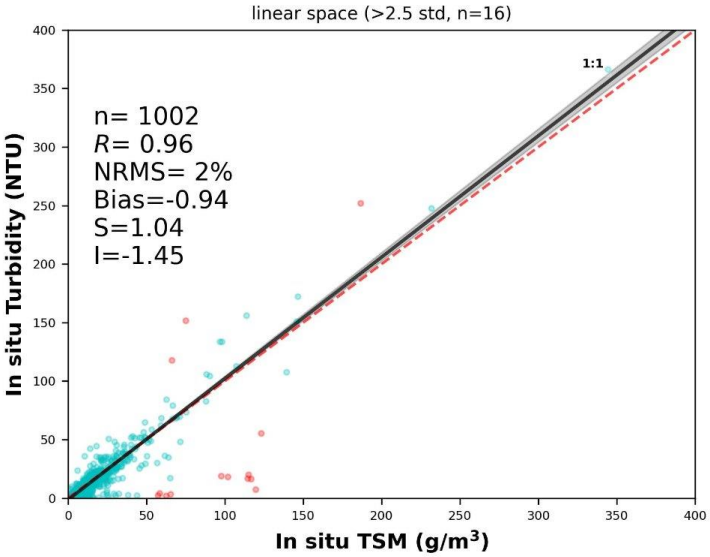


Figure 65. Relationship between in situ TSM and turbidity for the GLORIA dataset. Red points show outliers exceeding 2.5 standard deviations from the mean, that were removed

4.4.5 Comparison with in situ data – Trophic State Index (v1.3)

The chlorophyll-*a* concentration is the input parameter for the trophic state product and can be validated against in situ data. Figure 66 shows the results of the validation for chlorophyll-*a*

algorithms performed in the GloboLakes project using in situ concentration and reflectance data. This is shown for reference as the best-attainable result when using satellite imagery. The scatter plots show the chlorophyll-*a* concentration separated by the different water types and corresponding chlorophyll algorithms. The dataset of satellite matchups and in situ observations is notably smaller and necessitates validation over a ±7-day matchup window. These results, shown in Figure 67 and which do not cover all water types, should be interpreted as worst-case. These show a slight systematic underestimation, and a coefficient of determination R^2=0.62 over the observed range (0-70 mg m^{-3}).

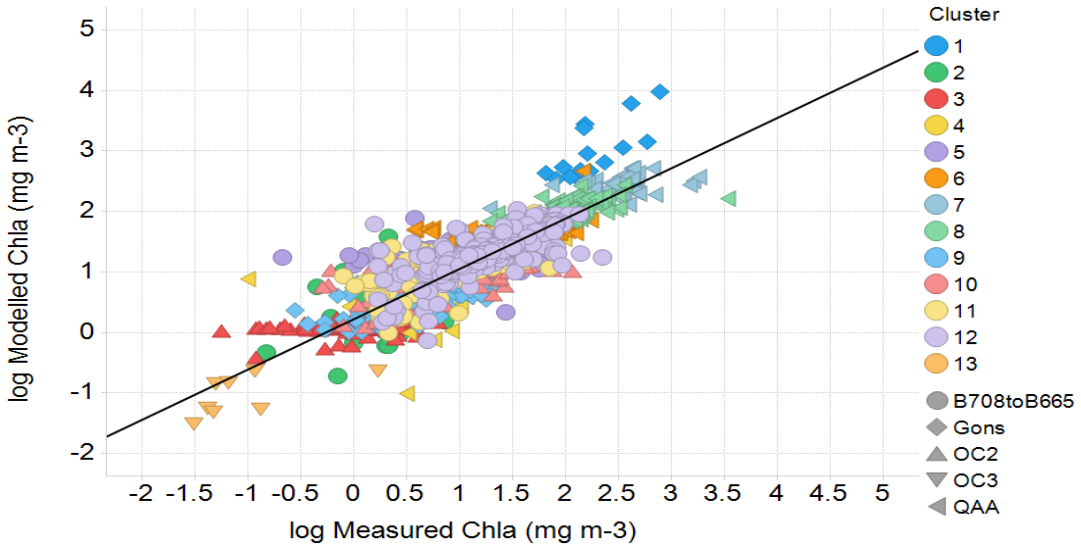


Figure 66: Performance of chlorophyll-a retrieval across all optical water types (clusters 1-13) and associated algorithms following tuning of each algorithm, for each OWT, against the Limnades database (results courtesy University of Stirling, GloboLakes project, Neil et al. 2019)

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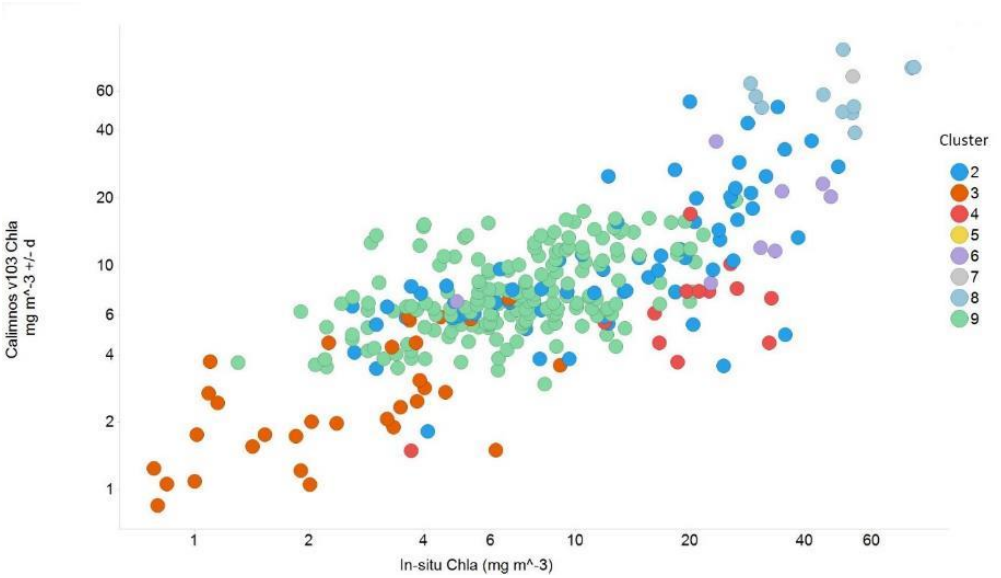


Figure 67: Satellite matchup analysis for chlorophyll-a retrieval in the Calimnos-MERIS processing chain. Linear regression analysis provides R^2=0.62, slope=0.82, intercept=1.16, n=350. To obtain sufficient matchup points, a matchup window of ±7 days is used. Temporal variation in this timeframe can be significant, contributing to scatter in the observed relationship. The result should be interpreted as worst-case

Two examples for a time series of chlorophyll-a concentration in Lake Michigan (USA) are given in Figure 69 where the positions of the stations are shown in the map of a small subset of western Lake Michigan in Figure 68. They show the temporal evolution for a station very close to the shoreline with high chlorophyll values in both datasets (up to 20 mg m^{-3}) and for a station in the transition zone to clear waters with a chlorophyll concentration below 1 mg m^{-3}. For the latter, the chlorophyll concentration is slightly higher in the satellite products than in the in situ data.

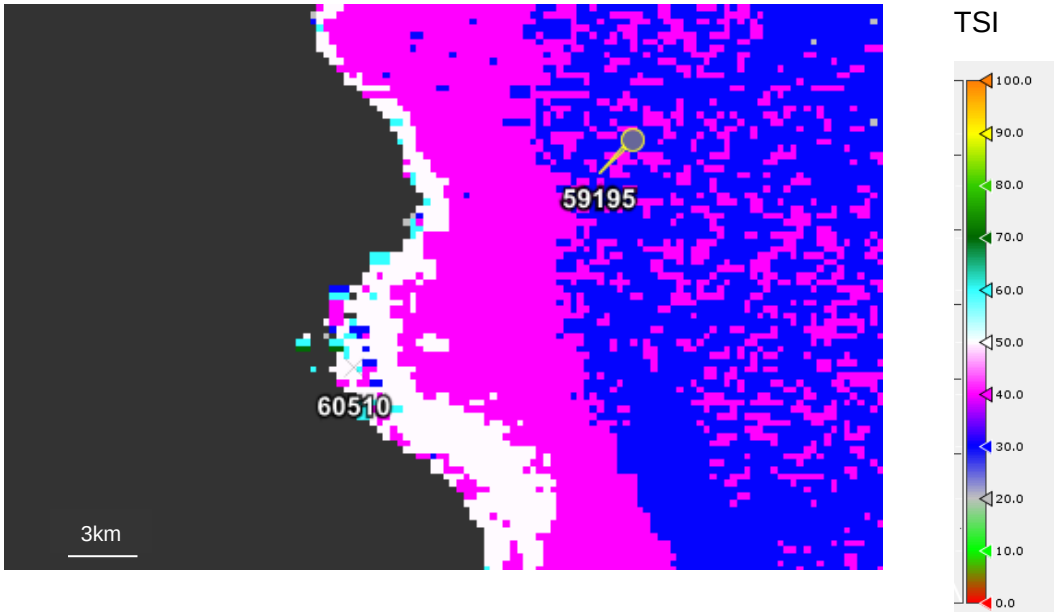


Figure 68: Trophic state index map for Lake Michigan (20161101) with stations of time series extractions close to the coast at Milwaukee and within the transition to clear waters in the centre of the lake

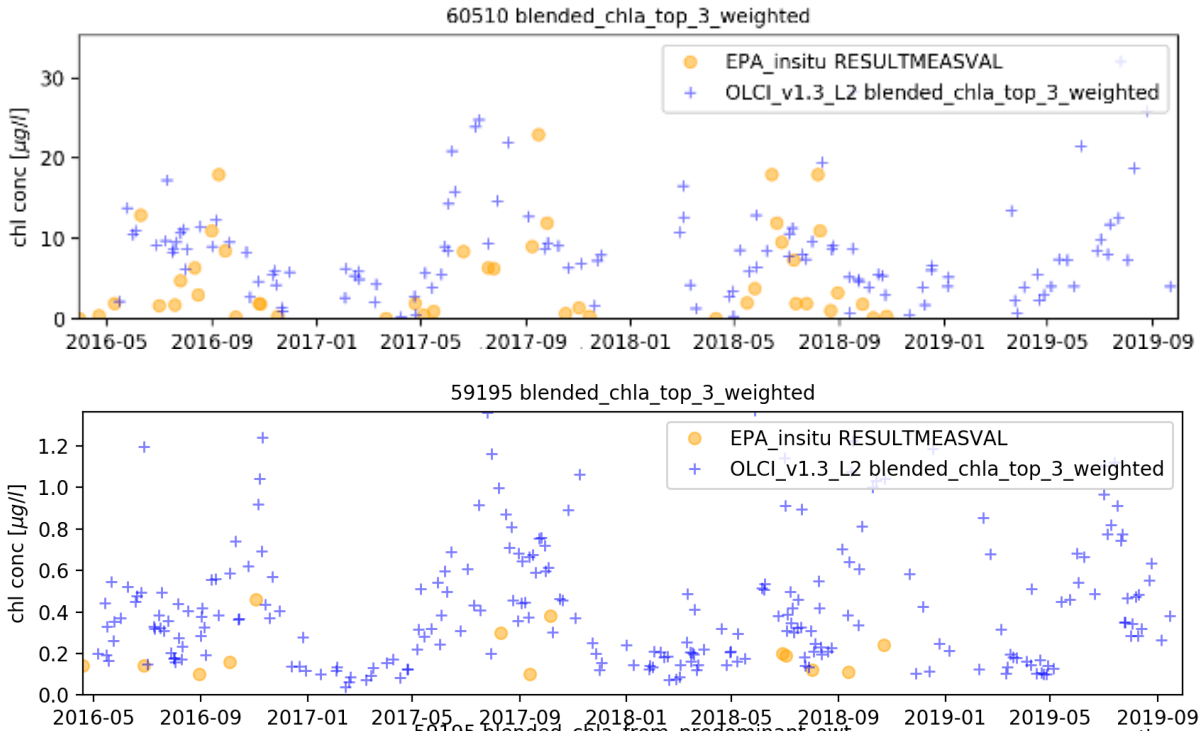


Figure 69: Chlorophyll-a concentration time series at two stations in Lake Michigan from the Calimnos-OLCI processing chain (L2, no temporal aggregation of OLCI products). Coastal station

(upper panel), station in transition zone to clear waters (lower panel). In situ data: US Data bases STORET (http://www3.epa.gov/storet/) and WQP (http://waterqualitydata.us/portal/)

A comparison of the dekad products for TSI against chlorophyll-a measurements from the US data bases (STORET) for water quality data from monitoring programs is shown in Figure 70 for Lake Michigan and Lake Champlain. The shaded area in the figure indicates the definition of TSI based on chlorophyll-a concentration. This result indicates that TSI from satellite may be overestimated based on the corresponding in situ chlorophyll-*a* concentration. It should be noted that the chlorophyll-a concentration range in these lakes is typically < 10 mg m^{-3} which is a range for which chlorophyll-*a* determination is primarily based on the tuned OC2 algorithm. The use of the blue reflectance band in this algorithm is subject to sensitivity to variability dissolved organic matter.

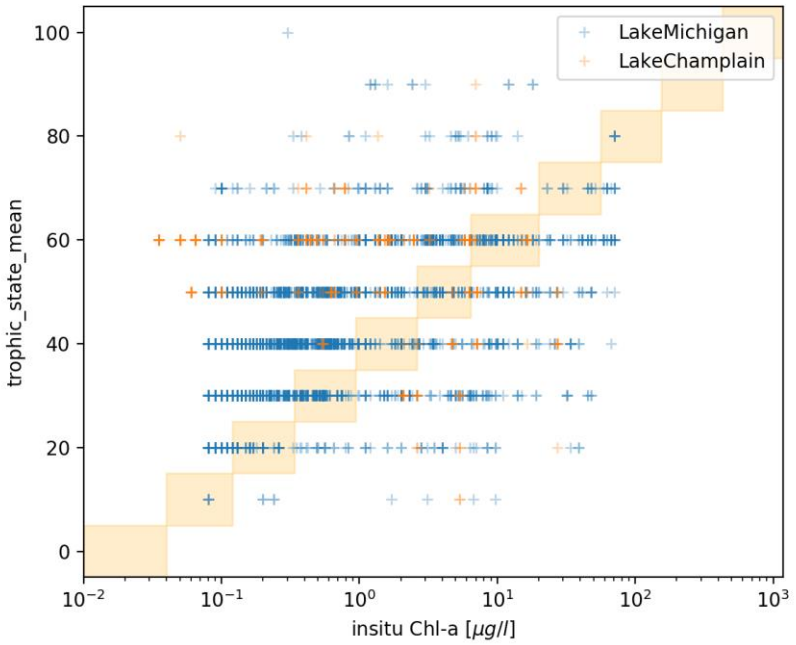


Figure 70: Satellite matchup analysis for trophic state retrieval in the Calimnos-OLCI processing chain (10-day mean). Yellow areas show the definitions of trophic state index and the corresponding ranges of chlorophyll-a concentration. In situ data are measured in Lake Michigan (blue crosses) and Lake Champlain (orange crosses). In situ data: US data bases STORET (http://www3.epa.gov/storet/) and WQP (http://waterqualitydata.us/portal/). Trophic state classes range from 0 and 100, separated in 10 categories corresponding to chlorophyll-a concentration following Carlson (1977)

Finally, time series of TSI for two different stations in Lake Vänern are shown in Figure 42. The TSI in both cases is in agreement in both datasets. However, variability is higher in the satellite products.

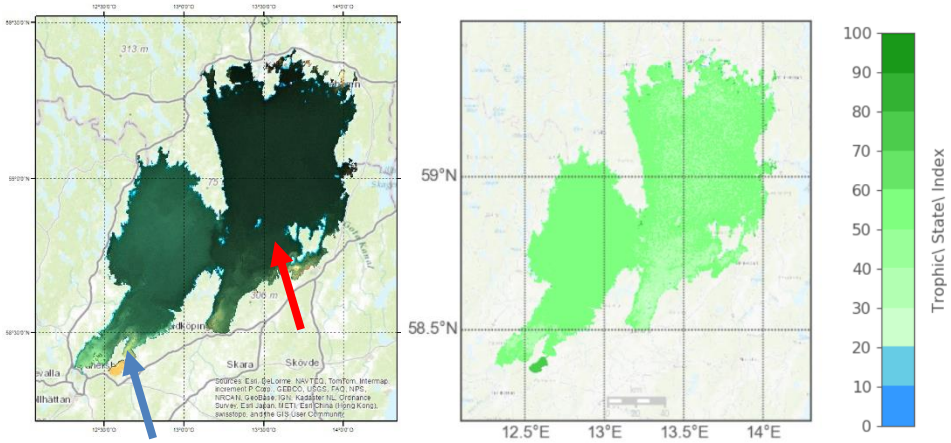


Figure 71: Lake Vänern (left: RGB from lake water reflectance, right: TSI) and position of stations for time series shown in Figure 72: Time series in Lake Vänern showing the consistency with Trophic State Index based on in situ chlorophyll-a concentration. The in situ data are collected within the framework of the coordinated Swedish environmental monitoring program(red arrow for upper panel, blue arrow for lower panel)

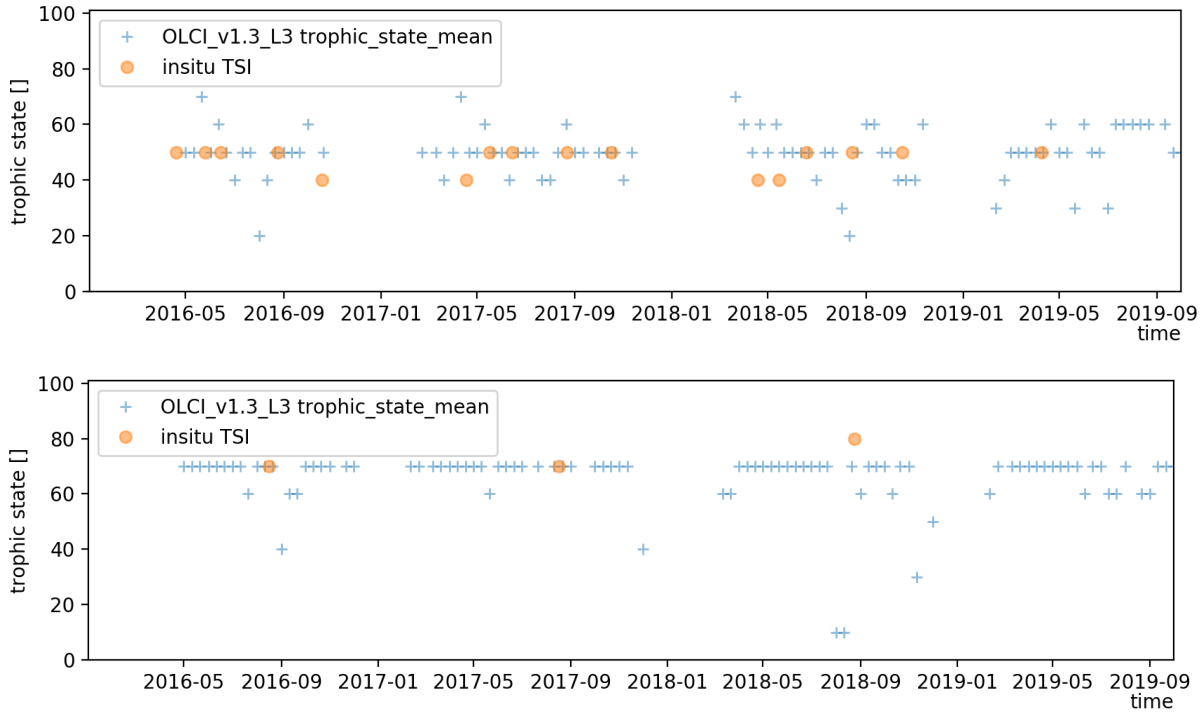


Figure 72: Time series in Lake Vänern showing the consistency with Trophic State Index based on in situ chlorophyll-a concentration. The in situ data are collected within the framework of the coordinated Swedish environmental monitoring programme

4.4.6 Outlook

Validation of the full set of products will be carried out again when v3.0 is introduced to the service. The timeframe for the upgrade to v3.0 is closely linked to developments in the ESA Lakes_cci where algorithm recalibration is carried out.

5 CONCLUSIONS

5.1 SUMMARY

The assessment of sample lakes shows that the products are temporally and, to a large extent, spatially consistent. Seasonal patterns are mostly as expected. Turbidity occasionally shows spatial patterns expected to be caused by a) the merging of algorithms and b) temporal averaging. The first effect should be addressed by further optimizing the optical water type library and associated, tuned algorithms, which does rely on availability of in situ data. The second effect is expected to be a dampening of short-term variation and cannot be avoided in 10-day products, although variability within the observation range (such as the standard deviation) could be added to the product suite. The comparison between the products and in situ data show similar value ranges while the in situ data support relatively few direct comparisons.

The following list provides an overview on the performance of the products with respect to the requirements.

Requirement	**LWQ product specification**
Spatial resolution: 100m and/or 300m and/or 1km.	300m products are included in the current version of the quality assessment. 1km products were discontinued in 2018.
Location accuracy 1/3 of the at-nadir instantaneous field of view.	Fulfilled according to Bicheron et al. 2005 (AMORGOS processing step in the *Calimnos* processing chain).
Coverage: global windows	Fulfilled.
Lake Water Reflectance with an associated accuracy requirement of 30%.	According to the validation performed within GloboLakes, the accuracy of the LSR differs per waveband, with the detrended (unbiased) normalised root-mean-square-error (dNRMSE%) in the range of 15-109%. Best performance is seen in the red and NIR bands with errors of 25%, 19%, 17%, and 15% in bands 665, 709, 754, and 779nm, respectively. These scores are decisive for retrieval of the chlorophyll-a and TSM estimates in a wide range of inland water types. The average dNRMSE% across the spectrum is 46.6%, owing to larger errors in the blue wavebands.
Dekads: days 1 to 10, days 11 to 20 and days 21 to end of month for each month of the year	Fulfilled.

5.2 LIMITATIONS AND KNOWN ISSUES

We first note that some of the spectral bands serve their primary purpose in atmospheric correction and may have been assigned negative values. All results from atmospheric correction are nevertheless included to support the widest possible use of the data. There is a systematic underestimation of Reflectance in the best-performing (in terms of linearity) atmospheric correction processor.

The most crucial assumption in the *Calimnos* processing chain is that optical water types which have been defined from a large set of *in situ* data from optically complex waters (lakes, reservoirs, lagoons, estuaries, and coastal areas) can be assigned successfully to each satellite observation (pixel) containing open water. In practise, atmospheric correction may have systematic errors for some water types, leading to low membership scores for these water types. This, in turn, implies that suboptimal reflectance algorithms for chlorophyll-a and suspended matter may be selected. Solutions for this issue are being explored.

If the pixel identification and thus the flagging of erroneous pixels does not work properly, e.g. at cloud borders, thin clouds or lake ice coverage, the water-leaving reflectance and subsequent water quality parameter retrieval may fail. Users interested in lake averages rather than maps are advised to mask a buffer of one or more pixels from shorelines, where such effects are most difficult to control and flag. An upgrade for the lake ice detection was introduced with version 1.4.0 (in production since June 2020). Version 1.3.0 data (before June 2020) are affected by less performant ice detection, although a static ice mask was applied to mask locations where ice is likely to occur.

The spatial blending of different algorithms and/or the averaging of different, very inhomogeneous days under partial cloud cover can lead to visually inconsistent maps. Lakes may show patchy patterns where coverage on different days of observation was incomplete, and if optical conditions in the lake changed over this period. This is considered normal behaviour for the dekadal product.

The radiometric correction of OLCI on Sentinel 3A is implemented based on system vicarious calibration results. OLCI-B currently follows pre-flight calibrations while results from the tandem flights are pending. OLCI-B gains will be applied from LWQ product version 3.0.

Chlorophyll-a estimation for concentrations of less than 10 mg m^{-3} remains problematic. The state-of-the-art remains an empirically tuned blue-green band ratio algorithm. However, splitting the calibration data into different optical water types may resolve some of the overestimation that is currently observed in large clear lakes with moderate concentrations of dissolved organic matter.

Version 2.0 will solve some of the issues with pixel identification. Version 3.0 will solve issues with high turbid waters and new *IdePix* version will solve not detected ice pixel. Version 3.0 will need a full reprocessing of OLCI timeseries to keep consistency.

6 RECOMMENDATIONS

Manual inspection of all products for more than 4000 water bodies is not feasible and will require extensive in situ data or local knowledge. The validation of the products is therefore based on a small sample of well-studied areas. Users of these products are advised to inspect the results for their area of interest before generating derivative products. This inspection could include, for example, histograms to identify outliers. Users are also advised to take into account the number of observations underlying the results. Where observations are sparse, having a small number of satellite passes to cover a large water body can lead to visual inconsistencies that do not reflect the state of the water body at any particular time – this is the nature of creating aggregate products.

The state-of-the-art in algorithms for optical-biogeochemical remote sensing of inland water products is stronger for turbid, eutrophic and large lakes than it is for smaller and clearer lakes. The state-of-the-art supports the use of remote sensing for trend detection and determining locations and timing of complementary water sampling. For other interpretations of the data it will be advisable to involve remote sensing experts, e.g. working directly with the reflectance data.

Expert users are encouraged to take part in the calibration and validation of these products, which increasingly addresses the global scale. The spatiotemporal coverage and quality of the global lake water products can only be improved if the algorithms underlying these products can be accurately adjusted to waters of each optical type (and in some cases, new water types may need to be defined). We point interested users to the Limnades initiative (https://www.globolakes.stir.ac.uk/limnades.html), from where many of the presented validation results are derived.

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